

2024, Mar. 2024 Release LOC:Acute Adult**General Surgical**Utilization Benchmarks: *Multiple options available, see table below for options***Overview****Select Day****Pre-op Day** ⁽¹⁾**Operative Day** ⁽¹⁵⁾**Post-op Day 1** ⁽¹²⁵⁾**Post-op Day 2** ⁽¹⁵²⁾**Post-op Day 3-21****Post-op Day 22****Discharge Screens** ⁽¹⁹⁰⁾**Utilization Benchmarks****Notes**

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Overview**Instruction:** This subset is for patients requiring hospitalization for the following:

- Preoperative pain management at the Observation level of care prior to an ambulatory procedure for a traumatic injury
- Preoperative pain management and/or intravenous (IV) fluid management at the Acute level of care prior to a designated inpatient surgery for a traumatic injury
- Preoperative care required at the Acute level of care prior to surgery. Post surgical care criteria for those patients who require continued stay are located at the Acute level of care on Operative Day
- Postoperative management of ambulatory procedure complications can be found at the Observation level of care on Operative Day and Post-op Day 1
- Patients requiring rehabilitation consult due to a postoperative functional impairment can be found at the Observation level of care on Operative Day
- Postoperative management of an ambulatory procedure identified as high risk per medical practitioner documentation can be found at the Acute level of care on Operative Day
- Preoperative and postoperative management of solid organ transplants

Inpatient surgical procedures are categorized as follows:

- Moderate stay: expected length of stay 3-5 days
- Long stay: expected length of stay ≥ 6 days

To determine the appropriate category for a given surgical procedure, refer to the examples provided in the notes attached to criteria.

Evaluation and Treatment:

The complexity of the surgical procedure and the presence of comorbidities will influence the rate of postoperative progression. Management of the surgical patient includes:

- Assessment and monitoring of respiratory status, intake and output, laboratory values, pain, and wound appearance
- Pain control
- Activity or mobility progression
- Diet advancement
- Medication administration

- Wound care
- Discharge planning

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Pre-op Day, One:** ⁽¹⁾

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽²⁾● Traumatic injury and, **Both:** ⁽³⁾

- Ambulatory surgery scheduled within 24h

● Pain, severe, **Both:**

- Unresponsive to ≥ 2 doses of analgesic prior to admit decision

- Requiring additional analgesic ≥ 2 doses after admission

● **ACUTE, One:** ⁽⁴⁾

- Solid organ transplant to be performed within 24h of admission

● High risk for thromboembolism, **Both:** ^(5, 6, 7)● Finding, ≥ **One:**● Atrial fibrillation and, ≥ **One:** ⁽⁸⁾⁽⁹⁾ CHADS₂-VASc (congestive heart failure, hypertension, age years, diabetes mellitus, stroke) score > 2

- Rheumatic valvular disease

- Stroke or TIA within last 3 mos

● Mechanical heart valve and, ≥ **One:**

- Mitral valve

- Older aortic valve (e.g., caged-ball, tilting disc)

- Stroke or TIA within last 6 mos

● Venous thromboembolism (VTE) and, ≥ **One:**

- Within last 3 mos

- Severe thrombophilia ⁽¹⁰⁾● Anticoagulant, **One:**

- LMWH

- Fondaparinux

- Unfractionated heparin ⁽¹¹⁾

- Post cardiac catheterization and urgent cardiac surgery scheduled within 24h

● Traumatic injury and, **Both:** ⁽¹²⁾- Designated inpatient surgery scheduled within 24h ⁽¹³⁾● Intervention, ≥ **One:**- IV fluid ⁽¹⁴⁾

- Analgesic ≥ 4x/24h or continuous

● **Operative Day, One:** ⁽¹⁵⁾

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽²⁾● Post ambulatory surgery or procedure and complication, ≥ **One:**● Arrhythmia, **All:**

- Antiarrhythmic (includes PO)

- Continuous cardiac monitoring (excludes Holter)

- Worsening of a previously controlled arrhythmia

● Bleeding, ≥ **One:**● Active bleeding and, ≥ **One:**

- Blood product transfusion

- Factor replacement

- Hct or Hb monitoring at least 2x/24h

● Excessive intraoperative blood loss and, **Both:**

- Hct or Hb monitoring at least 2x/24h

● Intervention, ≥ **One:**

- Blood product transfusion

- IV fluid ⁽¹⁴⁾
- Hematuria and, ≥ **One**:
 - Bladder irrigation, continuous or cyclic
 - Blood product transfusion
 - Hct or Hb monitoring at least 2x/24h
 - IV fluid ⁽¹⁴⁾
- Delayed recovery from anesthesia and, ≥ **One**: ⁽¹⁶⁾
 - Assessment and monitoring at least 6x/24h and, ≥ **One**:
 - Breathing
 - Level of consciousness
 - Mobility
 - Sensation
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(17, 18)
- Inability to remove sheath post cardiac catheterization and, ≥ **One**: ⁽¹⁹⁾
 - ACT > 180 sec
 - PTT > 60 sec
- Post open fracture repair and prophylactic anti-infective ^(20, 21, 22)
- Pain uncontrolled and, **Both**:
 - Pain assessment ⁽²³⁾
 - Requiring change in medication, dose, or frequency
- Spinal headache and, ≥ **One**:
 - Epidural blood patch performed within last 24h ⁽²⁴⁾
 - IV caffeine and fluid ⁽²⁵⁾
- Unable to void and requiring urinary catheter
- Vomiting, protracted and, **Both**: ⁽²⁶⁾
 - Antiemetic, ≥ **One**:
 - Antiemetic ≥ 3x/24h
 - Serotonin antagonist ≥ 2x/24h
 - IV fluid ⁽¹⁴⁾
- Post-operative functional impairment requiring rehabilitation initiation ≤ 24h ^(27, 28)
- **ACUTE**, ≥ **One**: ⁽⁴⁾
 - Surgery and designated inpatient setting or prior authorization ⁽²⁹⁾
 - Ambulatory procedure (excludes low risk procedure) and high risk for complications requiring inpatient management, **Both**: ^(30, 31, 32)
 - Physician documentation supports the need for post procedure management for at least 2d ⁽³⁰⁾
 - Comorbid or high risk medical history, ≥ **One**: ^(33, 34)
 - American Society of Anesthesiologists (ASA) class III or IV ^(35, 36)
 - Severe valvular heart disease ⁽³⁷⁾
 - Ascites within 30d of surgery ⁽³⁸⁾
 - COPD (excludes asthma, diffuse interstitial fibrosis, or sarcoidosis), ≥ **One**:
 - Functional disability from COPD ⁽³⁹⁾
 - Chronic bronchodilator therapy (includes oral or inhaled agents)
 - FEV1 of < 75%(0.75) of predicted ⁽⁴⁰⁾
 - Diabetes Mellitus and HbA1c ≥ 7%(0.07) ⁽⁴¹⁾
 - ESRD and chronic dialysis ^(42, 43)
 - Congestive heart failure exacerbation within 30d of surgery ⁽⁴⁴⁾
 - Immunocompromised ⁽⁴⁵⁾
 - Malnutrition ^(46, 47)
 - BMI ≥ 40 kg/m² ⁽⁴⁸⁾
 - Coagulopathy ⁽⁴⁹⁾
 - Frailty, **Both**: ^(50, 51)
 - Age ≥ 65 yrs
 - Finding, ≥ **One**:

- Frailty, active diagnosis ⁽⁵²⁾
- Frailty characteristics, ≥ **Two**: ⁽⁵³⁾
 - Chronic condition(s), ≥ **One**:
 - Dementia or cognitive impairment ⁽⁵⁴⁾
 - Multimorbidity, ≥ 2 chronic illnesses ^(55, 56)
 - Functional impairment ^(57, 58)
 - Malnutrition ⁽⁵⁹⁾
 - Polypharmacy, ≥ 5 chronic medications ^(60, 61)
- High risk for thromboembolism, **Both**: ^(5, 6, 7)
 - Finding, ≥ **One**:
 - Atrial fibrillation and, ≥ **One**: ⁽⁸⁾
 - ⑨ CHADS₂-VASc (congestive heart failure, hypertension, age years, diabetes mellitus, stroke) score > 2
 - Rheumatic valvular disease
 - Stroke or TIA within last 3 mos
 - Mechanical heart valve and, ≥ **One**:
 - Mitral valve
 - Older aortic valve (e.g., caged-ball, tilting disc)
 - Stroke or TIA within last 6 mos
 - Venous thromboembolism (VTE) and, ≥ **One**:
 - Within last 3 mos
 - Severe thrombophilia ⁽¹⁰⁾
 - Anticoagulant, **One**:
 - LMWH
 - Fondaparinux
 - Unfractionated heparin ⁽¹¹⁾
 - Post external fixation of open fracture awaiting internal fixation ≤ 5d ^(62, 63)
- **INTERMEDIATE**, ≥ **One**: ⁽⁶⁴⁾
 - Troponin positive (use **Acute Coronary Syndrome subset**) ^(65, 66)
 - Bleeding and, **Both**:
 - Finding, ≥ **One**:
 - Chest pain
 - Dyspnea
 - Heart rate > 100/min, sustained, and orthostatic hypotension and, **Both**: ⁽⁶⁷⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
 - Blood pressure, ≥ **One**:
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Blood product transfusion
 - Complex wound care ≤ 5d and, **Both**: ⁽⁶⁸⁾
 - ≥ 3x/24h
 - ≥ 30 min in duration
 - Hyperkalemia, **All**:
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, ≥ **One**: ⁽⁶⁹⁾
 - Peaked T waves
 - PVCs > 6/min
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
 - Hypokalemia, **All**: ^(71, 72)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min

- Potassium repletion
- Antiarrhythmic and hemodynamically stable
- Neurological impairment or condition, **Both:** ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h, ≤ 2d ⁽⁷⁴⁾
- NIPPV ⁽⁷⁵⁾
- Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁷⁾
- Post neurosurgery and neurological assessment every 3-4h, ≤ 2d ⁽⁷⁴⁾
- Post procedure continuous cardiac monitoring (excludes Holter) ⁽⁷⁶⁾
- Risk factor, **Both:**
 - Designated inpatient setting ⁽⁷⁷⁾
- Finding, ≥ **One:**
 - COPD and, **All:**
 - Bronchodilator ≥ 4x/24h ⁽⁷⁸⁾
 - Corticosteroid (includes PO or inhaled) ⁽⁷⁹⁾
 - Respiratory interventions every 3-4h ⁽⁸⁰⁾
 - New York Heart Association Functional Classification (NYHA) Class III or IV and continuous cardiac monitoring (excludes Holter) ⁽⁸¹⁾
 - Pre-existing VAD and continuous cardiac monitoring (excludes Holter)
 - Sleep apnea requiring NIPPV ^(75, 82)
- **CRITICAL, ≥ One:** ⁽⁸³⁾
 - Troponin positive (**use Acute Coronary Syndrome subset**) ^(65, 66)
- Solid organ transplant, **Both:**
 - Organ, ≥ **One:**
 - Heart within the last 5d
 - Heart-lung within the last 6d
 - Liver within the last 3d
 - Lung within the last 6d
 - Pancreas within the last 2d
 - Small bowel within the last 10d
 - Expected post-op course, **All:**
 - Immunosuppressant (includes PO) ^(84, 85)
 - NPO and IV fluid ⁽¹⁴⁾
 - Intake and output monitoring
 - Assessment of graft function ⁽⁸⁶⁾
 - Pain assessment, **One:** ⁽²³⁾
 - Patient denies pain
 - Pain controlled ⁽⁸⁷⁾
 - Uncontrolled pain requiring change in medication, dose, or frequency (includes PO)
- Acute humoral rejection and apheresis ^(88, 89)
- Acute kidney injury (AKI) and, **Both:** ⁽⁹⁰⁾
 - Finding, ≥ **One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁹¹⁾
 - Widening QRS
 - Sodium, **One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - Calcium chloride
 - Calcium gluconate

- Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Diuretic, continuous
- Dialysis, **One**:
 - Hemodialysis $\leq 7d$ since initiation
 - Peritoneal dialysis $\leq 7d$ since initiation
 - IV fluid resuscitation $\leq 2d$ ⁽⁹²⁾
- CRRT ⁽⁹³⁾
- Arrhythmia, $\geq$ **One**:
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
- Antiarrhythmic and, **Both**:
 - Arrhythmia, $\geq$ **One**:
 - Recurrent arrhythmia ⁽⁹⁴⁾
 - Refractory arrhythmia ⁽⁹⁴⁾
 - Symptomatic, $\geq$ **One**:
 - Chest pain
 - Heart failure and, $\geq$ **One**:
 - Arterial $PO_2 < 56$ mmHg(7.4 kPa)
 - O_2 sat $< 89\%$ (0.89) and $<$ baseline
 - Hemodynamic instability, $\geq$ **One**:
 - Heart rate $> 120/\text{min}$, sustained ⁽⁶⁷⁾
 - Systolic blood pressure, $\geq$ **One**:
 - < 90 mmHg
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽¹⁸⁾
 - Labile ⁽⁹⁵⁾
 - MAP < 65 mmHg
- Bronchodilator, $\geq$ **One**: ⁽⁹⁶⁾
 - Continuous
 - Every 1-2h
- Cardiac arrest and post resuscitation care $\leq 2d$
- Cerebral edema and, $\geq$ **One**:
 - ICP monitoring (includes OR placement)
 - Corticosteroid
 - Hyperosmolar therapy, $\geq$ **One**: ⁽⁹⁷⁾
 - Mannitol
 - Hypertonic saline $\geq 3\%$ (0.03) solution
- Complex wound care and, $\geq$ **One**:
 - Every 1-2h
 - Requiring procedural sedation ⁽⁹⁸⁾
- Compromised airway with artificial airway placement (excludes nasal trumpet) and monitoring every 1-2h
- DIC, **All**: ⁽⁹⁹⁾
 - Fibrinogen degradation product (FDP) increasing
 - Fibrinogen decreasing
 - Blood product transfusion
- Digit or limb reimplantation and monitoring every 1-2h
- External ventricular or lumbar drainage and neurological assessment every 1-2h ⁽⁷⁴⁾
- ECMO or ECLS ⁽¹⁰⁰⁾
- Hemodynamic instability, **All**: ⁽¹⁰¹⁾
 - Initial treatment, $\geq$ **One**:
 - ≥ 2 units blood product transfusion
 - ≥ 2 liters IV fluid ⁽⁹²⁾

- ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
- Findings after initial treatment, $\geq$ **One**:
 - Heart rate $> 120/\text{min}$
 - Systolic blood pressure < 100 mmHg and $<$ baseline
 - MAP < 65 mmHg
 - Requiring ongoing resuscitation
- Hemodynamic monitoring, invasive ⁽¹⁰²⁾
- Hypercalcemia, **Both**: ^(103, 104)
 - Calcium ≥ 14.0 mg/dL(3.5 mmol/L)
- Intervention, $\geq$ **One**:
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
- Treatment, **Both**:
 - Calcitonin
 - Saline hydration ^(14, 105)
- Hypocalcemia, **Both**: ^(104, 106)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Hyperkalemia, **Both**: ⁽⁶⁹⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁹¹⁾
 - Widening QRS
 - Intervention, $\geq$ **One**:
 - Calcium chloride or gluconate
 - Dialysis, **One**:
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Hypokalemia, **All**: ^(71, 72, 107)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - Electrocardiogram (ECG) findings, $\geq$ **One**: ⁽¹⁰⁷⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽¹⁰⁸⁾
 - T and U wave fusion ⁽¹⁰⁹⁾
 - U wave ⁽¹¹⁰⁾
 - Potassium repletion ⁽⁷²⁾
- Hypomagnesemia, **Both**:
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- Hypernatremia, **Both**: ⁽¹¹¹⁾
 - Sodium > 160 mEq/L(160 mmol/L)
 - Intervention, $\geq$ **One**:
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(14, 112)
- Hyponatremia, **Both**: ⁽¹¹³⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ⁽¹¹⁴⁾
 - Intervention, $\geq$ **One**:
 - Conivaptan or tolvaptan (includes PO) ⁽¹¹⁵⁾
 - Diuretic $\geq 2\times/24\text{h}$
 - Fluid restriction

- Sodium chloride 3%(0.03) ⁽¹¹⁶⁾
- Hypophosphatemia, **Both:** ⁽¹¹⁷⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- IABP ⁽¹¹⁸⁾
- Major procedure requiring post surgical care ⁽¹¹⁹⁾
- Malignant hyperthermia and, **Both:** ⁽¹²⁰⁾
 - Dantrolene
 - Temperature regulation and monitoring ⁽¹²¹⁾
- Mechanical ventilation
- Necrotizing fasciitis and, ≥ **Two:** ⁽¹²²⁾
 - Anti-infectives ≥ 2 drugs
 - Complex dressing every 2-4h requiring procedural sedation ⁽⁹⁸⁾
 - Return to OR for additional debridement, scheduled or performed within 24h
- Neurological impairment or condition, **Both:** ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 1-2h, ≤ 2d ⁽⁷⁴⁾
- NIPPV and, ≥ **One:** ⁽⁷⁵⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Post neurosurgery and neurological assessment every 1-2h ⁽⁷⁴⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, new ^(123, 124)
- **Post-op Day 1, One:** ⁽¹²⁵⁾
 - **OBSERVATION, One:** ⁽²⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹²⁶⁾
 - Functional status, **One:** ⁽²⁷⁾
 - Able to perform ADLs or return to baseline ⁽¹⁸⁾
 - Functional impairment and discharge to home, **One:** ⁽¹²⁷⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Functional impairment and discharge to inpatient facility for rehabilitative services
 - Pain controlled or manageable ⁽⁸⁷⁾
 - Tolerating PO or nutritional route established ^(128, 129)
 - Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹³⁰⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One:** ⁽¹³⁰⁾
 - Arrhythmia controlled
 - Bleeding controlled or resolved
 - Open fracture repair and, **Both:**
 - Prophylactic anti-infective completed
 - Wound without signs of infection
 - Passing urine without urinary retention
 - Postoperative vomiting resolved
 - Recovered from anesthesia, **All:**
 - Mobility and coordination at baseline
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽¹³¹⁾
 - Sensation intact
 - Stable level of consciousness
 - Sheath removed
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**

- Post ambulatory surgery or procedure and complication requiring management, ≥ **One**:
 - Arrhythmia, **All**:
 - Antiarrhythmic (includes PO)
 - Continuous cardiac monitoring (excludes Holter)
 - Worsening of a previously controlled arrhythmia
 - Bleeding, ≥ **One**:
 - Active bleeding and, ≥ **One**:
 - Blood product transfusion
 - Factor replacement
 - Hct or Hb monitoring at least 2x/24h
 - Excessive intraoperative blood loss and, **Both**:
 - Hct or Hb monitoring at least 2x/24h
 - Intervention, ≥ **One**:
 - Blood product transfusion
 - IV fluid ⁽¹⁴⁾
 - Hematuria and, ≥ **One**:
 - Bladder irrigation, continuous or cyclic
 - Blood product transfusion
 - Hct or Hb monitoring at least 2x/24h
 - IV fluid ⁽¹⁴⁾
 - Delayed recovery from anesthesia and, ≥ **One**: ⁽¹⁶⁾
 - Assessment and monitoring at least 6x/24h and, ≥ **One**:
 - Breathing
 - Level of consciousness
 - Mobility
 - Sensation
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(17, 18)
 - Inability to remove sheath post cardiac catheterization and, ≥ **One**: ⁽¹⁹⁾
 - ACT > 180 sec
 - PTT > 60 sec
 - Post open fracture repair and prophylactic anti-infective ^(20, 21, 22)
 - Pain uncontrolled and, **Both**:
 - Pain assessment ⁽²³⁾
 - Requiring change in medication, dose, or frequency
 - Spinal headache and, ≥ **One**:
 - Epidural blood patch performed within last 24h ⁽²⁴⁾
 - IV caffeine and fluid ⁽²⁵⁾
 - Unable to void and requiring urinary catheter
 - Vomiting, protracted and, **Both**: ⁽²⁶⁾
 - Antiemetic, ≥ **One**:
 - Antiemetic ≥ 3x/24h
 - Serotonin antagonist ≥ 2x/24h
 - IV fluid ⁽¹⁴⁾
- ACUTE, **One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽¹²⁶⁾
 - Afebrile
 - Able to perform ADLs or return to baseline ⁽¹⁸⁾
 - Pain controlled or manageable ⁽⁸⁷⁾
 - Tolerating PO or nutritional route established ^(128, 129)
 - Complication or comorbidity, **One**:
 - No complication or active comorbidity relevant to this episode of care ⁽¹³⁰⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One**: ⁽¹³⁰⁾
 - Bleeding controlled or resolved

- Chest tube removed or Heimlich valve
- Electrolytes within acceptable limits ⁽¹³¹⁾
- Heart failure resolving
- Passing urine without urinary retention
- O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽¹³¹⁾
- VAD, established
- **Partial responder**, not clinically stable for discharge and requires continued stay, **One**:
 - Post Ambulatory procedure and high risk for complications determined on Operative Day requiring post operative management at least 2d ⁽¹³²⁾
- Routine post-op stay, **One**:
 - Moderate stay within the last 5d ⁽¹³³⁾
 - Long stay within the last 6d ⁽¹³⁴⁾
 - Post external fixation of open fracture awaiting internal fixation ≤ 5d ^(62, 63)
- Solid organ transplant, ≥ **One**:
 - Heart within the last 11d
 - Heart-lung within the last 15d
 - Kidney within the last 6d
 - Liver within the last 11d
 - Lung within the last 13d
 - Pancreas within the last 8d
 - Small bowel within the last 18d
- Post surgery or invasive procedure and complication, ≥ **One**:
 - Bleeding actual or suspected and, **Both**:
 - Finding, ≥ **One**:
 - Heart rate 100-120/min, sustained ⁽⁶⁷⁾
 - Hct or Hb decreased within last 24h
 - Orthostatic hypotension, **Both**:
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
 - Blood pressure, ≥ **One**:
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Intervention, ≥ **One**:
 - Blood product transfusion
 - Return to OR within last 24h
 - Requiring Hct or Hb monitoring at least 2x/24h, ≤ 2d
- Chest tube and, ≥ **One**: ⁽¹³⁵⁾
 - Suction, continuous
 - Drainage ≥ 250 mL/24h
 - Repositioning within last 24h ⁽¹³⁶⁾
 - Water seal within last 24h
- Heart failure and, **Both**: ⁽¹³⁷⁾
 - Dyspnea > baseline and, ≥ **One**: ⁽¹⁸⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁸⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽¹³⁸⁾
 - Intervention, ≥ **One**: ⁽¹³⁹⁾
 - Diuretic, **One**: ⁽¹⁴⁰⁾
 - ≥ 2x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One**: ⁽¹⁴¹⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹⁸⁾
 - Creatinine increased from baseline ⁽¹⁸⁾
 - Dialysis
- Hypercalcemia, **Both**: ^(103, 104)

- Calcium 11.1-13.9 mg/dL(2.78-3.48 mmol/L)
- **Intervention, ≥ One:**
 - Bisphosphonate ⁽¹⁴²⁾
 - Corticosteroid (includes PO) ⁽¹⁴²⁾
 - Saline hydration ^(14, 105)
- **Hypocalcemia, Both:** ^(104, 106)
 - Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L)
 - Calcium repletion
- **Hyperkalemia, All:**
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(143, 144)
- **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
 - IV fluid ⁽¹⁴⁾
 - Potassium binding agent (includes PO or PR) ⁽¹⁴⁵⁾
- **Hypokalemia, All:** ^(71, 72)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - No electrocardiogram (ECG) changes ^(143, 144)
 - Potassium repletion (includes PO)
- **Hypermagnesemia, Both:**
 - Magnesium > 2.5 mg/dL(1.03 mmol/L)
- **Intervention, ≥ One:**
 - Diuretic ≥ 2x/24h (includes PO)
 - IV fluid ⁽¹⁴⁾
- **Hypomagnesemia, Both:**
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Magnesium repletion
- **Hypernatremia, Both:** ⁽¹¹¹⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L)
 - IV fluid ^(14, 112)
- **Hyponatremia or SIADH, Both:** ⁽¹⁴⁶⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
- **Intervention, ≥ One:**
 - Diuretic and oral sodium supplement
 - Fluid restriction
- **Medication administration, ≥ One:**
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
 - Sodium chloride 0.9%(0.009) ⁽¹¹⁶⁾
- **Hypophosphatemia, Both:** ⁽¹¹⁷⁾
 - Phosphate 1.0-2.0 mg/dL(0.32-0.65 mmol/L)
 - Phosphate repletion (includes PO)
- **Ileus and, One:** ⁽¹⁴⁷⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and IV fluid ≤ 5d ⁽¹⁴⁾
- Inability to void requiring urinary catheter
- **Inadequate oral intake and, ≥ One:** ⁽¹⁴⁸⁾
 - IV fluid ⁽¹⁴⁾
 - Advancing diet as tolerated
- **Pain uncontrolled and, Both:**
 - Pain assessment ⁽²³⁾

- Requiring change in medication, dose, or frequency ^(141, 149)
- Respiratory impairment (including pneumonia) and, **Both:**
 - Finding, ≥ **One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁵⁰⁾
 - Inability to clear secretions ≤ 24h
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁸⁾
 - Intervention, ≥ **One:**
 - Anti-infective (includes PO)
 - Requiring supplemental oxygen ⁽¹⁷⁾
 - Respiratory interventions or incentive spirometry at least 6x/24h ⁽⁸⁰⁾
- Spinal headache and, ≥ **One:**
 - Epidural blood patch performed within last 24h ⁽²⁴⁾
 - IV caffeine and fluid ⁽²⁵⁾
 - Post Critical care ≤ 24h
- INTERMEDIATE, ≥ **One:**
 - Arrhythmia, ≥ **One:**
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽¹⁵¹⁾
 - Bleeding and, **Both:**
 - Finding, ≥ **One:**
 - Chest pain
 - Dyspnea
 - Heart rate > 100/min, sustained, and orthostatic hypotension and, **Both:** ⁽⁶⁷⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
 - Blood pressure, ≥ **One:**
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Blood product transfusion
 - Complex wound care ≤ 5d and, **Both:** ⁽⁶⁸⁾
 - ≥ 3x/24h
 - ≥ 30 min in duration
 - External ventricular or lumbar drainage and neurological assessment every 3-4h ⁽⁷⁴⁾
 - Hyperkalemia, **All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, ≥ **One:** ⁽⁶⁹⁾
 - Peaked T waves
 - PVCs > 6/min
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
 - Hypokalemia, **All:** ^(71, 72)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
 - Neurological impairment or condition, **Both:** ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h, ≤ 2d ⁽⁷⁴⁾
 - NIPPV ⁽⁷⁵⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁷⁾
 - Permanent pacemaker placement within last 24h
 - Post neurosurgery and neurological assessment every 3-4h, ≤ 2d ⁽⁷⁴⁾

● Risk factor, **Both**:

- Designated inpatient setting ⁽⁷⁷⁾

● Finding, ≥ **One**:● COPD and, **All**:

- Bronchodilator ≥ 4x/24h ⁽⁷⁸⁾
- Corticosteroid (includes PO or inhaled) ⁽⁷⁹⁾
- Respiratory interventions every 3-4h ⁽⁸⁰⁾

- New York Heart Association Functional Classification (NYHA) Class III or IV and continuous cardiac monitoring (excludes Holter) ⁽⁸¹⁾

- Pre-existing VAD and continuous cardiac monitoring (excludes Holter)

- Sleep apnea requiring NIPPV ^(75, 82)

● **CRITICAL**, ≥ **One**:● Solid organ transplant, **Both**:● Organ, ≥ **One**:

- Heart within the last 5d
- Heart-lung within the last 6d
- Liver within the last 3d
- Lung within the last 6d
- Pancreas within the last 2d
- Small bowel within the last 10d

● Expected post-op course, **All**:

- Immunosuppressant (includes PO) ^(84, 85)
- NPO and IV fluid ⁽¹⁴⁾
- Intake and output monitoring
- Assessment of graft function ⁽⁸⁶⁾

● Pain assessment, **One**: ⁽²³⁾

- Patient denies pain
- Pain controlled ⁽⁸⁷⁾
- Uncontrolled pain requiring change in medication, dose, or frequency (includes PO)

- Acute humoral rejection and apheresis ^(88, 89)

● Acute kidney injury (AKI) and, **Both**: ⁽⁹⁰⁾● Finding, ≥ **One**:● Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:

- Loss of P wave
- Multifocal PVCs
- Neuromuscular deficit ⁽⁹¹⁾
- Widening QRS

● Sodium, **One**:

- < 120 mEq/L(120 mmol/L) and volume overload
- > 160 mEq/L(160 mmol/L)

● Intervention, ≥ **One**:

- Calcium chloride
- Calcium gluconate
- Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Diuretic, continuous

● Dialysis, **One**:

- Hemodialysis ≤ 7d since initiation
- Peritoneal dialysis ≤ 7d since initiation
- IV fluid resuscitation ≤ 2d ⁽⁹²⁾

- CRRT ⁽⁹³⁾

● Arrhythmia, ≥ **One**:

- Urgent electrical cardioversion performed within last 24h
- Defibrillation performed within last 24h

- Temporary pacemaker insertion performed within last 2d
- Antiarrhythmic and, **Both:**
 - Arrhythmia, ≥ **One:**
 - Recurrent arrhythmia ⁽⁹⁴⁾
 - Refractory arrhythmia ⁽⁹⁴⁾
 - Symptomatic, ≥ **One:**
 - Chest pain
 - Heart failure and, ≥ **One:**
 - Arterial $\text{Po}_2 < 56 \text{ mmHg}$ (7.4 kPa)
 - $\text{O}_2 \text{ sat} < 89\%$ (0.89) and $< \text{baseline}$
 - Hemodynamic instability, ≥ **One:**
 - Heart rate $> 120/\text{min}$, sustained ⁽⁶⁷⁾
 - Systolic blood pressure, ≥ **One:**
 - $< 90 \text{ mmHg}$
 - $< 110 \text{ mmHg}$ with chronic hypertension
 - $> 30 \text{ mmHg}$ decrease from baseline ⁽¹⁸⁾
 - Labile ⁽⁹⁵⁾
 - $\text{MAP} < 65 \text{ mmHg}$
- Bronchodilator, ≥ **One:** ⁽⁹⁶⁾
 - Continuous
 - Every 1-2h
- Cardiac arrest and post resuscitation care $\leq 2\text{d}$
- Cerebral edema and, ≥ **One:**
 - ICP monitoring (includes OR placement)
 - Corticosteroid
- Hyperosmolar therapy, ≥ **One:** ⁽⁹⁷⁾
 - Mannitol
 - Hypertonic saline $\geq 3\%$ (0.03) solution
- Complex wound care and, ≥ **One:**
 - Every 1-2h
 - Requiring procedural sedation ⁽⁹⁸⁾
- Compromised airway with artificial airway placement (excludes nasal trumpet) and monitoring every 1-2h
- DIC and blood product transfusion ⁽⁹⁹⁾
- Digit or limb reimplantation and monitoring every 1-2h
- External ventricular or lumbar drainage and neurological assessment every 1-2h ⁽⁷⁴⁾
- ECMO or ECLS ⁽¹⁰⁰⁾
- Hemodynamic instability persisting after $\geq 1\text{L}$ of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** ⁽¹⁰¹⁾
 - Finding, ≥ **One:**
 - Heart rate $> 120/\text{min}$
 - Systolic blood pressure $< 100 \text{ mmHg}$ and $< \text{baseline}$
 - $\text{MAP} < 65 \text{ mmHg}$
 - Intervention, ≥ **One:**
 - Blood product transfusion
 - IV fluid resuscitation $\geq 1\text{L} \leq 24\text{h}$ ⁽⁹²⁾
- Hemodynamic monitoring, invasive ⁽¹⁰²⁾
- Hypercalcemia, **Both:** ^(103, 104)
 - Calcium $\geq 14.0 \text{ mg/dL}$ (3.5 mmol/L)
- Intervention, ≥ **One:**
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
- Treatment, **Both:**
 - Calcitonin

- Saline hydration ^(14, 105)
- Hypocalcemia, **Both:** ^(104, 106)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Hyperkalemia, **Both:** ⁽⁶⁹⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁹¹⁾
 - Widening QRS
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - Dialysis, **One:**
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Hypokalemia, **All:** ^(71, 72, 107)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - Electrocardiogram (ECG) findings, ≥ **One:** ⁽¹⁰⁷⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽¹⁰⁸⁾
 - T and U wave fusion ⁽¹⁰⁹⁾
 - U wave ⁽¹¹⁰⁾
 - Potassium repletion ⁽⁷²⁾
- Hypomagnesemia, **Both:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- Hypernatremia, **Both:** ⁽¹¹¹⁾
 - Sodium > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(14, 112)
- Hyponatremia, **Both:** ⁽¹¹³⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ⁽¹¹⁴⁾
 - Intervention, ≥ **One:**
 - Conivaptan or tolvaptan (includes PO) ⁽¹¹⁵⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽¹¹⁶⁾
- Hypophosphatemia, **Both:** ⁽¹¹⁷⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- IABP ⁽¹¹⁸⁾
- Malignant hyperthermia and, **Both:** ⁽¹²⁰⁾
 - Dantrolene
 - Temperature regulation and monitoring ⁽¹²¹⁾
- Mechanical ventilation
- Necrotizing fasciitis and, ≥ **Two:** ⁽¹²²⁾
 - Anti-infectives ≥ 2 drugs
 - Complex dressing every 2-4h requiring procedural sedation ⁽⁹⁸⁾
 - Return to OR for additional debridement, scheduled or performed within 24h
- Neurological impairment or condition, **Both:** ⁽⁷³⁾

- Acute onset or deterioration
- Neurological assessment every 1-2h, $\leq 2d$ ⁽⁷⁴⁾
- NIPPV and, $\geq$ **One:** ⁽⁷⁵⁾
 - Arterial $PO_2 \leq 39$ mmHg(5.2 kPa)
 - Arterial or venous $Pco_2 \geq 60$ mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous $pH \leq 7.24$
 - No hx of COPD
- Post neurosurgery and neurological assessment every 1-2h ⁽⁷⁴⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, new ^(123, 124)
- **Post-op Day 2, One:** ⁽¹⁵²⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹²⁶⁾
 - Able to perform ADLs or return to baseline ⁽¹⁸⁾
 - Pain controlled or manageable ⁽⁸⁷⁾
 - Tolerating PO or nutritional route established ^(128, 129)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹³⁰⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:** ⁽¹³⁰⁾
 - Arrhythmia controlled
 - Bleeding controlled or resolved
 - **Open fracture repair and, Both:**
 - Prophylactic anti-infective completed
 - Wound without signs of infection
 - Passing urine without urinary retention
 - Postoperative vomiting resolved
 - **Recovered from anesthesia, All:**
 - Mobility and coordination at baseline
 - O_2 sat $\geq 92\%(0.92)$ or within acceptable limits ⁽¹³¹⁾
 - Sensation intact
 - Stable level of consciousness
 - Sheath removed
 - **Non-responder**, not clinically stable for discharge and exceeds post-op days at the Observation level of care for this condition, **One:**
 - Use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
- **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹²⁶⁾
 - Afebrile
 - Able to perform ADLs or return to baseline ⁽¹⁸⁾
 - Pain controlled or manageable ⁽⁸⁷⁾
 - Tolerating PO or nutritional route established ^(128, 129)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹³⁰⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:** ⁽¹³⁰⁾
 - **AKI, $\geq$ One:**
 - Kidney function improving or within acceptable limits ⁽¹³¹⁾
 - Outpatient dialysis regimen established
 - Anticoagulation regimen established ^(153, 154, 155)
 - Bleeding controlled or resolved
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽¹⁵⁶⁾
 - Chest tube removed or Heimlich valve
 - Electrolytes within acceptable limits ⁽¹³¹⁾

- Functional impairment, **One:** ⁽²⁷⁾
 - Discharge to home, **One:** ⁽¹²⁷⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Heart failure resolving
 - Neurological stability ⁽¹⁵⁷⁾
 - Passing urine without urinary retention
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽¹³¹⁾
 - VAD, established
 - Vital signs within acceptable limits ⁽¹³¹⁾
 - Wound regimen established and adequate healing
- **Partial responder**, not clinically stable for discharge and requires continued stay, **One:** ⁽¹⁵⁸⁾
 - Routine post-op stay, **One:**
 - Moderate stay within the last 5d ⁽¹³³⁾
 - Long stay within the last 6d ⁽¹³⁴⁾
 - Post external fixation of open fracture awaiting internal fixation ≤ 5d ^(62, 63)
 - Solid organ transplant, ≥ **One:**
 - Heart within the last 11d
 - Heart-lung within the last 15d
 - Kidney within the last 6d
 - Liver within the last 11d
 - Lung within the last 13d
 - Pancreas within the last 8d
 - Small bowel within the last 18d
 - Complication or comorbid condition, ≥ **One:**
 - Abnormal vital signs, **Both:**
 - Finding, ≥ **One:**
 - Heart rate > 100/min and > baseline ⁽¹⁸⁾
 - Systolic blood pressure > 140 mmHg and > baseline ^(18, 159)
 - Medication adjustment (includes PO) ⁽¹⁴¹⁾
 - Bleeding actual or suspected and, **Both:**
 - Finding, ≥ **One:**
 - Heart rate 100-120/min, sustained ⁽⁶⁷⁾
 - Hct or Hb decreased within last 24h
 - Orthostatic hypotension, **Both:**
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
 - Blood pressure, ≥ **One:**
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Intervention, ≥ **One:**
 - Blood product transfusion
 - Return to OR within last 24h
 - Requiring Hct or Hb monitoring at least 2x/24h, ≤ 2d
 - *C. difficile* infection, actual or suspected and, **Both:** ⁽¹⁶⁰⁾
 - Anti-infective (includes PO) ≤ 5d ^(161, 162)
 - Inadequate oral intake ⁽¹⁶³⁾
 - Chest tube and, ≥ **One:** ⁽¹³⁵⁾
 - Suction, continuous
 - Drainage ≥ 250 mL/24h
 - Repositioning within last 24h ⁽¹³⁶⁾
 - Water seal within last 24h
 - Fever and anti-infective and, **One:**

- Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ ⁽¹⁵⁰⁾
- Functional impairment (new) requiring rehabilitation initiation $\leq 24\text{h}$ and, $\geq$ **One:** ^(164, 165)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
- Graft dysfunction and immunosuppressant and, $\geq$ **One:** ^(84, 85, 166)
 - IV
 - Conversion from IV to PO $\leq 2\text{d}$ ⁽¹⁶⁷⁾
 - Medication adjustment (includes PO) ^(168, 169)
- Heart failure and, **Both:** ⁽¹³⁷⁾
 - Dyspnea $>$ baseline and, $\geq$ **One:** ⁽¹⁸⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline ⁽¹⁸⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽¹³⁸⁾
 - Intervention, $\geq$ **One:** ⁽¹³⁹⁾
 - Diuretic, **One:** ⁽¹⁴⁰⁾
 - $\geq 2\text{x}/24\text{h}$ (includes PO)
 - At least $1\text{x}/24\text{h}$ and requires daily medication adjustment (includes PO) and, $\geq$ **One:** ⁽¹⁴¹⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹⁸⁾
 - Creatinine increased from baseline ⁽¹⁸⁾
 - Dialysis
- Hypercalcemia, **Both:** ^(103, 104)
 - Calcium $11.1\text{-}13.9\text{ mg/dL}(2.78\text{-}3.48\text{ mmol/L})$
 - Intervention, $\geq$ **One:**
 - Bisphosphonate ⁽¹⁴²⁾
 - Corticosteroid (includes PO) ⁽¹⁴²⁾
 - Saline hydration ^(14, 105)
- Hypocalcemia, **Both:** ^(104, 106)
 - Calcium $5.0\text{-}7.4\text{ mg/dL}(1.25\text{-}1.85\text{ mmol/L})$
 - Calcium repletion
- Hyperkalemia, **All:**
 - Potassium $> 5.5\text{ mEq/L}(5.5\text{ mmol/L})$
 - No electrocardiogram (ECG) changes ^(143, 144)
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose $50\%(0.50)$ with insulin ⁽⁷⁰⁾
 - IV fluid ⁽¹⁴⁾
 - Potassium binding agent (includes PO or PR) ⁽¹⁴⁵⁾
- Hypokalemia, **All:** ^(71, 72)
 - Potassium $< 3.0\text{ mEq/L}(3.0\text{ mmol/L})$
 - No electrocardiogram (ECG) changes ^(143, 144)
 - Potassium repletion (includes PO)
- Hypermagnesemia, **Both:**
 - Magnesium $> 2.5\text{ mg/dL}(1.03\text{ mmol/L})$
 - Intervention, $\geq$ **One:**
 - Diuretic $\geq 2\text{x}/24\text{h}$ (includes PO)
 - IV fluid ⁽¹⁴⁾
- Hypomagnesemia, **Both:**
 - Magnesium $1.0\text{-}1.4\text{ mg/dL}(0.41\text{-}0.58\text{ mmol/L})$
 - Magnesium repletion

- **Hypernatremia, Both:** ⁽¹¹¹⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L)
 - IV fluid ^(14, 112)
- **Hyponatremia or SIADH, Both:** ⁽¹⁴⁶⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
 - **Intervention, ≥ One:**
 - Diuretic and oral sodium supplement
 - Fluid restriction
 - **Medication administration, ≥ One:**
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
 - Sodium chloride 0.9%(0.009) ⁽¹¹⁶⁾
- **Hypophosphatemia, Both:** ⁽¹¹⁷⁾
 - Phosphate 1.0-2.0 mg/dL(0.32-0.65 mmol/L)
 - Phosphate repletion (includes PO)
- **Ileus and, One:** ⁽¹⁴⁷⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and IV fluid ≤ 5d ⁽¹⁴⁾
- **Immunosuppressant level abnormal or signs of toxicity and medication adjustment (includes PO)** ^(168, 169)
- **Inadequate oral intake and, ≥ One:** ⁽¹⁴⁸⁾
 - IV fluid ⁽¹⁴⁾
 - Advancing diet as tolerated
- **Neurological impairment or condition and, Both:** ⁽⁷³⁾
 - **Finding, ≥ One:**
 - Acute onset
 - Deterioration
 - Neurological assessment at least 3x/24h, ≤ 2d ⁽⁷⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(17, 18)
- **Pain uncontrolled and, Both:**
 - Pain assessment ⁽²³⁾
 - Requiring change in medication, dose, or frequency ^(141, 149)
- **Respiratory impairment (including pneumonia) and, Both:**
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁵⁰⁾
 - Inability to clear secretions ≤ 24h
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁸⁾
 - **Intervention, ≥ One:**
 - Anti-infective (includes PO)
 - Requiring supplemental oxygen ⁽¹⁷⁾
 - Respiratory interventions or incentive spirometry at least 6x/24h ⁽⁸⁰⁾
- **Spinal headache and, ≥ One:**
 - Epidural blood patch performed within last 24h ⁽²⁴⁾
 - IV caffeine and fluid ⁽²⁵⁾
- **Therapeutic anticoagulation, including DVT and, One:**
 - LMWH ≤ 3d ⁽¹⁷⁰⁾
 - UFH, continuous ≤ 5d ⁽¹¹⁾
 - HIT, suspected or by history and anticoagulation ⁽¹⁷¹⁾
 - **Pretreatment or bridging anticoagulation, All:** ⁽¹⁷²⁾
 - Patient or caregiver unable to administer
 - Outpatient services unavailable ⁽¹⁷³⁾
 - **Intervention, One:**
 - INR non-therapeutic and anticoagulation (includes PO) ⁽¹⁷⁴⁾

- Anticoagulation $\leq 5d$ since initiation ^(175, 176)
- Wound, $\geq$ **One**:
 - Complex wound care $\leq 5d$ and, $\geq$ **One**: ⁽⁶⁸⁾
 - Wound care $\geq 3x/24h$ and, $\geq$ **One**:
 - Requiring parenteral analgesic
 - ≥ 15 min in duration
 - Wound management teaching $\leq 24h$ ⁽¹⁷⁷⁾
 - Return to OR within last 24h ⁽¹⁷⁸⁾
 - Post Critical care $\leq 24h$
- **INTERMEDIATE**, $\geq$ **One**:
 - Arrhythmia, $\geq$ **One**:
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One**:
 - No proarrhythmic risk $\leq 24h$
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) $\leq 3d$ ⁽¹⁵¹⁾
 - Bleeding and, **Both**:
 - Finding, $\geq$ **One**:
 - Chest pain
 - Dyspnea
 - Heart rate $> 100/min$, sustained, and orthostatic hypotension and, **Both**: ⁽⁶⁷⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
 - Blood pressure, $\geq$ **One**:
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Blood product transfusion
 - Complex wound care $\leq 5d$ and, **Both**: ⁽⁶⁸⁾
 - $\geq 3x/24h$
 - ≥ 30 min in duration
 - External ventricular or lumbar drainage and neurological assessment every 3-4h ⁽⁷⁴⁾
 - Hyperkalemia, **All**:
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, $\geq$ **One**: ⁽⁶⁹⁾
 - Peaked T waves
 - PVCs $> 6/min$
 - Intervention, $\geq$ **One**:
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
 - Hypokalemia, **All**: ^(71, 72)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs $> 6/min$
 - Potassium repletion
 - Neurological impairment or condition, **Both**: ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h, $\leq 2d$ ⁽⁷⁴⁾
 - NIPPV ⁽⁷⁵⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽¹⁷⁾
 - Permanent pacemaker placement within last 24h
 - Post neurosurgery and neurological assessment every 3-4h, $\leq 2d$ ⁽⁷⁴⁾
 - Risk factor, **Both**:
 - Designated inpatient setting ⁽⁷⁷⁾
 - Finding, $\geq$ **One**:
 - COPD and, **All**:
 - Bronchodilator $\geq 4x/24h$ ⁽⁷⁸⁾

- Corticosteroid (includes PO or inhaled) ⁽⁷⁹⁾
- Respiratory interventions every 3-4h ⁽⁸⁰⁾
- New York Heart Association Functional Classification (NYHA) Class III or IV and continuous cardiac monitoring (excludes Holter) ⁽⁸¹⁾
- Pre-existing VAD and continuous cardiac monitoring (excludes Holter)
- Sleep apnea requiring NIPPV ^(75, 82)

● **CRITICAL, ≥ One:**

● Solid organ transplant, **Both:**

● Organ, ≥ **One:**

- Heart within the last 5d
- Heart-lung within the last 6d
- Liver within the last 3d
- Lung within the last 6d
- Small bowel within the last 10d

● Expected post-op course, **All:**

- Immunosuppressant (includes PO) ^(84, 85)
- NPO and IV fluid ⁽¹⁴⁾
- Intake and output monitoring
- Assessment of graft function ⁽⁸⁶⁾

● Pain assessment, **One:** ⁽²³⁾

- Patient denies pain
- Pain controlled ⁽⁸⁷⁾
- Uncontrolled pain requiring change in medication, dose, or frequency (includes PO)

- Acute humoral rejection and apheresis ^(88, 89)

● Acute kidney injury (AKI) and, **Both:** ⁽⁹⁰⁾

● Finding, ≥ **One:**

● Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**

- Loss of P wave
- Multifocal PVCs
- Neuromuscular deficit ⁽⁹¹⁾
- Widening QRS

● Sodium, **One:**

- < 120 mEq/L(120 mmol/L) and volume overload
- > 160 mEq/L(160 mmol/L)

● Intervention, ≥ **One:**

- Calcium chloride
- Calcium gluconate
- Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Diuretic, continuous

● Dialysis, **One:**

- Hemodialysis ≤ 7d since initiation
- Peritoneal dialysis ≤ 7d since initiation
- IV fluid resuscitation ≤ 2d ⁽⁹²⁾

- CRRT ⁽⁹³⁾

● Arrhythmia, ≥ **One:**

- Urgent electrical cardioversion performed within last 24h
- Defibrillation performed within last 24h
- Temporary pacemaker insertion performed within last 2d

● Antiarrhythmic and, **Both:**

● Arrhythmia, ≥ **One:**

- Recurrent arrhythmia ⁽⁹⁴⁾
- Refractory arrhythmia ⁽⁹⁴⁾

● Symptomatic, ≥ **One:**

- Chest pain
- Heart failure and, ≥ **One**:
 - Arterial $\text{Po}_2 < 56 \text{ mmHg}(7.4 \text{ kPa})$
 - $\text{O}_2 \text{ sat} < 89\%(0.89)$ and $< \text{baseline}$
- Hemodynamic instability, ≥ **One**:
 - Heart rate $> 120/\text{min}$, sustained ⁽⁶⁷⁾
 - Systolic blood pressure, ≥ **One**:
 - $< 90 \text{ mmHg}$
 - $< 110 \text{ mmHg}$ with chronic hypertension
 - $> 30 \text{ mmHg}$ decrease from baseline ⁽¹⁸⁾
 - Labile ⁽⁹⁵⁾
 - MAP $< 65 \text{ mmHg}$
- Bronchodilator, ≥ **One**: ⁽⁹⁶⁾
 - Continuous
 - Every 1-2h
- Cardiac arrest and post resuscitation care $\leq 2\text{d}$
- Cerebral edema and, ≥ **One**:
 - ICP monitoring (includes OR placement)
 - Corticosteroid
 - Hyperosmolar therapy, ≥ **One**: ⁽⁹⁷⁾
 - Mannitol
 - Hypertonic saline $\geq 3\%(0.03)$ solution
- Complex wound care and, ≥ **One**:
 - Every 1-2h
 - Requiring procedural sedation ⁽⁹⁸⁾
- Compromised airway with artificial airway placement (excludes nasal trumpet) and monitoring every 1-2h
- DIC and blood product transfusion ⁽⁹⁹⁾
- Digit or limb reimplantation and monitoring every 1-2h
- External ventricular or lumbar drainage and neurological assessment every 1-2h ⁽⁷⁴⁾
- ECMO or ECLS ⁽¹⁰⁰⁾
- Hemodynamic instability persisting after $\geq 1\text{L}$ of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both**: ⁽¹⁰¹⁾
 - Finding, ≥ **One**:
 - Heart rate $> 120/\text{min}$
 - Systolic blood pressure $< 100 \text{ mmHg}$ and $< \text{baseline}$
 - MAP $< 65 \text{ mmHg}$
 - Intervention, ≥ **One**:
 - Blood product transfusion
 - IV fluid resuscitation $\geq 1\text{L} \leq 24\text{h}$ ⁽⁹²⁾
- Hemodynamic monitoring, invasive ⁽¹⁰²⁾
- Hypercalcemia, **Both**: ^(103, 104)
 - Calcium $\geq 14.0 \text{ mg/dL}(3.5 \text{ mmol/L})$
 - Intervention, ≥ **One**:
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - Treatment, **Both**:
 - Calcitonin
 - Saline hydration ^(14, 105)
- Hypocalcemia, **Both**: ^(104, 106)
 - Calcium $< 5.0 \text{ mg/dL}(1.25 \text{ mmol/L})$
 - Calcium repletion
- Hyperkalemia, **Both**: ⁽⁶⁹⁾
 - Potassium $> 6.0 \text{ mEq/L}(6.0 \text{ mmol/L})$ and, ≥ **One**:

- Loss of P wave
- Multifocal PVCs
- Neuromuscular deficit ⁽⁹¹⁾
- Widening QRS
- Intervention, ≥ **One**:
 - Calcium chloride or gluconate
- Dialysis, **One**:
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Hypokalemia, **All**: ^(71, 72, 107)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
- Electrocardiogram (ECG) findings, ≥ **One**: ⁽¹⁰⁷⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽¹⁰⁸⁾
 - T and U wave fusion ⁽¹⁰⁹⁾
 - U wave ⁽¹¹⁰⁾
 - Potassium repletion ⁽⁷²⁾
- Hypomagnesemia, **Both**:
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- Hyponatremia, **Both**: ⁽¹¹¹⁾
 - Sodium > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One**:
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(14, 112)
- Hyponatremia, **Both**: ⁽¹¹³⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ⁽¹¹⁴⁾
- Intervention, ≥ **One**:
 - Conivaptan or tolvaptan (includes PO) ⁽¹¹⁵⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽¹¹⁶⁾
- Hypophosphatemia, **Both**: ⁽¹¹⁷⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- IABP ⁽¹¹⁸⁾
- Mechanical ventilation
- Necrotizing fasciitis and, ≥ **Two**: ⁽¹²²⁾
 - Anti-infectives ≥ 2 drugs
 - Complex dressing every 2-4h requiring procedural sedation ⁽⁹⁸⁾
 - Return to OR for additional debridement, scheduled or performed within 24h
- Neurological impairment or condition, **Both**: ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 1-2h, ≤ 2d ⁽⁷⁴⁾
- NIPPV and, ≥ **One**: ⁽⁷⁵⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
- Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One**:
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, new ^(123, 124)

● **Post-op Day 3-21, One:**● **ACUTE, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹²⁶⁾

- Afebrile
- Able to perform ADLs or return to baseline ⁽¹⁸⁾
- Pain controlled or manageable ⁽⁸⁷⁾
- Tolerating PO or nutritional route established ^(128, 129)

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽¹³⁰⁾

● **Complication or comorbidity and clinically stable for discharge, ≥ One:** ⁽¹³⁰⁾● **AKI, ≥ One:**

- Kidney function improving or within acceptable limits ⁽¹³¹⁾
- Outpatient dialysis regimen established

● **Anticoagulation regimen established** ^(153, 154, 155)● **Bleeding controlled or resolved**● **C. difficile infection, symptoms resolving, and tolerating PO** ⁽¹⁵⁶⁾● **Chest tube removed or Heimlich valve**● **Electrolytes within acceptable limits** ⁽¹³¹⁾● **Functional impairment, One:** ⁽²⁷⁾● **Discharge to home, One:** ⁽¹²⁷⁾

- Patient or caregiver able to safely manage impairment
- Home care services arranged

● **Discharge to inpatient facility for rehabilitative services**● **Graft function established and stable**● **Heart failure resolving**● **Immunosuppressant regimen tolerated and levels within acceptable limits** ⁽¹³¹⁾● **Neurological stability** ⁽¹⁵⁷⁾● **Passing urine without urinary retention**● **O₂ sat ≥ 92%(0.92) or within acceptable limits** ⁽¹³¹⁾● **Vital signs within acceptable limits** ⁽¹³¹⁾● **Wound regimen established and adequate healing**● **Partial responder**, not clinically stable for discharge and requires continued stay, **One:** ⁽¹⁵⁸⁾● **Routine post-op stay, One:**● **Moderate stay within the last 5d** ⁽¹³³⁾● **Long stay within the last 6d** ⁽¹³⁴⁾● **Post external fixation of open fracture awaiting internal fixation ≤ 5d** ^(62, 63)● **Solid organ transplant, ≥ One:**● **Heart within the last 11d**● **Heart-lung within the last 15d**● **Kidney within the last 6d**● **Liver within the last 11d**● **Lung within the last 13d**● **Pancreas within the last 8d**● **Small bowel within the last 18d**● **Complication or comorbid condition, ≥ One:**● **Abnormal vital signs, Both:**● **Finding, ≥ One:**● **Heart rate > 100/min and > baseline** ⁽¹⁸⁾● **Systolic blood pressure > 140 mmHg and > baseline** ^(18, 159)● **Medication adjustment (includes PO)** ⁽¹⁴¹⁾● **Bleeding actual or suspected and, Both:**● **Finding, ≥ One:**● **Heart rate 100-120/min, sustained** ⁽⁶⁷⁾

- Hct or Hb decreased within last 24h
- Orthostatic hypotension, **Both:**
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
- Blood pressure, ≥ **One:**
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
- Intervention, ≥ **One:**
 - Blood product transfusion
 - Return to OR within last 24h
 - Requiring Hct or Hb monitoring at least 2x/24h, ≤ 2d
- *C. difficile* infection, actual or suspected and, **Both:** ⁽¹⁶⁰⁾
 - Anti-infective (includes PO) ≤ 5d ^(161, 162)
 - Inadequate oral intake ⁽¹⁶³⁾
- Chest tube and, ≥ **One:** ⁽¹³⁵⁾
 - Suction, continuous
 - Drainage ≥ 250 mL/24h
 - Repositioning within last 24h ⁽¹³⁶⁾
 - Water seal within last 24h
- Fever and anti-infective and, **One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁵⁰⁾
- Fever, new onset and, **Both:** ⁽¹⁷⁹⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁵⁰⁾
- Intervention, ≥ **One:**
 - Culture pending ≤ 2d
 - Drug-induced fever suspected and precipitating drug discontinued ≤ 3d ⁽¹⁸⁰⁾
 - Imaging study ≤ 24h
- Graft dysfunction and immunosuppressant and, ≥ **One:** ^(84, 85, 166)
 - IV
 - Conversion from IV to PO ≤ 2d ⁽¹⁶⁷⁾
 - Medication adjustment (includes PO) ^(168, 169)
- Heart failure and, **Both:** ⁽¹³⁷⁾
 - Dyspnea > baseline and, ≥ **One:** ⁽¹⁸⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁸⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽¹³⁸⁾
 - Intervention, ≥ **One:** ⁽¹³⁹⁾
 - Diuretic, **One:** ⁽¹⁴⁰⁾
 - ≥ 2x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** ⁽¹⁴¹⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹⁸⁾
 - Creatinine increased from baseline ⁽¹⁸⁾
 - Dialysis
- Hypercalcemia, **Both:** ^(103, 104)
 - Calcium 11.1-13.9 mg/dL(2.78-3.48 mmol/L)
 - Intervention, ≥ **One:**
 - Bisphosphonate ⁽¹⁴²⁾
 - Corticosteroid (includes PO) ⁽¹⁴²⁾
 - Saline hydration ^(14, 105)
- Hypocalcemia, **Both:** ^(104, 106)
 - Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L)
 - Calcium repletion
- Hyperkalemia, **All:**
 - Potassium > 5.5 mEq/L(5.5 mmol/L)

- No electrocardiogram (ECG) changes ^(143, 144)
- **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
 - IV fluid ⁽¹⁴⁾
 - Potassium binding agent (includes PO or PR) ⁽¹⁴⁵⁾
- **Hypokalemia, All:** ^(71, 72)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - No electrocardiogram (ECG) changes ^(143, 144)
 - Potassium repletion (includes PO)
- **Hypermagnesemia, Both:**
 - Magnesium > 2.5 mg/dL(1.03 mmol/L)
- **Intervention, ≥ One:**
 - Diuretic ≥ 2x/24h (includes PO)
 - IV fluid ⁽¹⁴⁾
- **Hypomagnesemia, Both:**
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Magnesium repletion
- **Hypernatremia, Both:** ⁽¹¹¹⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L)
 - IV fluid ^(14, 112)
- **Hyponatremia or SIADH, Both:** ⁽¹⁴⁶⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
- **Intervention, ≥ One:**
 - Diuretic and oral sodium supplement
 - Fluid restriction
- **Medication administration, ≥ One:**
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
- Sodium chloride 0.9%(0.009) ⁽¹¹⁶⁾
- **Hypophosphatemia, Both:** ⁽¹¹⁷⁾
 - Phosphate 1.0-2.0 mg/dL(0.32-0.65 mmol/L)
 - Phosphate repletion (includes PO)
- **Ileus and, One:** ⁽¹⁴⁷⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and IV fluid ≤ 5d ⁽¹⁴⁾
- **Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ One:** ^(164, 165)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
- Immunosuppressant level abnormal or signs of toxicity and medication adjustment (includes PO) ^(168, 169)
- **Inadequate oral intake and, ≥ One:** ⁽¹⁴⁸⁾
 - IV fluid ⁽¹⁴⁾
 - Advancing diet as tolerated
- **Neurological impairment or condition and, Both:** ⁽⁷³⁾
 - **Finding, ≥ One:**
 - Acute onset
 - Deterioration
 - Neurological assessment at least 3x/24h, ≤ 2d ⁽⁷⁴⁾

- O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(17, 18)
- Pain uncontrolled and, **Both:**
 - Pain assessment ⁽²³⁾
 - Requiring change in medication, dose, or frequency ^(141, 149)
- Respiratory impairment (including pneumonia) and, **Both:**
 - Finding, ≥ **One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁵⁰⁾
 - Inability to clear secretions ≤ 24h
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁸⁾
 - Intervention, ≥ **One:**
 - Anti-infective (includes PO)
 - Requiring supplemental oxygen ⁽¹⁷⁾
 - Respiratory interventions or incentive spirometry at least 6x/24h ⁽⁸⁰⁾
- Spinal headache and, ≥ **One:**
 - Epidural blood patch performed within last 24h ⁽²⁴⁾
 - IV caffeine and fluid ⁽²⁵⁾
- Therapeutic anticoagulation, including DVT and, **One:**
 - LMWH ≤ 3d ⁽¹⁷⁰⁾
 - UFH, continuous ≤ 5d ⁽¹¹⁾
 - HIT and anticoagulation ^(171, 181)
- Pretreatment or bridging anticoagulation, **All:** ⁽¹⁷²⁾
 - Patient or caregiver unable to administer
 - Outpatient services unavailable ⁽¹⁷³⁾
- Intervention, **One:**
 - INR non-therapeutic and anticoagulation (includes PO) ⁽¹⁷⁴⁾
 - Anticoagulation ≤ 5d since initiation ^(175, 176)
- Wound, ≥ **One:**
 - Complex wound care ≤ 5d and, ≥ **One:** ⁽⁶⁸⁾
 - Wound care ≥ 3x/24h and, ≥ **One:**
 - Requiring parenteral analgesic
 - ≥ 15 min in duration
 - Wound management teaching ≤ 24h ⁽¹⁷⁷⁾
 - Return to OR within last 24h ⁽¹⁷⁸⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Arrhythmia, ≥ **One:**
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽¹⁵¹⁾
 - Bleeding and, **Both:**
 - Finding, ≥ **One:**
 - Chest pain
 - Dyspnea
 - Heart rate > 100/min, sustained, and orthostatic hypotension and, **Both:** ⁽⁶⁷⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁶⁷⁾
 - Blood pressure, ≥ **One:**
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - Blood product transfusion
 - Complex wound care ≤ 5d and, **Both:** ⁽⁶⁸⁾
 - ≥ 3x/24h
 - ≥ 30 min in duration

- External ventricular or lumbar drainage and neurological assessment every 3-4h ⁽⁷⁴⁾
- **Hyperkalemia, All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - **Electrocardiogram (ECG) changes, $\geq$ One:** ⁽⁶⁹⁾
 - Peaked T waves
 - PVCs $> 6/\text{min}$
 - **Intervention, $\geq$ One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- **Hypokalemia, All:** ^(71, 72)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs $> 6/\text{min}$
 - Potassium repletion
- **Neurological impairment or condition, Both:** ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h, $\leq 2\text{d}$ ⁽⁷⁴⁾
- NIPPV ⁽⁷⁵⁾
- Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽¹⁷⁾
- Permanent pacemaker placement within last 24h
- **VAD and, $\geq$ One:** ^(123, 124)
 - **Anticoagulation, $\geq$ One:** ^(182, 183, 184)
 - Direct thrombin inhibitor
 - GP IIb or IIIa inhibitor
 - Heparin
 - Thrombolytic
 - **Bleeding actual or suspected and, $\geq$ One:**
 - Hct or Hb monitoring at least 3x/24h, $\leq 3\text{d}$
 - Volume expander or blood product ⁽¹⁸⁵⁾
 - **Inadequate nutrition, $\geq$ One:**
 - Calorie count $\leq 3\text{d}$
 - Requiring daily tube feeding adjustment
 - TPN initiation $\leq 24\text{h}$
 - **Infection actual or suspected and, Both:**
 - **Finding, $\geq$ One:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽¹⁵⁰⁾
 - WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - WBC $\leq 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - ANC $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$ ⁽¹⁸⁶⁾
 - Anti-infective
 - **Post acute facility unavailable and, $\geq$ One:**
 - Functional impairment requiring therapy ⁽¹⁶⁴⁾
 - Inadequate social or family support
 - Patient or caregiver unable to demonstrate functional ability to maintain and operate VAD
 - Post Critical care $\leq 24\text{h}$
- **CRITICAL, $\geq$ One:**
 - **Solid organ transplant, Both:**
 - **Organ, $\geq$ One:**
 - Heart within the last 5d
 - Heart-lung within the last 6d
 - Liver within the last 3d
 - Lung within the last 6d
 - Small bowel within the last 10d

- Expected post-op course, **All**:
 - Immunosuppressant (includes PO) ^(84, 85)
- Hydration or nutrition, ≥ **One**:
 - IV fluid ⁽¹⁴⁾
 - Advancing diet as tolerated or nutritional route established ⁽¹²⁸⁾
 - Intake and output monitoring
 - Assessment of graft function ⁽⁸⁶⁾
- Pain assessment, **One**: ⁽²³⁾
 - Patient denies pain
 - Pain controlled ⁽⁸⁷⁾
 - Uncontrolled pain requiring change in medication, dose, or frequency (includes PO)
- Liver transplant complication and, ≥ **One**:
 - Roux-en-Y choledochojejunostomy performed within last 2d
 - Arterial or venous thrombectomy performed within last 2d
 - Intra-arterial or venous thrombolysis performed within last 24h
- Acute humoral rejection and apheresis ^(88, 89)
- Acute kidney injury (AKI) and, **Both**: ⁽⁹⁰⁾
 - Finding, ≥ **One**:
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁹¹⁾
 - Widening QRS
 - Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One**:
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
 - Diuretic, continuous
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - IV fluid resuscitation ≤ 2d ⁽⁹²⁾
- CRRT ⁽⁹³⁾
- Arrhythmia, ≥ **One**:
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
- Antiarrhythmic and, **Both**:
 - Arrhythmia, ≥ **One**:
 - Recurrent arrhythmia ⁽⁹⁴⁾
 - Refractory arrhythmia ⁽⁹⁴⁾
 - Symptomatic, ≥ **One**:
 - Chest pain
 - Heart failure and, ≥ **One**:
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - O₂ sat < 89%(0.89) and < baseline
 - Hemodynamic instability, ≥ **One**:
 - Heart rate > 120/min, sustained ⁽⁶⁷⁾
 - Systolic blood pressure, ≥ **One**:
 - < 90 mmHg

- < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽¹⁸⁾
 - Labile ⁽⁹⁵⁾
 - MAP < 65 mmHg
- Bronchodilator, ≥ **One:** ⁽⁹⁶⁾
 - Continuous
 - Every 1-2h
- Cardiac arrest and post resuscitation care ≤ 2d
- Cerebral edema and, ≥ **One:**
 - ICP monitoring (includes OR placement)
 - Corticosteroid
- Hyperosmolar therapy, ≥ **One:** ⁽⁹⁷⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
- Complex wound care and, ≥ **One:**
 - Every 1-2h
 - Requiring procedural sedation ⁽⁹⁸⁾
- Compromised airway with artificial airway placement (excludes nasal trumpet) and monitoring every 1-2h
- DIC and blood product transfusion ⁽⁹⁹⁾
- External ventricular or lumbar drainage and neurological assessment every 1-2h ⁽⁷⁴⁾
- ECMO or ECLS ⁽¹⁰⁰⁾
- Hemodynamic instability persisting after ≥1L of IVF resuscitation or ≥1 unit blood product transfusion and, **Both:** ⁽¹⁰¹⁾
 - Finding, ≥ **One:**
 - Heart rate > 120/min
 - Systolic blood pressure < 100 mmHg and < baseline
 - MAP < 65 mmHg
 - Intervention, ≥ **One:**
 - Blood product transfusion
 - IV fluid resuscitation ≥1L ≤ 24h ⁽⁹²⁾
- Hemodynamic monitoring, invasive ⁽¹⁰²⁾
- Hypercalcemia, **Both:** ^(103, 104)
 - Calcium ≥ 14.0 mg/dL(3.5 mmol/L)
- Intervention, ≥ **One:**
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
- Treatment, **Both:**
 - Calcitonin
 - Saline hydration ^(14, 105)
- Hypocalcemia, **Both:** ^(104, 106)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Hyperkalemia, **Both:** ⁽⁶⁹⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁹¹⁾
 - Widening QRS
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - Dialysis, **One:**
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal

- Glucose 50%(0.50) with insulin ⁽⁷⁰⁾
- Hypokalemia, **All:** ^(71, 72, 107)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - Electrocardiogram (ECG) findings, ≥ **One:** ⁽¹⁰⁷⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽¹⁰⁸⁾
 - T and U wave fusion ⁽¹⁰⁹⁾
 - U wave ⁽¹¹⁰⁾
 - Potassium repletion ⁽⁷²⁾
- Hypomagnesemia, **Both:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- Hypernatremia, **Both:** ⁽¹¹¹⁾
 - Sodium > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - CRRT ⁽⁹³⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ^(14, 112)
- Hyponatremia, **Both:** ⁽¹¹³⁾
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ⁽¹¹⁴⁾
 - Intervention, ≥ **One:**
 - Conivaptan or tolvaptan (includes PO) ⁽¹¹⁵⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽¹¹⁶⁾
- Hypophosphatemia, **Both:** ⁽¹¹⁷⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- IABP ⁽¹¹⁸⁾
- Mechanical ventilation
- Necrotizing fasciitis and, ≥ **Two:** ⁽¹²²⁾
 - Anti-infectives ≥ 2 drugs
 - Complex dressing every 2-4h requiring procedural sedation ⁽⁹⁸⁾
 - Return to OR for additional debridement, scheduled or performed within 24h
- Neurological impairment or condition, **Both:** ⁽⁷³⁾
 - Acute onset or deterioration
 - Neurological assessment every 1-2h, ≤ 2d ⁽⁷⁴⁾
- NIPPV and, ≥ **One:** ⁽⁷⁵⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, **One:** ^(123, 124)
 - Instability, ≥ **One:** ⁽¹⁸⁷⁾
 - Blood pressure, ≥ **One:**
 - < baseline and inadequate perfusion ⁽¹⁸⁸⁾
 - Mean arterial pressure (MAP) < 70 mmHg and continuous flow device ⁽¹⁸⁹⁾
 - Systolic < 90 mmHg and pulsatile flow device
 - Decreased level of consciousness
 - Decreased VAD flows and inadequate perfusion ⁽¹⁸⁸⁾
 - O₂ sat < 89%(0.89)
 - Percutaneous placement

● **Post-op Day 22, One:**● **ACUTE, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹²⁶⁾

- Afebrile
- Able to perform ADLs or return to baseline ⁽¹⁸⁾
- Pain controlled or manageable ⁽⁸⁷⁾
- Tolerating PO or nutritional route established ^(128, 129)

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽¹³⁰⁾

● **Complication or comorbidity and clinically stable for discharge, ≥ One:** ⁽¹³⁰⁾● **AKI, ≥ One:**

- Kidney function improving or within acceptable limits ⁽¹³¹⁾
- Outpatient dialysis regimen established
- Anticoagulation regimen established ^(153, 154, 155)
- Bleeding controlled or resolved
- *C. difficile* infection, symptoms resolving, and tolerating PO ⁽¹⁵⁶⁾
- Chest tube removed or Heimlich valve
- Electrolytes within acceptable limits ⁽¹³¹⁾

● **Functional impairment, One:** ⁽²⁷⁾● **Discharge to home, One:** ⁽¹²⁷⁾

- Patient or caregiver able to safely manage impairment
- Home care services arranged
- Discharge to inpatient facility for rehabilitative services
- Graft function established and stable
- Heart failure resolving
- Immunosuppressant regimen tolerated and levels within acceptable limits ⁽¹³¹⁾
- Neurological stability ⁽¹⁵⁷⁾
- O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽¹³¹⁾
- Passing urine without urinary retention
- Wound regimen established and adequate healing

● **Non-responder**, not clinically stable for discharge and exceeds post-op days at the Acute level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

● **INTERMEDIATE, One:**● **Non-responder**, not clinically stable for discharge and exceeds post-op days at the Intermediate level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

● **CRITICAL, One:**● **Non-responder**, not clinically stable for discharge and exceeds post-op days at the Critical level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

Discharge, Level of Care, ≥ One: ⁽¹⁹⁰⁾● **HOME, Both:**● **Level of care appropriateness, Both:**

- Home environment safe and accessible ⁽¹⁹¹⁾
- Patient or caregiver demonstrates ability to manage care

● **Surgical discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions ⁽¹⁹²⁾
- Education on the importance of, **All:** ⁽¹⁹³⁾

- Wound care
- Activity restrictions
- When to seek care
- Medication reconciliation ⁽¹⁹⁴⁾
- Follow-up care planned ⁽¹⁹⁵⁾
- Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽¹⁹¹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(196, 197)
 - Skilled services, ≥ **One:** ⁽¹⁹⁸⁾
 - Skilled nursing ⁽¹⁹⁹⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽²⁰⁰⁾
 - Surgical discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽¹⁹²⁾
 - Education on the importance of, **All:** ⁽¹⁹³⁾
 - Wound care
 - Activity restrictions
 - When to seek care
 - Medication reconciliation ⁽¹⁹⁴⁾
 - Follow-up care planned ⁽¹⁹⁵⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽²⁰¹⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽²⁰²⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽²⁰³⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽¹⁹⁹⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(204, 205)
 - Functional impairments requiring at least supervision ⁽²⁰⁶⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹⁹⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽¹⁹⁹⁾
 - Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(204, 205)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽²⁰⁶⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹⁹⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- SUBACUTE REHAB, **Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(204, 205)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽²⁰⁶⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹⁹⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- ACUTE REHAB, **Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(204, 205)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽²⁰⁶⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽²⁰⁷⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹⁹⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- LONG TERM ACUTE CARE, **Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽²⁰⁸⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(209, 210)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽²¹¹⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹⁹⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	LOS (days)	Type
001 HEART TRANSPLANT OR IMPLANT OF HEART ASSIST SYSTEM WITH MCC	30.3	CMS GMLOS

Condition or Procedure	LOS (days)	Type
001 HEART TRANSPLANT OR IMPLANT OF HEART ASSIST SYSTEM WITH MCC	27.6	InterQual
002 HEART TRANSPLANT OR IMPLANT OF HEART ASSIST SYSTEM WITHOUT MCC	9.6	CMS GMLOS
003 ECMO OR TRACHEOSTOMY WITH MV >96 HOURS OR PRINCIPAL DIAGNOSIS EXCEPT FACE, MOUTH AND NECK WITH MAJOR O.R. PROCEDURES	26.7	CMS GMLOS
003 ECMO OR TRACHEOSTOMY WITH MV >96 HOURS OR PRINCIPAL DIAGNOSIS EXCEPT FACE, MOUTH AND NECK WITH MAJOR O.R. PROCEDURES	32.5	InterQual
004 TRACHEOSTOMY WITH MV >96 HOURS OR PRINCIPAL DIAGNOSIS EXCEPT FACE, MOUTH AND NECK WITHOUT MAJOR O.R. PROCEDURES	26.1	CMS GMLOS
004 TRACHEOSTOMY WITH MV >96 HOURS OR PRINCIPAL DIAGNOSIS EXCEPT FACE, MOUTH AND NECK WITHOUT MAJOR O.R. PROCEDURES	31.9	InterQual
005 LIVER TRANSPLANT WITH MCC OR INTESTINAL TRANSPLANT	14.8	CMS GMLOS
005 LIVER TRANSPLANT WITH MCC OR INTESTINAL TRANSPLANT	23.6	InterQual
006 LIVER TRANSPLANT WITHOUT MCC	7.7	InterQual
006 LIVER TRANSPLANT WITHOUT MCC	7.8	CMS GMLOS
007 LUNG TRANSPLANT	18.5	CMS GMLOS
007 LUNG TRANSPLANT	19.6	InterQual
008 SIMULTANEOUS PANCREAS AND KIDNEY TRANSPLANT	8.7	CMS GMLOS
008 SIMULTANEOUS PANCREAS AND KIDNEY TRANSPLANT	13.4	InterQual
010 PANCREAS TRANSPLANT	9.5	CMS GMLOS
011 TRACHEOSTOMY FOR FACE, MOUTH AND NECK DIAGNOSES OR LARYNGECTOMY WITH MCC	12	InterQual
011 TRACHEOSTOMY FOR FACE, MOUTH AND NECK DIAGNOSES OR LARYNGECTOMY WITH MCC	11.5	CMS GMLOS
012 TRACHEOSTOMY FOR FACE, MOUTH AND NECK DIAGNOSES OR LARYNGECTOMY WITH CC	8.6	InterQual
012 TRACHEOSTOMY FOR FACE, MOUTH AND NECK DIAGNOSES OR LARYNGECTOMY WITH CC	8.5	CMS GMLOS

Condition or Procedure	LOS (days)	Type
013 TRACHEOSTOMY FOR FACE, MOUTH AND NECK DIAGNOSES OR LARYNGECTOMY WITHOUT CC/MCC	6.6	InterQual
013 TRACHEOSTOMY FOR FACE, MOUTH AND NECK DIAGNOSES OR LARYNGECTOMY WITHOUT CC/MCC	6.3	CMS GMLOS
020 INTRACRANIAL VASCULAR PROCEDURES WITH PRINCIPAL DIAGNOSIS HEMORRHAGE WITH MCC	13.8	InterQual
020 INTRACRANIAL VASCULAR PROCEDURES WITH PRINCIPAL DIAGNOSIS HEMORRHAGE WITH MCC	10.2	CMS GMLOS
021 INTRACRANIAL VASCULAR PROCEDURES WITH PRINCIPAL DIAGNOSIS HEMORRHAGE WITH CC	7.9	InterQual
021 INTRACRANIAL VASCULAR PROCEDURES WITH PRINCIPAL DIAGNOSIS HEMORRHAGE WITH CC	6.3	CMS GMLOS
022 INTRACRANIAL VASCULAR PROCEDURES WITH PRINCIPAL DIAGNOSIS HEMORRHAGE WITHOUT CC/MCC	2.5	CMS GMLOS
022 INTRACRANIAL VASCULAR PROCEDURES WITH PRINCIPAL DIAGNOSIS HEMORRHAGE WITHOUT CC/MCC	4.3	InterQual
023 CRANIOTOMY WITH MAJOR DEVICE IMPLANT OR ACUTE COMPLEX CNS PRINCIPAL DIAGNOSIS WITH MCC OR CHEMOTHERAPY IMPLANT OR EPILEPSY WITH NEUROSTIMULATOR	7.5	CMS GMLOS
023 CRANIOTOMY WITH MAJOR DEVICE IMPLANT OR ACUTE COMPLEX CNS PRINCIPAL DIAGNOSIS WITH MCC OR CHEMOTHERAPY IMPLANT OR EPILEPSY WITH NEUROSTIMULATOR	10.2	InterQual
024 CRANIOTOMY WITH MAJOR DEVICE IMPLANT OR ACUTE COMPLEX CNS PRINCIPAL DIAGNOSIS WITHOUT MCC	4	CMS GMLOS
024 CRANIOTOMY WITH MAJOR DEVICE IMPLANT OR ACUTE COMPLEX CNS PRINCIPAL DIAGNOSIS WITHOUT MCC	4.5	InterQual
025 CRANIOTOMY AND ENDOVASCULAR INTRACRANIAL PROCEDURES WITH MCC	7.3	InterQual
025 CRANIOTOMY AND ENDOVASCULAR INTRACRANIAL PROCEDURES WITH MCC	6.6	CMS GMLOS
026 CRANIOTOMY AND ENDOVASCULAR INTRACRANIAL PROCEDURES WITH CC	4	InterQual
026 CRANIOTOMY AND ENDOVASCULAR INTRACRANIAL PROCEDURES WITH CC	3.1	CMS GMLOS
027 CRANIOTOMY AND ENDOVASCULAR INTRACRANIAL PROCEDURES WITHOUT CC/MCC	1.6	CMS GMLOS

Condition or Procedure	LOS (days)	Type
027 CRANIOTOMY AND ENDOVASCULAR INTRACRANIAL PROCEDURES WITHOUT CC/MCC	2	InterQual
028 SPINAL PROCEDURES WITH MCC	11.2	InterQual
028 SPINAL PROCEDURES WITH MCC	10.2	CMS GMLOS
029 SPINAL PROCEDURES WITH CC OR SPINAL NEUROSTIMULATORS	5.7	InterQual
029 SPINAL PROCEDURES WITH CC OR SPINAL NEUROSTIMULATORS	4.9	CMS GMLOS
030 SPINAL PROCEDURES WITHOUT CC/MCC	3.6	InterQual
030 SPINAL PROCEDURES WITHOUT CC/MCC	2.7	CMS GMLOS
031 VENTRICULAR SHUNT PROCEDURES WITH MCC	7	CMS GMLOS
031 VENTRICULAR SHUNT PROCEDURES WITH MCC	7.1	InterQual
032 VENTRICULAR SHUNT PROCEDURES WITH CC	2.7	CMS GMLOS
032 VENTRICULAR SHUNT PROCEDURES WITH CC	3.3	InterQual
033 VENTRICULAR SHUNT PROCEDURES WITHOUT CC/MCC	1.6	CMS GMLOS
033 VENTRICULAR SHUNT PROCEDURES WITHOUT CC/MCC	1.8	InterQual
034 CAROTID ARTERY STENT PROCEDURES WITH MCC	4.9	CMS GMLOS
034 CAROTID ARTERY STENT PROCEDURES WITH MCC	6.6	InterQual
035 CAROTID ARTERY STENT PROCEDURES WITH CC	2.4	InterQual
035 CAROTID ARTERY STENT PROCEDURES WITH CC	2	CMS GMLOS
036 CAROTID ARTERY STENT PROCEDURES WITHOUT CC/MCC	1.2	CMS GMLOS
036 CAROTID ARTERY STENT PROCEDURES WITHOUT CC/MCC	1.2	InterQual
037 EXTRACRANIAL PROCEDURES WITH MCC	6.8	InterQual
037 EXTRACRANIAL PROCEDURES WITH MCC	5.1	CMS GMLOS
038 EXTRACRANIAL PROCEDURES WITH CC	1.9	CMS GMLOS
038 EXTRACRANIAL PROCEDURES WITH CC	2.5	InterQual

Condition or Procedure	LOS (days)	Type
039 EXTRACRANIAL PROCEDURES WITHOUT CC/MCC	1.2	InterQual
039 EXTRACRANIAL PROCEDURES WITHOUT CC/MCC	1.2	CMS GMLOS
040 PERIPHERAL, CRANIAL NERVE AND OTHER NERVOUS SYSTEM PROCEDURES WITH MCC	9.1	InterQual
040 PERIPHERAL, CRANIAL NERVE AND OTHER NERVOUS SYSTEM PROCEDURES WITH MCC	7.4	CMS GMLOS
041 PERIPHERAL, CRANIAL NERVE AND OTHER NERVOUS SYSTEM PROCEDURES WITH CC OR PERIPHERAL NEUROSTIMULATOR	3.9	CMS GMLOS
041 PERIPHERAL, CRANIAL NERVE AND OTHER NERVOUS SYSTEM PROCEDURES WITH CC OR PERIPHERAL NEUROSTIMULATOR	4.7	InterQual
042 PERIPHERAL, CRANIAL NERVE AND OTHER NERVOUS SYSTEM PROCEDURES WITHOUT CC/MCC	2.4	CMS GMLOS
042 PERIPHERAL, CRANIAL NERVE AND OTHER NERVOUS SYSTEM PROCEDURES WITHOUT CC/MCC	3.9	InterQual
135 SINUS AND MASTOID PROCEDURES WITH CC/MCC	4.2	CMS GMLOS
135 SINUS AND MASTOID PROCEDURES WITH CC/MCC	3.5	InterQual
136 SINUS AND MASTOID PROCEDURES WITHOUT CC/MCC	5.6	InterQual
136 SINUS AND MASTOID PROCEDURES WITHOUT CC/MCC	1.3	CMS GMLOS
137 MOUTH PROCEDURES WITH CC/MCC	3.7	CMS GMLOS
137 MOUTH PROCEDURES WITH CC/MCC	4.8	InterQual
138 MOUTH PROCEDURES WITHOUT CC/MCC	1.6	CMS GMLOS
138 MOUTH PROCEDURES WITHOUT CC/MCC	2.7	InterQual
139 SALIVARY GLAND PROCEDURES	3.6	InterQual
139 SALIVARY GLAND PROCEDURES	2.2	CMS GMLOS
163 MAJOR CHEST PROCEDURES WITH MCC	7.5	CMS GMLOS
163 MAJOR CHEST PROCEDURES WITH MCC	8.4	InterQual
164 MAJOR CHEST PROCEDURES WITH CC	3.7	CMS GMLOS
164 MAJOR CHEST PROCEDURES WITH CC	4.2	InterQual

Condition or Procedure	LOS (days)	Type
165 MAJOR CHEST PROCEDURES WITHOUT CC/MCC	2.2	CMS GMLOS
165 MAJOR CHEST PROCEDURES WITHOUT CC/MCC	3.1	InterQual
166 OTHER RESPIRATORY SYSTEM O.R. PROCEDURES WITH MCC	8.4	CMS GMLOS
166 OTHER RESPIRATORY SYSTEM O.R. PROCEDURES WITH MCC	9.9	InterQual
167 OTHER RESPIRATORY SYSTEM O.R. PROCEDURES WITH CC	3.5	CMS GMLOS
167 OTHER RESPIRATORY SYSTEM O.R. PROCEDURES WITH CC	4.7	InterQual
168 OTHER RESPIRATORY SYSTEM O.R. PROCEDURES WITHOUT CC/MCC	2.1	InterQual
168 OTHER RESPIRATORY SYSTEM O.R. PROCEDURES WITHOUT CC/MCC	1.8	CMS GMLOS
216 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITH CARDIAC CATHETERIZATION WITH MCC	11.2	CMS GMLOS
216 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITH CARDIAC CATHETERIZATION WITH MCC	14.2	InterQual
217 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITH CARDIAC CATHETERIZATION WITH CC	5.2	CMS GMLOS
217 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITH CARDIAC CATHETERIZATION WITH CC	8	InterQual
218 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITH CARDIAC CATHETERIZATION WITHOUT CC/MCC	2.2	CMS GMLOS
218 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITH CARDIAC CATHETERIZATION WITHOUT CC/MCC	3.7	InterQual
219 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITHOUT CARDIAC CATHETERIZATION WITH MCC	8.7	CMS GMLOS
219 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITHOUT CARDIAC CATHETERIZATION WITH MCC	9.4	InterQual
220 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITHOUT CARDIAC CATHETERIZATION WITH CC	5.7	CMS GMLOS

Condition or Procedure	LOS (days)	Type
220 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITHOUT CARDIAC CATHETERIZATION WITH CC	5.9	InterQual
221 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITHOUT CARDIAC CATHETERIZATION WITHOUT CC/MCC	3.3	CMS GMLOS
221 CARDIAC VALVE AND OTHER MAJOR CARDIOTHORACIC PROCEDURES WITHOUT CARDIAC CATHETERIZATION WITHOUT CC/MCC	4.5	InterQual
228 OTHER CARDIOTHORACIC PROCEDURES WITH MCC	7	InterQual
228 OTHER CARDIOTHORACIC PROCEDURES WITH MCC	6.5	CMS GMLOS
229 OTHER CARDIOTHORACIC PROCEDURES WITHOUT MCC	3.9	InterQual
229 OTHER CARDIOTHORACIC PROCEDURES WITHOUT MCC	2.6	CMS GMLOS
231 CORONARY BYPASS WITH PTCA WITH MCC	6.5	InterQual
231 CORONARY BYPASS WITH PTCA WITH MCC	10.3	CMS GMLOS
232 CORONARY BYPASS WITH PTCA WITHOUT MCC	7.1	InterQual
232 CORONARY BYPASS WITH PTCA WITHOUT MCC	7.5	CMS GMLOS
233 CORONARY BYPASS WITH CARDIAC CATHETERIZATION OR OPEN ABLATION WITH MCC	11.4	InterQual
233 CORONARY BYPASS WITH CARDIAC CATHETERIZATION OR OPEN ABLATION WITH MCC	11.4	CMS GMLOS
234 CORONARY BYPASS WITH CARDIAC CATHETERIZATION OR OPEN ABLATION WITHOUT MCC	7.8	InterQual
234 CORONARY BYPASS WITH CARDIAC CATHETERIZATION OR OPEN ABLATION WITHOUT MCC	8.1	CMS GMLOS
235 CORONARY BYPASS WITHOUT CARDIAC CATHETERIZATION WITH MCC	8.2	CMS GMLOS
235 CORONARY BYPASS WITHOUT CARDIAC CATHETERIZATION WITH MCC	8.5	InterQual
236 CORONARY BYPASS WITHOUT CARDIAC CATHETERIZATION WITHOUT MCC	5.8	CMS GMLOS
236 CORONARY BYPASS WITHOUT CARDIAC CATHETERIZATION WITHOUT MCC	5.9	InterQual

Condition or Procedure	LOS (days)	Type
239 AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS EXCEPT UPPER LIMB AND TOE WITH MCC	11.1	CMS GMLOS
239 AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS EXCEPT UPPER LIMB AND TOE WITH MCC	13.4	InterQual
240 AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS EXCEPT UPPER LIMB AND TOE WITH CC	7.1	CMS GMLOS
240 AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS EXCEPT UPPER LIMB AND TOE WITH CC	8.1	InterQual
241 AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS EXCEPT UPPER LIMB AND TOE WITHOUT CC/MCC	4.2	CMS GMLOS
241 AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS EXCEPT UPPER LIMB AND TOE WITHOUT CC/MCC	3.6	InterQual
242 PERMANENT CARDIAC PACEMAKER IMPLANT WITH MCC	5.4	InterQual
242 PERMANENT CARDIAC PACEMAKER IMPLANT WITH MCC	5.1	CMS GMLOS
243 PERMANENT CARDIAC PACEMAKER IMPLANT WITH CC	3.2	InterQual
243 PERMANENT CARDIAC PACEMAKER IMPLANT WITH CC	2.9	CMS GMLOS
244 PERMANENT CARDIAC PACEMAKER IMPLANT WITHOUT CC/MCC	2.1	CMS GMLOS
244 PERMANENT CARDIAC PACEMAKER IMPLANT WITHOUT CC/MCC	2.5	InterQual
245 AICD GENERATOR PROCEDURES	4	InterQual
245 AICD GENERATOR PROCEDURES	4.1	CMS GMLOS
250 PERCUTANEOUS CARDIOVASCULAR PROCEDURES WITHOUT INTRALUMINAL DEVICE WITH MCC	3.8	InterQual
250 PERCUTANEOUS CARDIOVASCULAR PROCEDURES WITHOUT INTRALUMINAL DEVICE WITH MCC	3.4	CMS GMLOS
251 PERCUTANEOUS CARDIOVASCULAR PROCEDURES WITHOUT INTRALUMINAL DEVICE WITHOUT MCC	2.2	InterQual

Condition or Procedure	LOS (days)	Type
251 PERCUTANEOUS CARDIOVASCULAR PROCEDURES WITHOUT INTRALUMINAL DEVICE WITHOUT MCC	2	CMS GMLOS
252 OTHER VASCULAR PROCEDURES WITH MCC	6.8	InterQual
252 OTHER VASCULAR PROCEDURES WITH MCC	5.4	CMS GMLOS
253 OTHER VASCULAR PROCEDURES WITH CC	4.6	InterQual
253 OTHER VASCULAR PROCEDURES WITH CC	3.8	CMS GMLOS
254 OTHER VASCULAR PROCEDURES WITHOUT CC/MCC	2.7	InterQual
254 OTHER VASCULAR PROCEDURES WITHOUT CC/MCC	1.9	CMS GMLOS
255 UPPER LIMB AND TOE AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS WITH MCC	7	CMS GMLOS
255 UPPER LIMB AND TOE AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS WITH MCC	7.6	InterQual
256 UPPER LIMB AND TOE AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS WITH CC	5	CMS GMLOS
256 UPPER LIMB AND TOE AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS WITH CC	5.3	InterQual
257 UPPER LIMB AND TOE AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS WITHOUT CC/MCC	3.4	CMS GMLOS
257 UPPER LIMB AND TOE AMPUTATION FOR CIRCULATORY SYSTEM DISORDERS WITHOUT CC/MCC	4.1	InterQual
258 CARDIAC PACEMAKER DEVICE REPLACEMENT WITH MCC	4.7	CMS GMLOS
258 CARDIAC PACEMAKER DEVICE REPLACEMENT WITH MCC	5.1	InterQual
259 CARDIAC PACEMAKER DEVICE REPLACEMENT WITHOUT MCC	2.4	CMS GMLOS
259 CARDIAC PACEMAKER DEVICE REPLACEMENT WITHOUT MCC	2.7	InterQual
260 CARDIAC PACEMAKER REVISION EXCEPT DEVICE REPLACEMENT WITH MCC	5.7	CMS GMLOS
260 CARDIAC PACEMAKER REVISION EXCEPT DEVICE REPLACEMENT WITH MCC	6.8	InterQual

Condition or Procedure	LOS (days)	Type
261 CARDIAC PACEMAKER REVISION EXCEPT DEVICE REPLACEMENT WITH CC	3.8	InterQual
261 CARDIAC PACEMAKER REVISION EXCEPT DEVICE REPLACEMENT WITH CC	2.9	CMS GMLOS
262 CARDIAC PACEMAKER REVISION EXCEPT DEVICE REPLACEMENT WITHOUT CC/MCC	2.3	CMS GMLOS
262 CARDIAC PACEMAKER REVISION EXCEPT DEVICE REPLACEMENT WITHOUT CC/MCC	2.6	InterQual
263 VEIN LIGATION AND STRIPPING	4.6	CMS GMLOS
263 VEIN LIGATION AND STRIPPING	3	InterQual
264 OTHER CIRCULATORY SYSTEM O.R. PROCEDURES	8.2	InterQual
264 OTHER CIRCULATORY SYSTEM O.R. PROCEDURES	6.9	CMS GMLOS
265 AICD LEAD PROCEDURES	4	CMS GMLOS
265 AICD LEAD PROCEDURES	4.2	InterQual
266 ENDOVASCULAR CARDIAC VALVE REPLACEMENT AND SUPPLEMENT PROCEDURES WITH MCC	2.7	CMS GMLOS
266 ENDOVASCULAR CARDIAC VALVE REPLACEMENT AND SUPPLEMENT PROCEDURES WITH MCC	3.5	InterQual
267 ENDOVASCULAR CARDIAC VALVE REPLACEMENT AND SUPPLEMENT PROCEDURES WITHOUT MCC	1.3	CMS GMLOS
267 ENDOVASCULAR CARDIAC VALVE REPLACEMENT AND SUPPLEMENT PROCEDURES WITHOUT MCC	1.3	InterQual
268 AORTIC AND HEART ASSIST PROCEDURES EXCEPT PULSATION BALLOON WITH MCC	8.9	InterQual
268 AORTIC AND HEART ASSIST PROCEDURES EXCEPT PULSATION BALLOON WITH MCC	6.1	CMS GMLOS
269 AORTIC AND HEART ASSIST PROCEDURES EXCEPT PULSATION BALLOON WITHOUT MCC	1.7	InterQual
269 AORTIC AND HEART ASSIST PROCEDURES EXCEPT PULSATION BALLOON WITHOUT MCC	1.5	CMS GMLOS
270 OTHER MAJOR CARDIOVASCULAR PROCEDURES WITH MCC	8.3	InterQual

Condition or Procedure	LOS (days)	Type
270 OTHER MAJOR CARDIOVASCULAR PROCEDURES WITH MCC	6.5	CMS GMLOS
271 OTHER MAJOR CARDIOVASCULAR PROCEDURES WITH CC	4.9	InterQual
271 OTHER MAJOR CARDIOVASCULAR PROCEDURES WITH CC	4	CMS GMLOS
272 OTHER MAJOR CARDIOVASCULAR PROCEDURES WITHOUT CC/MCC	2.3	InterQual
272 OTHER MAJOR CARDIOVASCULAR PROCEDURES WITHOUT CC/MCC	1.9	CMS GMLOS
273 PERCUTANEOUS AND OTHER INTRACARDIAC PROCEDURES WITH MCC	5.1	InterQual
273 PERCUTANEOUS AND OTHER INTRACARDIAC PROCEDURES WITH MCC	3.6	CMS GMLOS
274 PERCUTANEOUS AND OTHER INTRACARDIAC PROCEDURES WITHOUT MCC	1.2	CMS GMLOS
274 PERCUTANEOUS AND OTHER INTRACARDIAC PROCEDURES WITHOUT MCC	1.5	InterQual
326 STOMACH, ESOPHAGEAL AND DUODENAL PROCEDURES WITH MCC	9.4	CMS GMLOS
326 STOMACH, ESOPHAGEAL AND DUODENAL PROCEDURES WITH MCC	9.8	InterQual
327 STOMACH, ESOPHAGEAL AND DUODENAL PROCEDURES WITH CC	4.3	CMS GMLOS
327 STOMACH, ESOPHAGEAL AND DUODENAL PROCEDURES WITH CC	4.5	InterQual
328 STOMACH, ESOPHAGEAL AND DUODENAL PROCEDURES WITHOUT CC/MCC	2.1	CMS GMLOS
328 STOMACH, ESOPHAGEAL AND DUODENAL PROCEDURES WITHOUT CC/MCC	2.2	InterQual
329 MAJOR SMALL AND LARGE BOWEL PROCEDURES WITH MCC	9.8	CMS GMLOS
329 MAJOR SMALL AND LARGE BOWEL PROCEDURES WITH MCC	10.8	InterQual
330 MAJOR SMALL AND LARGE BOWEL PROCEDURES WITH CC	5.1	CMS GMLOS
330 MAJOR SMALL AND LARGE BOWEL PROCEDURES WITH CC	5.4	InterQual

Condition or Procedure	LOS (days)	Type
331 MAJOR SMALL AND LARGE BOWEL PROCEDURES WITHOUT CC/MCC	2.9	CMS GMLOS
331 MAJOR SMALL AND LARGE BOWEL PROCEDURES WITHOUT CC/MCC	3.1	InterQual
332 RECTAL RESECTION WITH MCC	6.7	CMS GMLOS
333 RECTAL RESECTION WITH CC	3.8	InterQual
333 RECTAL RESECTION WITH CC	3.4	CMS GMLOS
334 RECTAL RESECTION WITHOUT CC/MCC	2.6	InterQual
334 RECTAL RESECTION WITHOUT CC/MCC	2.3	CMS GMLOS
335 PERITONEAL ADHESIOLYSIS WITH MCC	9.5	InterQual
335 PERITONEAL ADHESIOLYSIS WITH MCC	8.9	CMS GMLOS
336 PERITONEAL ADHESIOLYSIS WITH CC	5.5	CMS GMLOS
336 PERITONEAL ADHESIOLYSIS WITH CC	5.8	InterQual
337 PERITONEAL ADHESIOLYSIS WITHOUT CC/MCC	3.7	InterQual
337 PERITONEAL ADHESIOLYSIS WITHOUT CC/MCC	3.4	CMS GMLOS
344 MINOR SMALL AND LARGE BOWEL PROCEDURES WITH MCC	6.1	InterQual
344 MINOR SMALL AND LARGE BOWEL PROCEDURES WITH MCC	7.1	CMS GMLOS
345 MINOR SMALL AND LARGE BOWEL PROCEDURES WITH CC	4.2	CMS GMLOS
345 MINOR SMALL AND LARGE BOWEL PROCEDURES WITH CC	4.1	InterQual
346 MINOR SMALL AND LARGE BOWEL PROCEDURES WITHOUT CC/MCC	3.5	InterQual
346 MINOR SMALL AND LARGE BOWEL PROCEDURES WITHOUT CC/MCC	2.9	CMS GMLOS
347 ANAL AND STOMAL PROCEDURES WITH MCC	4	InterQual
347 ANAL AND STOMAL PROCEDURES WITH MCC	5.6	CMS GMLOS
348 ANAL AND STOMAL PROCEDURES WITH CC	3.1	CMS GMLOS
348 ANAL AND STOMAL PROCEDURES WITH CC	3.5	InterQual

Condition or Procedure	LOS (days)	Type
349 ANAL AND STOMAL PROCEDURES WITHOUT CC/MCC	3.4	InterQual
349 ANAL AND STOMAL PROCEDURES WITHOUT CC/MCC	1.8	CMS GMLOS
350 INGUINAL AND FEMORAL HERNIA PROCEDURES WITH MCC	5.4	InterQual
350 INGUINAL AND FEMORAL HERNIA PROCEDURES WITH MCC	4.8	CMS GMLOS
351 INGUINAL AND FEMORAL HERNIA PROCEDURES WITH CC	3.5	InterQual
351 INGUINAL AND FEMORAL HERNIA PROCEDURES WITH CC	3	CMS GMLOS
352 INGUINAL AND FEMORAL HERNIA PROCEDURES WITHOUT CC/MCC	2	CMS GMLOS
352 INGUINAL AND FEMORAL HERNIA PROCEDURES WITHOUT CC/MCC	1.9	InterQual
353 HERNIA PROCEDURES EXCEPT INGUINAL AND FEMORAL WITH MCC	6	CMS GMLOS
353 HERNIA PROCEDURES EXCEPT INGUINAL AND FEMORAL WITH MCC	6.4	InterQual
354 HERNIA PROCEDURES EXCEPT INGUINAL AND FEMORAL WITH CC	3.5	CMS GMLOS
354 HERNIA PROCEDURES EXCEPT INGUINAL AND FEMORAL WITH CC	3.7	InterQual
355 HERNIA PROCEDURES EXCEPT INGUINAL AND FEMORAL WITHOUT CC/MCC	2.3	CMS GMLOS
355 HERNIA PROCEDURES EXCEPT INGUINAL AND FEMORAL WITHOUT CC/MCC	2.5	InterQual
356 OTHER DIGESTIVE SYSTEM O.R. PROCEDURES WITH MCC	8	CMS GMLOS
356 OTHER DIGESTIVE SYSTEM O.R. PROCEDURES WITH MCC	8.7	InterQual
357 OTHER DIGESTIVE SYSTEM O.R. PROCEDURES WITH CC	4.4	CMS GMLOS
357 OTHER DIGESTIVE SYSTEM O.R. PROCEDURES WITH CC	5.2	InterQual
358 OTHER DIGESTIVE SYSTEM O.R. PROCEDURES WITHOUT CC/MCC	2.9	InterQual

Condition or Procedure	LOS (days)	Type
358 OTHER DIGESTIVE SYSTEM O.R. PROCEDURES WITHOUT CC/MCC	2.5	CMS GMLOS
405 PANCREAS, LIVER AND SHUNT PROCEDURES WITH MCC	9	CMS GMLOS
405 PANCREAS, LIVER AND SHUNT PROCEDURES WITH MCC	10	InterQual
406 PANCREAS, LIVER AND SHUNT PROCEDURES WITH CC	5	CMS GMLOS
406 PANCREAS, LIVER AND SHUNT PROCEDURES WITH CC	5.4	InterQual
407 PANCREAS, LIVER AND SHUNT PROCEDURES WITHOUT CC/MCC	3.5	CMS GMLOS
407 PANCREAS, LIVER AND SHUNT PROCEDURES WITHOUT CC/MCC	4	InterQual
408 BILIARY TRACT PROCEDURES EXCEPT ONLY CHOLECYSTECTOMY WITH OR WITHOUT C.D.E. WITH MCC	8.1	CMS GMLOS
408 BILIARY TRACT PROCEDURES EXCEPT ONLY CHOLECYSTECTOMY WITH OR WITHOUT C.D.E. WITH MCC	7.6	InterQual
409 BILIARY TRACT PROCEDURES EXCEPT ONLY CHOLECYSTECTOMY WITH OR WITHOUT C.D.E. WITH CC	5.1	InterQual
409 BILIARY TRACT PROCEDURES EXCEPT ONLY CHOLECYSTECTOMY WITH OR WITHOUT C.D.E. WITH CC	4.6	CMS GMLOS
410 BILIARY TRACT PROCEDURES EXCEPT ONLY CHOLECYSTECTOMY WITH OR WITHOUT C.D.E. WITHOUT CC/MCC	3.1	CMS GMLOS
410 BILIARY TRACT PROCEDURES EXCEPT ONLY CHOLECYSTECTOMY WITH OR WITHOUT C.D.E. WITHOUT CC/MCC	3.7	InterQual
411 CHOLECYSTECTOMY WITH C.D.E. WITH MCC	6.2	CMS GMLOS
412 CHOLECYSTECTOMY WITH C.D.E. WITH CC	4.1	CMS GMLOS
413 CHOLECYSTECTOMY WITH C.D.E. WITHOUT CC/MCC	2.5	CMS GMLOS
414 CHOLECYSTECTOMY EXCEPT BY LAPAROSCOPE WITHOUT C.D.E. WITH MCC	7.6	CMS GMLOS
414 CHOLECYSTECTOMY EXCEPT BY LAPAROSCOPE WITHOUT C.D.E. WITH MCC	7.7	InterQual

Condition or Procedure	LOS (days)	Type
415 CHOLECYSTECTOMY EXCEPT BY LAPAROSCOPE WITHOUT C.D.E. WITH CC	4.7	CMS GMLOS
415 CHOLECYSTECTOMY EXCEPT BY LAPAROSCOPE WITHOUT C.D.E. WITH CC	4.8	InterQual
416 CHOLECYSTECTOMY EXCEPT BY LAPAROSCOPE WITHOUT C.D.E. WITHOUT CC/MCC	3.1	InterQual
416 CHOLECYSTECTOMY EXCEPT BY LAPAROSCOPE WITHOUT C.D.E. WITHOUT CC/MCC	2.8	CMS GMLOS
417 LAPAROSCOPIC CHOLECYSTECTOMY WITHOUT C.D.E. WITH MCC	4.9	InterQual
417 LAPAROSCOPIC CHOLECYSTECTOMY WITHOUT C.D.E. WITH MCC	4.9	CMS GMLOS
418 LAPAROSCOPIC CHOLECYSTECTOMY WITHOUT C.D.E. WITH CC	3.5	InterQual
418 LAPAROSCOPIC CHOLECYSTECTOMY WITHOUT C.D.E. WITH CC	3.4	CMS GMLOS
419 LAPAROSCOPIC CHOLECYSTECTOMY WITHOUT C.D.E. WITHOUT CC/MCC	2.4	InterQual
419 LAPAROSCOPIC CHOLECYSTECTOMY WITHOUT C.D.E. WITHOUT CC/MCC	2.3	CMS GMLOS
420 HEPATOBILIARY DIAGNOSTIC PROCEDURES WITH MCC	3	InterQual
420 HEPATOBILIARY DIAGNOSTIC PROCEDURES WITH MCC	7.1	CMS GMLOS
421 HEPATOBILIARY DIAGNOSTIC PROCEDURES WITH CC	5.5	InterQual
421 HEPATOBILIARY DIAGNOSTIC PROCEDURES WITH CC	3.7	CMS GMLOS
422 HEPATOBILIARY DIAGNOSTIC PROCEDURES WITHOUT CC/MCC	2.4	CMS GMLOS
422 HEPATOBILIARY DIAGNOSTIC PROCEDURES WITHOUT CC/MCC	3.1	InterQual
423 OTHER HEPATOBILIARY OR PANCREAS O.R. PROCEDURES WITH MCC	7.4	InterQual
423 OTHER HEPATOBILIARY OR PANCREAS O.R. PROCEDURES WITH MCC	8	CMS GMLOS
424 OTHER HEPATOBILIARY OR PANCREAS O.R. PROCEDURES WITH CC	4.4	InterQual

Condition or Procedure	LOS (days)	Type
424 OTHER HEPATOBILIARY OR PANCREAS O.R. PROCEDURES WITH CC	4.5	CMS GMLOS
425 OTHER HEPATOBILIARY OR PANCREAS O.R. PROCEDURES WITHOUT CC/MCC	2.4	CMS GMLOS
425 OTHER HEPATOBILIARY OR PANCREAS O.R. PROCEDURES WITHOUT CC/MCC	3.3	InterQual
453 COMBINED ANTERIOR AND POSTERIOR SPINAL FUSION WITH MCC	8.1	InterQual
453 COMBINED ANTERIOR AND POSTERIOR SPINAL FUSION WITH MCC	7.3	CMS GMLOS
454 COMBINED ANTERIOR AND POSTERIOR SPINAL FUSION WITH CC	4	InterQual
454 COMBINED ANTERIOR AND POSTERIOR SPINAL FUSION WITH CC	3.7	CMS GMLOS
455 COMBINED ANTERIOR AND POSTERIOR SPINAL FUSION WITHOUT CC/MCC	2.4	InterQual
455 COMBINED ANTERIOR AND POSTERIOR SPINAL FUSION WITHOUT CC/MCC	2.3	CMS GMLOS
456 SPINAL FUSION EXCEPT CERVICAL WITH SPINAL CURVATURE, MALIGNANCY, INFECTION OR EXTENSIVE FUSIONS WITH MCC	10.6	CMS GMLOS
456 SPINAL FUSION EXCEPT CERVICAL WITH SPINAL CURVATURE, MALIGNANCY, INFECTION OR EXTENSIVE FUSIONS WITH MCC	11.2	InterQual
457 SPINAL FUSION EXCEPT CERVICAL WITH SPINAL CURVATURE, MALIGNANCY, INFECTION OR EXTENSIVE FUSIONS WITH CC	5.3	CMS GMLOS
457 SPINAL FUSION EXCEPT CERVICAL WITH SPINAL CURVATURE, MALIGNANCY, INFECTION OR EXTENSIVE FUSIONS WITH CC	6.2	InterQual
458 SPINAL FUSION EXCEPT CERVICAL WITH SPINAL CURVATURE, MALIGNANCY, INFECTION OR EXTENSIVE FUSIONS WITHOUT CC/MCC	3	CMS GMLOS
458 SPINAL FUSION EXCEPT CERVICAL WITH SPINAL CURVATURE, MALIGNANCY, INFECTION OR EXTENSIVE FUSIONS WITHOUT CC/MCC	3.3	InterQual
459 SPINAL FUSION EXCEPT CERVICAL WITH MCC	7.6	CMS GMLOS
459 SPINAL FUSION EXCEPT CERVICAL WITH MCC	8.5	InterQual
460 SPINAL FUSION EXCEPT CERVICAL WITHOUT MCC	3.1	InterQual

Condition or Procedure	LOS (days)	Type
460 SPINAL FUSION EXCEPT CERVICAL WITHOUT MCC	2.8	CMS GMLOS
461 BILATERAL OR MULTIPLE MAJOR JOINT PROCEDURES OF LOWER EXTREMITY WITH MCC	7.4	CMS GMLOS
461 BILATERAL OR MULTIPLE MAJOR JOINT PROCEDURES OF LOWER EXTREMITY WITH MCC	5.6	InterQual
462 BILATERAL OR MULTIPLE MAJOR JOINT PROCEDURES OF LOWER EXTREMITY WITHOUT MCC	2.9	InterQual
462 BILATERAL OR MULTIPLE MAJOR JOINT PROCEDURES OF LOWER EXTREMITY WITHOUT MCC	2.3	CMS GMLOS
463 WOUND DEBRIDEMENT AND SKIN GRAFT EXCEPT HAND FOR MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS WITH MCC	11.1	CMS GMLOS
463 WOUND DEBRIDEMENT AND SKIN GRAFT EXCEPT HAND FOR MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS WITH MCC	5.5	InterQual
464 WOUND DEBRIDEMENT AND SKIN GRAFT EXCEPT HAND FOR MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS WITH CC	6.9	InterQual
464 WOUND DEBRIDEMENT AND SKIN GRAFT EXCEPT HAND FOR MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS WITH CC	5.6	CMS GMLOS
465 WOUND DEBRIDEMENT AND SKIN GRAFT EXCEPT HAND FOR MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS WITHOUT CC/MCC	2.4	CMS GMLOS
465 WOUND DEBRIDEMENT AND SKIN GRAFT EXCEPT HAND FOR MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS WITHOUT CC/MCC	2.7	InterQual
466 REVISION OF HIP OR KNEE REPLACEMENT WITH MCC	7.2	CMS GMLOS
466 REVISION OF HIP OR KNEE REPLACEMENT WITH MCC	6.3	InterQual
467 REVISION OF HIP OR KNEE REPLACEMENT WITH CC	3.2	CMS GMLOS
467 REVISION OF HIP OR KNEE REPLACEMENT WITH CC	4	InterQual
468 REVISION OF HIP OR KNEE REPLACEMENT WITHOUT CC/MCC	2	InterQual
468 REVISION OF HIP OR KNEE REPLACEMENT WITHOUT CC/MCC	1.6	CMS GMLOS

Condition or Procedure	LOS (days)	Type
469 MAJOR HIP AND KNEE JOINT REPLACEMENT OR REATTACHMENT OF LOWER EXTREMITY WITH MCC OR TOTAL ANKLE REPLACEMENT	3.5	CMS GMLOS
469 MAJOR HIP AND KNEE JOINT REPLACEMENT OR REATTACHMENT OF LOWER EXTREMITY WITH MCC OR TOTAL ANKLE REPLACEMENT	4.2	InterQual
470 MAJOR HIP AND KNEE JOINT REPLACEMENT OR REATTACHMENT OF LOWER EXTREMITY WITHOUT MCC	1.7	CMS GMLOS
470 MAJOR HIP AND KNEE JOINT REPLACEMENT OR REATTACHMENT OF LOWER EXTREMITY WITHOUT MCC	1.8	InterQual
471 CERVICAL SPINAL FUSION WITH MCC	8.3	InterQual
471 CERVICAL SPINAL FUSION WITH MCC	7.1	CMS GMLOS
472 CERVICAL SPINAL FUSION WITH CC	2.9	InterQual
472 CERVICAL SPINAL FUSION WITH CC	2.6	CMS GMLOS
473 CERVICAL SPINAL FUSION WITHOUT CC/MCC	1.6	InterQual
473 CERVICAL SPINAL FUSION WITHOUT CC/MCC	1.7	CMS GMLOS
474 AMPUTATION FOR MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DISORDERS WITH MCC	10.1	CMS GMLOS
474 AMPUTATION FOR MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DISORDERS WITH MCC	11.7	InterQual
475 AMPUTATION FOR MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DISORDERS WITH CC	7.2	InterQual
475 AMPUTATION FOR MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DISORDERS WITH CC	5.9	CMS GMLOS
476 AMPUTATION FOR MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DISORDERS WITHOUT CC/MCC	3.7	InterQual
476 AMPUTATION FOR MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DISORDERS WITHOUT CC/MCC	2.7	CMS GMLOS
480 HIP AND FEMUR PROCEDURES EXCEPT MAJOR JOINT WITH MCC	7.4	InterQual
480 HIP AND FEMUR PROCEDURES EXCEPT MAJOR JOINT WITH MCC	6.5	CMS GMLOS
481 HIP AND FEMUR PROCEDURES EXCEPT MAJOR JOINT WITH CC	5.1	InterQual

Condition or Procedure	LOS (days)	Type
481 HIP AND FEMUR PROCEDURES EXCEPT MAJOR JOINT WITH CC	4.5	CMS GMLOS
482 HIP AND FEMUR PROCEDURES EXCEPT MAJOR JOINT WITHOUT CC/MCC	3.2	CMS GMLOS
482 HIP AND FEMUR PROCEDURES EXCEPT MAJOR JOINT WITHOUT CC/MCC	3.6	InterQual
483 MAJOR JOINT OR LIMB REATTACHMENT PROCEDURES OF UPPER EXTREMITIES	1.6	InterQual
483 MAJOR JOINT OR LIMB REATTACHMENT PROCEDURES OF UPPER EXTREMITIES	1.7	CMS GMLOS
485 KNEE PROCEDURES WITH PRINCIPAL DIAGNOSIS OF INFECTION WITH MCC	8.3	InterQual
485 KNEE PROCEDURES WITH PRINCIPAL DIAGNOSIS OF INFECTION WITH MCC	8.3	CMS GMLOS
486 KNEE PROCEDURES WITH PRINCIPAL DIAGNOSIS OF INFECTION WITH CC	5.6	InterQual
486 KNEE PROCEDURES WITH PRINCIPAL DIAGNOSIS OF INFECTION WITH CC	5	CMS GMLOS
487 KNEE PROCEDURES WITH PRINCIPAL DIAGNOSIS OF INFECTION WITHOUT CC/MCC	3.8	InterQual
487 KNEE PROCEDURES WITH PRINCIPAL DIAGNOSIS OF INFECTION WITHOUT CC/MCC	3.5	CMS GMLOS
488 KNEE PROCEDURES WITHOUT PRINCIPAL DIAGNOSIS OF INFECTION WITH CC/MCC	4.3	InterQual
488 KNEE PROCEDURES WITHOUT PRINCIPAL DIAGNOSIS OF INFECTION WITH CC/MCC	3.1	CMS GMLOS
489 KNEE PROCEDURES WITHOUT PRINCIPAL DIAGNOSIS OF INFECTION WITHOUT CC/MCC	1.5	CMS GMLOS
489 KNEE PROCEDURES WITHOUT PRINCIPAL DIAGNOSIS OF INFECTION WITHOUT CC/MCC	2	InterQual
492 LOWER EXTREMITY AND HUMERUS PROCEDURES EXCEPT HIP, FOOT AND FEMUR WITH MCC	7.4	InterQual
492 LOWER EXTREMITY AND HUMERUS PROCEDURES EXCEPT HIP, FOOT AND FEMUR WITH MCC	6.7	CMS GMLOS
493 LOWER EXTREMITY AND HUMERUS PROCEDURES EXCEPT HIP, FOOT AND FEMUR WITH CC	4.2	CMS GMLOS

Condition or Procedure	LOS (days)	Type
493 LOWER EXTREMITY AND HUMERUS PROCEDURES EXCEPT HIP, FOOT AND FEMUR WITH CC	4.9	InterQual
494 LOWER EXTREMITY AND HUMERUS PROCEDURES EXCEPT HIP, FOOT AND FEMUR WITHOUT CC/MCC	2.8	InterQual
494 LOWER EXTREMITY AND HUMERUS PROCEDURES EXCEPT HIP, FOOT AND FEMUR WITHOUT CC/MCC	2.8	CMS GMLOS
495 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES EXCEPT HIP AND FEMUR WITH MCC	7.7	CMS GMLOS
496 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES EXCEPT HIP AND FEMUR WITH CC	3.4	CMS GMLOS
496 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES EXCEPT HIP AND FEMUR WITH CC	4.4	InterQual
497 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES EXCEPT HIP AND FEMUR WITHOUT CC/MCC	1.8	CMS GMLOS
497 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES EXCEPT HIP AND FEMUR WITHOUT CC/MCC	2.4	InterQual
498 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES OF HIP AND FEMUR WITH CC/MCC	5.9	CMS GMLOS
498 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES OF HIP AND FEMUR WITH CC/MCC	6.5	InterQual
499 LOCAL EXCISION AND REMOVAL OF INTERNAL FIXATION DEVICES OF HIP AND FEMUR WITHOUT CC/MCC	1.9	CMS GMLOS
500 SOFT TISSUE PROCEDURES WITH MCC	7.8	CMS GMLOS
500 SOFT TISSUE PROCEDURES WITH MCC	8.7	InterQual
501 SOFT TISSUE PROCEDURES WITH CC	4.1	CMS GMLOS
501 SOFT TISSUE PROCEDURES WITH CC	4.6	InterQual
502 SOFT TISSUE PROCEDURES WITHOUT CC/MCC	2.3	CMS GMLOS
502 SOFT TISSUE PROCEDURES WITHOUT CC/MCC	3	InterQual
503 FOOT PROCEDURES WITH MCC	7.3	CMS GMLOS

Condition or Procedure	LOS (days)	Type
503 FOOT PROCEDURES WITH MCC	8.4	InterQual
504 FOOT PROCEDURES WITH CC	5.2	InterQual
504 FOOT PROCEDURES WITH CC	4.5	CMS GMLOS
505 FOOT PROCEDURES WITHOUT CC/MCC	2.6	CMS GMLOS
505 FOOT PROCEDURES WITHOUT CC/MCC	2.8	InterQual
506 MAJOR THUMB OR JOINT PROCEDURES	4	CMS GMLOS
506 MAJOR THUMB OR JOINT PROCEDURES	4	InterQual
507 MAJOR SHOULDER OR ELBOW JOINT PROCEDURES WITH CC/MCC	5.5	CMS GMLOS
507 MAJOR SHOULDER OR ELBOW JOINT PROCEDURES WITH CC/MCC	5.8	InterQual
508 MAJOR SHOULDER OR ELBOW JOINT PROCEDURES WITHOUT CC/MCC	2.3	CMS GMLOS
509 ARTHROSCOPY	3.3	CMS GMLOS
510 SHOULDER, ELBOW OR FOREARM PROCEDURES, EXCEPT MAJOR JOINT PROCEDURES WITH MCC	5	CMS GMLOS
510 SHOULDER, ELBOW OR FOREARM PROCEDURES, EXCEPT MAJOR JOINT PROCEDURES WITH MCC	5.3	InterQual
511 SHOULDER, ELBOW OR FOREARM PROCEDURES, EXCEPT MAJOR JOINT PROCEDURES WITH CC	3.6	InterQual
511 SHOULDER, ELBOW OR FOREARM PROCEDURES, EXCEPT MAJOR JOINT PROCEDURES WITH CC	3.4	CMS GMLOS
512 SHOULDER, ELBOW OR FOREARM PROCEDURES, EXCEPT MAJOR JOINT PROCEDURES WITHOUT CC/MCC	2.3	InterQual
512 SHOULDER, ELBOW OR FOREARM PROCEDURES, EXCEPT MAJOR JOINT PROCEDURES WITHOUT CC/MCC	2.2	CMS GMLOS
513 HAND OR WRIST PROCEDURES, EXCEPT MAJOR THUMB OR JOINT PROCEDURES WITH CC/MCC	3.9	CMS GMLOS
513 HAND OR WRIST PROCEDURES, EXCEPT MAJOR THUMB OR JOINT PROCEDURES WITH CC/MCC	2.9	InterQual
514 HAND OR WRIST PROCEDURES, EXCEPT MAJOR THUMB OR JOINT PROCEDURES WITHOUT CC/MCC	2.2	CMS GMLOS
514 HAND OR WRIST PROCEDURES, EXCEPT MAJOR THUMB OR JOINT PROCEDURES WITHOUT CC/MCC	2.3	InterQual

Condition or Procedure	LOS (days)	Type
515 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE O.R. PROCEDURES WITH MCC	7.9	InterQual
515 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE O.R. PROCEDURES WITH MCC	7	CMS GMLOS
516 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE O.R. PROCEDURES WITH CC	3.9	CMS GMLOS
516 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE O.R. PROCEDURES WITH CC	4.7	InterQual
517 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE O.R. PROCEDURES WITHOUT CC/MCC	2.3	CMS GMLOS
517 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE O.R. PROCEDURES WITHOUT CC/MCC	2.6	InterQual
518 BACK AND NECK PROCEDURES EXCEPT SPINAL FUSION WITH MCC OR DISC DEVICE OR NEUROSTIMULATOR	5.2	InterQual
518 BACK AND NECK PROCEDURES EXCEPT SPINAL FUSION WITH MCC OR DISC DEVICE OR NEUROSTIMULATOR	5.5	CMS GMLOS
519 BACK AND NECK PROCEDURES EXCEPT SPINAL FUSION WITH CC	3.4	InterQual
519 BACK AND NECK PROCEDURES EXCEPT SPINAL FUSION WITH CC	3.2	CMS GMLOS
520 BACK AND NECK PROCEDURES EXCEPT SPINAL FUSION WITHOUT CC/MCC	2.2	InterQual
520 BACK AND NECK PROCEDURES EXCEPT SPINAL FUSION WITHOUT CC/MCC	2	CMS GMLOS
521 HIP REPLACEMENT WITH PRINCIPAL DIAGNOSIS OF HIP FRACTURE WITH MCC	6.4	CMS GMLOS
521 HIP REPLACEMENT WITH PRINCIPAL DIAGNOSIS OF HIP FRACTURE WITH MCC	7.2	InterQual
522 HIP REPLACEMENT WITH PRINCIPAL DIAGNOSIS OF HIP FRACTURE WITHOUT MCC	4.1	CMS GMLOS
522 HIP REPLACEMENT WITH PRINCIPAL DIAGNOSIS OF HIP FRACTURE WITHOUT MCC	4.7	InterQual
573 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITH MCC	12.3	InterQual
573 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITH MCC	12.2	CMS GMLOS

Condition or Procedure	LOS (days)	Type
574 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITH CC	8	CMS GMLOS
574 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITH CC	8.4	InterQual
575 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITHOUT CC/MCC	4.4	CMS GMLOS
576 SKIN GRAFT EXCEPT FOR SKIN ULCER OR CELLULITIS WITH MCC	9.3	CMS GMLOS
576 SKIN GRAFT EXCEPT FOR SKIN ULCER OR CELLULITIS WITH MCC	7.9	InterQual
577 SKIN GRAFT EXCEPT FOR SKIN ULCER OR CELLULITIS WITH CC	4.6	CMS GMLOS
577 SKIN GRAFT EXCEPT FOR SKIN ULCER OR CELLULITIS WITH CC	5.2	InterQual
578 SKIN GRAFT EXCEPT FOR SKIN ULCER OR CELLULITIS WITHOUT CC/MCC	2.6	CMS GMLOS
578 SKIN GRAFT EXCEPT FOR SKIN ULCER OR CELLULITIS WITHOUT CC/MCC	3.6	InterQual
579 OTHER SKIN, SUBCUTANEOUS TISSUE AND BREAST PROCEDURES WITH MCC	9.3	InterQual
579 OTHER SKIN, SUBCUTANEOUS TISSUE AND BREAST PROCEDURES WITH MCC	8	CMS GMLOS
580 OTHER SKIN, SUBCUTANEOUS TISSUE AND BREAST PROCEDURES WITH CC	4.1	CMS GMLOS
580 OTHER SKIN, SUBCUTANEOUS TISSUE AND BREAST PROCEDURES WITH CC	4.4	InterQual
581 OTHER SKIN, SUBCUTANEOUS TISSUE AND BREAST PROCEDURES WITHOUT CC/MCC	2.3	InterQual
581 OTHER SKIN, SUBCUTANEOUS TISSUE AND BREAST PROCEDURES WITHOUT CC/MCC	2.1	CMS GMLOS
582 MASTECTOMY FOR MALIGNANCY WITH CC/MCC	2.4	CMS GMLOS
582 MASTECTOMY FOR MALIGNANCY WITH CC/MCC	2.8	InterQual
583 MASTECTOMY FOR MALIGNANCY WITHOUT CC/MCC	1.7	CMS GMLOS
583 MASTECTOMY FOR MALIGNANCY WITHOUT CC/MCC	2.2	InterQual
614 ADRENAL AND PITUITARY PROCEDURES WITH CC/MCC	3.2	InterQual

Condition or Procedure	LOS (days)	Type
614 ADRENAL AND PITUITARY PROCEDURES WITH CC/MCC	2.8	CMS GMLOS
615 ADRENAL AND PITUITARY PROCEDURES WITHOUT CC/MCC	1.8	InterQual
615 ADRENAL AND PITUITARY PROCEDURES WITHOUT CC/MCC	1.6	CMS GMLOS
616 AMPUTATION OF LOWER LIMB FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH MCC	10.2	InterQual
616 AMPUTATION OF LOWER LIMB FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH MCC	10.1	CMS GMLOS
617 AMPUTATION OF LOWER LIMB FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH CC	6.4	InterQual
617 AMPUTATION OF LOWER LIMB FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH CC	5.8	CMS GMLOS
618 AMPUTATION OF LOWER LIMB FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITHOUT CC/MCC	3.4	CMS GMLOS
619 O.R. PROCEDURES FOR OBESITY WITH MCC	2.6	CMS GMLOS
619 O.R. PROCEDURES FOR OBESITY WITH MCC	3	InterQual
620 O.R. PROCEDURES FOR OBESITY WITH CC	1.6	CMS GMLOS
620 O.R. PROCEDURES FOR OBESITY WITH CC	1.6	InterQual
621 O.R. PROCEDURES FOR OBESITY WITHOUT CC/MCC	1.2	InterQual
621 O.R. PROCEDURES FOR OBESITY WITHOUT CC/MCC	1.3	CMS GMLOS
622 SKIN GRAFTS AND WOUND DEBRIDEMENT FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH MCC	10.5	InterQual
622 SKIN GRAFTS AND WOUND DEBRIDEMENT FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH MCC	9.3	CMS GMLOS
623 SKIN GRAFTS AND WOUND DEBRIDEMENT FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH CC	6.2	InterQual

Condition or Procedure	LOS (days)	Type
623 SKIN GRAFTS AND WOUND DEBRIDEMENT FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITH CC	5.4	CMS GMLOS
624 SKIN GRAFTS AND WOUND DEBRIDEMENT FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITHOUT CC/MCC	3.3	InterQual
624 SKIN GRAFTS AND WOUND DEBRIDEMENT FOR ENDOCRINE, NUTRITIONAL AND METABOLIC DISORDERS WITHOUT CC/MCC	2.9	CMS GMLOS
625 THYROID, PARATHYROID AND THYROGLOSSAL PROCEDURES WITH MCC	5.1	CMS GMLOS
625 THYROID, PARATHYROID AND THYROGLOSSAL PROCEDURES WITH MCC	5.6	InterQual
626 THYROID, PARATHYROID AND THYROGLOSSAL PROCEDURES WITH CC	2	CMS GMLOS
626 THYROID, PARATHYROID AND THYROGLOSSAL PROCEDURES WITH CC	2.2	InterQual
627 THYROID, PARATHYROID AND THYROGLOSSAL PROCEDURES WITHOUT CC/MCC	1.3	InterQual
627 THYROID, PARATHYROID AND THYROGLOSSAL PROCEDURES WITHOUT CC/MCC	1.3	CMS GMLOS
628 OTHER ENDOCRINE, NUTRITIONAL AND METABOLIC O.R. PROCEDURES WITH MCC	8.4	CMS GMLOS
628 OTHER ENDOCRINE, NUTRITIONAL AND METABOLIC O.R. PROCEDURES WITH MCC	10	InterQual
629 OTHER ENDOCRINE, NUTRITIONAL AND METABOLIC O.R. PROCEDURES WITH CC	6.2	CMS GMLOS
629 OTHER ENDOCRINE, NUTRITIONAL AND METABOLIC O.R. PROCEDURES WITH CC	6.8	InterQual
630 OTHER ENDOCRINE, NUTRITIONAL AND METABOLIC O.R. PROCEDURES WITHOUT CC/MCC	2.1	CMS GMLOS
630 OTHER ENDOCRINE, NUTRITIONAL AND METABOLIC O.R. PROCEDURES WITHOUT CC/MCC	2.6	InterQual
652 KIDNEY TRANSPLANT	4.3	InterQual
652 KIDNEY TRANSPLANT	4.4	CMS GMLOS
653 MAJOR BLADDER PROCEDURES WITH MCC	10.1	CMS GMLOS
653 MAJOR BLADDER PROCEDURES WITH MCC	10.5	InterQual
654 MAJOR BLADDER PROCEDURES WITH CC	5.3	CMS GMLOS

Condition or Procedure	LOS (days)	Type
654 MAJOR BLADDER PROCEDURES WITH CC	5.5	InterQual
655 MAJOR BLADDER PROCEDURES WITHOUT CC/MCC	3.8	InterQual
655 MAJOR BLADDER PROCEDURES WITHOUT CC/MCC	3.3	CMS GMLOS
656 KIDNEY AND URETER PROCEDURES FOR NEOPLASM WITH MCC	5.2	CMS GMLOS
656 KIDNEY AND URETER PROCEDURES FOR NEOPLASM WITH MCC	5.6	InterQual
657 KIDNEY AND URETER PROCEDURES FOR NEOPLASM WITH CC	2.8	CMS GMLOS
657 KIDNEY AND URETER PROCEDURES FOR NEOPLASM WITH CC	3.1	InterQual
658 KIDNEY AND URETER PROCEDURES FOR NEOPLASM WITHOUT CC/MCC	1.8	InterQual
658 KIDNEY AND URETER PROCEDURES FOR NEOPLASM WITHOUT CC/MCC	1.7	CMS GMLOS
659 KIDNEY AND URETER PROCEDURES FOR NON-NEOPLASM WITH MCC	6.4	InterQual
659 KIDNEY AND URETER PROCEDURES FOR NON-NEOPLASM WITH MCC	5.9	CMS GMLOS
660 KIDNEY AND URETER PROCEDURES FOR NON-NEOPLASM WITH CC	2.9	CMS GMLOS
660 KIDNEY AND URETER PROCEDURES FOR NON-NEOPLASM WITH CC	3.1	InterQual
661 KIDNEY AND URETER PROCEDURES FOR NON-NEOPLASM WITHOUT CC/MCC	1.8	CMS GMLOS
661 KIDNEY AND URETER PROCEDURES FOR NON-NEOPLASM WITHOUT CC/MCC	2	InterQual
662 MINOR BLADDER PROCEDURES WITH MCC	7.1	CMS GMLOS
662 MINOR BLADDER PROCEDURES WITH MCC	5.5	InterQual
663 MINOR BLADDER PROCEDURES WITH CC	3.5	CMS GMLOS
663 MINOR BLADDER PROCEDURES WITH CC	3.2	InterQual
664 MINOR BLADDER PROCEDURES WITHOUT CC/MCC	1.8	CMS GMLOS
664 MINOR BLADDER PROCEDURES WITHOUT CC/MCC	1.9	InterQual

Condition or Procedure	LOS (days)	Type
665 PROSTATECTOMY WITH MCC	7.9	CMS GMLOS
665 PROSTATECTOMY WITH MCC	9	InterQual
666 PROSTATECTOMY WITH CC	4.5	InterQual
666 PROSTATECTOMY WITH CC	3.8	CMS GMLOS
667 PROSTATECTOMY WITHOUT CC/MCC	2	InterQual
667 PROSTATECTOMY WITHOUT CC/MCC	2	CMS GMLOS
668 TRANSURETHRAL PROCEDURES WITH MCC	7.9	InterQual
668 TRANSURETHRAL PROCEDURES WITH MCC	7.2	CMS GMLOS
669 TRANSURETHRAL PROCEDURES WITH CC	3.8	CMS GMLOS
669 TRANSURETHRAL PROCEDURES WITH CC	4.4	InterQual
670 TRANSURETHRAL PROCEDURES WITHOUT CC/MCC	2	CMS GMLOS
670 TRANSURETHRAL PROCEDURES WITHOUT CC/MCC	2.3	InterQual
671 URETHRAL PROCEDURES WITH CC/MCC	3.9	CMS GMLOS
671 URETHRAL PROCEDURES WITH CC/MCC	5.3	InterQual
672 URETHRAL PROCEDURES WITHOUT CC/MCC	1.5	CMS GMLOS
673 OTHER KIDNEY AND URINARY TRACT PROCEDURES WITH MCC	8.8	CMS GMLOS
673 OTHER KIDNEY AND URINARY TRACT PROCEDURES WITH MCC	10	InterQual
674 OTHER KIDNEY AND URINARY TRACT PROCEDURES WITH CC	6.1	CMS GMLOS
674 OTHER KIDNEY AND URINARY TRACT PROCEDURES WITH CC	7.3	InterQual
675 OTHER KIDNEY AND URINARY TRACT PROCEDURES WITHOUT CC/MCC	3.2	InterQual
675 OTHER KIDNEY AND URINARY TRACT PROCEDURES WITHOUT CC/MCC	2.7	CMS GMLOS
707 MAJOR MALE PELVIC PROCEDURES WITH CC/MCC	2.4	InterQual
707 MAJOR MALE PELVIC PROCEDURES WITH CC/MCC	2.2	CMS GMLOS

Condition or Procedure	LOS (days)	Type
708 MAJOR MALE PELVIC PROCEDURES WITHOUT CC/MCC	1.2	InterQual
708 MAJOR MALE PELVIC PROCEDURES WITHOUT CC/MCC	1.4	CMS GMLOS
709 PENIS PROCEDURES WITH CC/MCC	4	InterQual
709 PENIS PROCEDURES WITH CC/MCC	4.3	CMS GMLOS
710 PENIS PROCEDURES WITHOUT CC/MCC	10.4	InterQual
710 PENIS PROCEDURES WITHOUT CC/MCC	1.7	CMS GMLOS
711 TESTES PROCEDURES WITH CC/MCC	5.1	CMS GMLOS
711 TESTES PROCEDURES WITH CC/MCC	7.1	InterQual
712 TESTES PROCEDURES WITHOUT CC/MCC	2.6	CMS GMLOS
713 TRANSURETHRAL PROSTATECTOMY WITH CC/MCC	2.6	CMS GMLOS
713 TRANSURETHRAL PROSTATECTOMY WITH CC/MCC	3	InterQual
714 TRANSURETHRAL PROSTATECTOMY WITHOUT CC/MCC	1.4	InterQual
714 TRANSURETHRAL PROSTATECTOMY WITHOUT CC/MCC	1.5	CMS GMLOS
734 PELVIC EVISCERATION, RADICAL HYSTERECTOMY AND RADICAL VULVECTOMY WITH CC/MCC	3.6	InterQual
734 PELVIC EVISCERATION, RADICAL HYSTERECTOMY AND RADICAL VULVECTOMY WITH CC/MCC	3.4	CMS GMLOS
735 PELVIC EVISCERATION, RADICAL HYSTERECTOMY AND RADICAL VULVECTOMY WITHOUT CC/MCC	2	InterQual
735 PELVIC EVISCERATION, RADICAL HYSTERECTOMY AND RADICAL VULVECTOMY WITHOUT CC/MCC	1.6	CMS GMLOS
736 UTERINE AND ADNEXA PROCEDURES FOR OVARIAN OR ADNEXAL MALIGNANCY WITH MCC	7.9	CMS GMLOS
736 UTERINE AND ADNEXA PROCEDURES FOR OVARIAN OR ADNEXAL MALIGNANCY WITH MCC	7.9	InterQual
737 UTERINE AND ADNEXA PROCEDURES FOR OVARIAN OR ADNEXAL MALIGNANCY WITH CC	3.9	CMS GMLOS

Condition or Procedure	LOS (days)	Type
737 UTERINE AND ADNEXA PROCEDURES FOR OVARIAN OR ADNEXAL MALIGNANCY WITH CC	4	InterQual
738 UTERINE AND ADNEXA PROCEDURES FOR OVARIAN OR ADNEXAL MALIGNANCY WITHOUT CC/MCC	2.3	CMS GMLOS
738 UTERINE AND ADNEXA PROCEDURES FOR OVARIAN OR ADNEXAL MALIGNANCY WITHOUT CC/MCC	2.6	InterQual
739 UTERINE AND ADNEXA PROCEDURES FOR NON-OVARIAN AND NON-ADNEXAL MALIGNANCY WITH MCC	6.3	CMS GMLOS
739 UTERINE AND ADNEXA PROCEDURES FOR NON-OVARIAN AND NON-ADNEXAL MALIGNANCY WITH MCC	6.5	InterQual
740 UTERINE AND ADNEXA PROCEDURES FOR NON-OVARIAN AND NON-ADNEXAL MALIGNANCY WITH CC	3.4	InterQual
740 UTERINE AND ADNEXA PROCEDURES FOR NON-OVARIAN AND NON-ADNEXAL MALIGNANCY WITH CC	2.7	CMS GMLOS
741 UTERINE AND ADNEXA PROCEDURES FOR NON-OVARIAN AND NON-ADNEXAL MALIGNANCY WITHOUT CC/MCC	2	InterQual
741 UTERINE AND ADNEXA PROCEDURES FOR NON-OVARIAN AND NON-ADNEXAL MALIGNANCY WITHOUT CC/MCC	1.6	CMS GMLOS
742 UTERINE AND ADNEXA PROCEDURES FOR NON-MALIGNANCY WITH CC/MCC	2.7	CMS GMLOS
742 UTERINE AND ADNEXA PROCEDURES FOR NON-MALIGNANCY WITH CC/MCC	2.5	InterQual
743 UTERINE AND ADNEXA PROCEDURES FOR NON-MALIGNANCY WITHOUT CC/MCC	1.7	InterQual
743 UTERINE AND ADNEXA PROCEDURES FOR NON-MALIGNANCY WITHOUT CC/MCC	1.6	CMS GMLOS
746 VAGINA, CERVIX AND VULVA PROCEDURES WITH CC/MCC	4	InterQual
746 VAGINA, CERVIX AND VULVA PROCEDURES WITH CC/MCC	3.3	CMS GMLOS
747 VAGINA, CERVIX AND VULVA PROCEDURES WITHOUT CC/MCC	1.8	InterQual
747 VAGINA, CERVIX AND VULVA PROCEDURES WITHOUT CC/MCC	1.4	CMS GMLOS

Condition or Procedure	LOS (days)	Type
748 FEMALE REPRODUCTIVE SYSTEM RECONSTRUCTIVE PROCEDURES	1.5	InterQual
748 FEMALE REPRODUCTIVE SYSTEM RECONSTRUCTIVE PROCEDURES	1.4	CMS GMLOS
749 OTHER FEMALE REPRODUCTIVE SYSTEM O.R. PROCEDURES WITH CC/MCC	5.3	InterQual
749 OTHER FEMALE REPRODUCTIVE SYSTEM O.R. PROCEDURES WITH CC/MCC	5.5	CMS GMLOS
750 OTHER FEMALE REPRODUCTIVE SYSTEM O.R. PROCEDURES WITHOUT CC/MCC	7.6	InterQual
750 OTHER FEMALE REPRODUCTIVE SYSTEM O.R. PROCEDURES WITHOUT CC/MCC	2	CMS GMLOS
799 SPLENIC PROCEDURES WITH MCC	6.8	InterQual
799 SPLENIC PROCEDURES WITH MCC	7.7	CMS GMLOS
800 SPLENIC PROCEDURES WITH CC	4.2	CMS GMLOS
801 SPLENIC PROCEDURES WITHOUT CC/MCC	2.3	CMS GMLOS
801 SPLENIC PROCEDURES WITHOUT CC/MCC	2.7	InterQual
802 OTHER O.R. PROCEDURES OF THE BLOOD AND BLOOD FORMING ORGANS WITH MCC	7.7	InterQual
802 OTHER O.R. PROCEDURES OF THE BLOOD AND BLOOD FORMING ORGANS WITH MCC	7.7	CMS GMLOS
803 OTHER O.R. PROCEDURES OF THE BLOOD AND BLOOD FORMING ORGANS WITH CC	4.5	InterQual
803 OTHER O.R. PROCEDURES OF THE BLOOD AND BLOOD FORMING ORGANS WITH CC	3.8	CMS GMLOS
804 OTHER O.R. PROCEDURES OF THE BLOOD AND BLOOD FORMING ORGANS WITHOUT CC/MCC	2.2	InterQual
804 OTHER O.R. PROCEDURES OF THE BLOOD AND BLOOD FORMING ORGANS WITHOUT CC/MCC	1.7	CMS GMLOS
904 SKIN GRAFTS FOR INJURIES WITH CC/MCC	8.3	InterQual
904 SKIN GRAFTS FOR INJURIES WITH CC/MCC	6.9	CMS GMLOS
905 SKIN GRAFTS FOR INJURIES WITHOUT CC/MCC	3.2	CMS GMLOS
906 HAND PROCEDURES FOR INJURIES	2.9	InterQual
906 HAND PROCEDURES FOR INJURIES	2.9	CMS GMLOS

Condition or Procedure	LOS (days)	Type
907 OTHER O.R. PROCEDURES FOR INJURIES WITH MCC	6.9	CMS GMLOS
907 OTHER O.R. PROCEDURES FOR INJURIES WITH MCC	8	InterQual
908 OTHER O.R. PROCEDURES FOR INJURIES WITH CC	4.3	InterQual
908 OTHER O.R. PROCEDURES FOR INJURIES WITH CC	3.8	CMS GMLOS
909 OTHER O.R. PROCEDURES FOR INJURIES WITHOUT CC/MCC	2.6	InterQual
909 OTHER O.R. PROCEDURES FOR INJURIES WITHOUT CC/MCC	2.3	CMS GMLOS
927 EXTENSIVE BURNS OR FULL THICKNESS BURNS WITH MV >96 HOURS WITH SKIN GRAFT	26.4	CMS GMLOS
927 EXTENSIVE BURNS OR FULL THICKNESS BURNS WITH MV >96 HOURS WITH SKIN GRAFT	20.9	InterQual
928 FULL THICKNESS BURN WITH SKIN GRAFT OR INHALATION INJURY WITH CC/MCC	12.9	InterQual
928 FULL THICKNESS BURN WITH SKIN GRAFT OR INHALATION INJURY WITH CC/MCC	12.6	CMS GMLOS
929 FULL THICKNESS BURN WITH SKIN GRAFT OR INHALATION INJURY WITHOUT CC/MCC	6	CMS GMLOS
929 FULL THICKNESS BURN WITH SKIN GRAFT OR INHALATION INJURY WITHOUT CC/MCC	5.9	InterQual
933 EXTENSIVE BURNS OR FULL THICKNESS BURNS WITH MV >96 HOURS WITHOUT SKIN GRAFT	2.5	CMS GMLOS
934 FULL THICKNESS BURN WITHOUT SKIN GRAFT OR INHALATION INJURY	4.9	CMS GMLOS
934 FULL THICKNESS BURN WITHOUT SKIN GRAFT OR INHALATION INJURY	4.9	InterQual
955 CRANIOTOMY FOR MULTIPLE SIGNIFICANT TRAUMA	9.6	InterQual
955 CRANIOTOMY FOR MULTIPLE SIGNIFICANT TRAUMA	8.3	CMS GMLOS
956 LIMB REATTACHMENT, HIP AND FEMUR PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA	8.3	InterQual
956 LIMB REATTACHMENT, HIP AND FEMUR PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA	6.3	CMS GMLOS

Condition or Procedure	LOS (days)	Type
957 OTHER O.R. PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA WITH MCC	13.3	InterQual
957 OTHER O.R. PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA WITH MCC	10	CMS GMLOS
958 OTHER O.R. PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA WITH CC	7.6	InterQual
958 OTHER O.R. PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA WITH CC	6.7	CMS GMLOS
959 OTHER O.R. PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA WITHOUT CC/MCC	4.1	InterQual
959 OTHER O.R. PROCEDURES FOR MULTIPLE SIGNIFICANT TRAUMA WITHOUT CC/MCC	4	CMS GMLOS

Notes:**1:**

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. Patients undergoing surgical procedures have experienced a 30 day readmission rate between 5.7% and 12.8%. Reasons for these readmissions include post surgical complications, shorter lengths of stay, chronic comorbid conditions, and the American Society of Anesthesiologists class (Merkow et al., JAMA 2015, 313: 483-95; Morris et al., JAMA Surg 2014, 149: 348-54; Glance et al., JAMA Surg 2014, 149: 439-45; Dawes et al., J Am Coll Surg 2014, 219: 382-9; Lawson et al., Ann Surg 2013, 258: 10-8).

2:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

3:

Instruction: Patients with a traumatic injury who have an ambulatory surgical procedure scheduled within 24 hours and cannot be discharged before the procedure due to severe pain should be reviewed here.

4:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

5:

This criteria point applies to high-risk patients who require uninterrupted anticoagulation with a brief discontinuation prior to a surgical or invasive procedure. When patients are considered for uninterrupted anticoagulation or bridging therapy, the goal is to decrease the risk for thromboembolic events while minimizing the potential for bleeding. Procedures at high risk for bleeding may include, but are not limited to, urologic surgery, surgery on vascular organs (e.g., spleen or liver), cancer surgery, arthroplasty, or cranial, cardiac or spinal surgery (Douketis et al., Chest 2012, 141: e326S-50S).

6:

Patients in the moderate risk category may also be considered for bridging therapy based on additional assessment of their risk for stroke and the type of procedure being performed. Patients who are at moderate risk for thromboembolism include (Douketis et al., Chest 2012, 141: e326S-50S):

Mechanical heart valves:

- Bileaflet aortic valve and 1 or more of the following risk factors: atrial fibrillation, prior stroke or transient ischemic attack, hypertension, diabetes, heart failure, age ≥ 75)

Venous thromboembolism (VTE):

- Within the last 3-12 months
- Non-severe thrombophilia (e.g., heterozygous factor IV Leiden or prothrombin gene mutation)
- Recurrent VTE
- Active cancer (treated within 6 months or palliative)

Instruction: The need for "bridging therapy" in moderate risk has not been adequately defined and is based on the risk for bleeding, the risk of developing arterial embolism, as well as the surgical risk of the procedures being performed. These patients should be sent for secondary review.

7:

The CHADS₂ score evaluates a patient's stroke risk. Each item measured is assigned points as follows:

- Congestive heart failure = 1 point
- Hypertension = 1 point
- Age 75 or greater = 1 point
- Diabetes = 1 point
- Previous history of stroke or TIA = 2 points

A higher score correlates with an increased risk of stroke (Baron et al., N Engl J Med 2013, 368: 2113-24; Douketis et al., Chest 2012, 141: e326S-50S; Gage et al., JAMA 2001, 285: 2864-70).

8:

A trial related to the bridging of patients with atrial fibrillation demonstrated no difference in outcomes among low risk patients who received anticoagulant bridge therapy and those who did not. However, the conclusions were appropriate for low risk, predominantly outpatient procedures and were not applicable to inpatient care for patients where there is a high risk of bleeding and arterial thromboembolism (e.g., carotid endarterectomy, major cancer surgery, cardiac or neurosurgery) (Douketis et al., N Engl J Med 2015, 373: 823-33).

Instruction: These criteria are intended for patients requiring inpatient care for surgical procedures where there is a high risk of bleeding and arterial thromboembolism.

9:

Patients with atrial fibrillation should be considered for bridging therapy based on their risk for bleeding, the risk of developing arterial embolism, as well as the surgical risk of the procedures being performed. Including patients with a CHADS₂-VASc (Congestive heart failure, **H**igh blood pressure, **A**ge 75 or older, **D**iabetes, and previous **S**troke or transient ischemic attack) score greater than 2 assures that all high risk patients may be considered for bridging therapy (Douketis et al., Chest 2012, 141: e326S-50S).

10:

Patients with severe thrombophilia may be deficient in protein C, protein S, or antithrombin, have homozygous factor V Leiden or prothrombin gene mutations, or have antiphospholipid antibodies (Douketis et al., Chest 2022, 162: e207-e43).

11:

Previous practice suggested that patients receive 5 days of unfractionated heparin prior to vitamin K antagonist therapy. This shortened duration is expected to decrease the risk of heparin-induced thrombocytopenia. If the international normalized ratio (INR) becomes therapeutic (greater than 2.0) prior to 5 days of treatment it is acceptable to discontinue parenteral therapy prior to the patient receiving 5 days of treatment (Kearon et al., Chest 2016, 149: 315-52).

12:

Instruction: Patients with a traumatic injury who have a designated inpatient surgical procedure scheduled within 24 hours and require intravenous (IV) fluid or pain management can be reviewed here. Patients who require medical management of their injury before the surgical procedure may be reviewed in the General Trauma subset. Once the designated inpatient surgical procedure is scheduled to occur within 24 hours, the patient may be reviewed here.

13:

Instruction: The following resources can be used to determine when a procedure might be "Designated inpatient setting" for a traumatic injury:

- The procedure is listed on the InterQual® *Guidelines for Surgery and Procedures in the Inpatient Setting* list (located in Clinical Reference). **Note:** An asterisk "*" next to a procedure indicates the procedure may be performed in either the inpatient or outpatient setting. InterQual® recommends that hospitals consult with the responsible health plan prior to making a formal decision to determine the actual setting. At present, there is insufficient evidence to clearly define procedures marked with an "*" as being inpatient or outpatient due to variations in practice, the physician's experience, the presence of medical

comorbidities, the potential for significant complications, the need for prolonged monitoring, and the actual length of stay.

- The procedure is listed on the Centers for Medicare and Medicaid Services (CMS) Inpatient Only List, Addendum E (located in Clinical Reference)
- The procedure is on a list that organizational policy directs reviewers to apply

For criteria to review a procedure designated as outpatient, but being requested as inpatient due to a patient's clinical findings or risk factors, refer to the criteria "Ambulatory procedure (excludes low risk procedure) and high risk for complications requiring inpatient management."

14:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

15:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
 - Unexpected death
 - Transfer of the patient
 - Unexpected clinical improvement
 - Election of hospice care
- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
 - Surgical procedure or intervention found on the CMS Inpatient Only List
- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

16:

Delayed recovery from anesthesia includes:

- Delayed return of sensation (Regional)
- Deficit in mobility or coordination (Spinal)
- Hypercapnia or hypoventilation (General, Spinal)
- Mental status changes (General)
- Persistent hypotension (General, Spinal)
- Persistent vomiting (General)
- Urinary retention (General, Spinal)
- Unstable level of consciousness (General)

17:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

18:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

19:

Evidence in the medical literature to support this criterion or the effectiveness of this intervention or service is mixed or unclear. The criteria point reflects current best evidence and practice. It is the product of a peer review process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

20:

Instruction: Patients with an open fracture and delayed skin closure requiring a staged approach should be reviewed at the Acute level of care.

21:

Open fractures carry a higher risk of infection which can lead to impaired wound healing and nonunion; therefore, prophylactic systemic anti-infectives are recommended. For Gustilo Type I and II open fractures, anti-infectives with gram-positive coverage are recommended for 24 hours after internal fixation and wound repair. For Gustilo Type III open fractures, additional gram-negative coverage is recommended, and anti-infectives should continue for 24 hours after soft tissue coverage is achieved (Garner et al., J Am Acad Orthop Surg 2020, 28: 309-15; Hoff et al., J Trauma 2011, 70: 751-4).

22:

This criterion captures open fracture repair of the fibula, radius, ulna, hand, foot, digit, or facial fracture, which are not designated as inpatient procedures but may require prophylactic IV anti-infective post-procedure.

23:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

24:

Headache following lumbar puncture (LP) results from a persistent leak of cerebrospinal fluid at the site. The headache is typically worse when the patient is upright, and markedly relieved when supine. Most headaches will spontaneously resolve within a few days of the LP. Post LP headaches that do not resolve may benefit from placement of an epidural blood patch at the site of the leak, which usually relieves the symptoms (White, The lumbar epidural blood patch: A primer. 2019).

25:

Intravenous caffeine has demonstrated effectiveness in treating spinal headache. However, these findings should be interpreted with caution as the studies had small sample sizes and limited generalizability (Basurto Ona et al., Cochrane Database Syst Rev 2011: CD007887).

26:

Protracted vomiting refers to vomiting that is still active despite multiple doses of parenteral antiemetics.

27:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). The patient will require assistance with one or more limitations.

28:

Instruction: This criteria point is for the patient who needs to be evaluated by physical therapy (PT) and/or occupational therapy (OT) and set up with equipment before they can safely be discharged home as the result of an unplanned functional impairment that developed post ambulatory procedure. Patients who are having a procedure and expected to have a functional impairment as a result of the surgery (e.g., total knee replacement) should have the therapy evaluation as part of their pre-operative assessment and discharge plan and are excluded from this line of criteria.

29:

Instruction: The decision to admit a patient for a surgical procedure remains the responsibility of the treating provider. Documentation of the patient's clinical condition is essential to ensure the appropriate setting and level of care required. The following resources can be used to determine when a procedure might be "Designated inpatient setting":

- The procedure may have been prior authorized as inpatient by the patient's health plan
- The procedure is listed on the InterQual® *Guidelines for Surgery and Procedures in the Inpatient Setting* list (located in Clinical Reference). **Note:** An asterisk "*" next to a procedure indicates the procedure may

be performed in either the inpatient or outpatient setting. InterQual® recommends that hospitals consult with the responsible health plan prior to making a formal decision to determine the actual setting. At present, there is insufficient evidence to clearly define procedures marked with an "*" as being inpatient or outpatient due to variations in practice, the physician's experience, the presence of medical comorbidities, the potential for significant complications, the need for prolonged monitoring, and the actual length of stay.

- The procedure is listed on the Centers for Medicare and Medicaid Services (CMS) Inpatient Only List, Addendum E (located in Clinical Reference)
- The procedure is on a list that organizational policy directs reviewers to apply

For criteria to review a procedure designated as outpatient, but being requested as inpatient due to a patient's clinical findings or risk factors, refer to the criteria "Ambulatory procedure (excludes low risk procedure) and high risk for complications requiring inpatient management."

Note: These criteria are not intended to override a prior authorization determination.

30:

Instruction: To satisfy this criteria point, documentation in the patient's medical record should include the expectation that the patient will require at least 2 days of inpatient post-surgical management. The physician documentation validates the need for inpatient management of the patient post-operatively due to high risk comorbid conditions.

Application of this criteria point is appropriate for an ambulatory procedure that requires inpatient routine post-op management documented by the physician on or prior to admission. Determination of the appropriate setting (inpatient versus outpatient) is a clinical decision best made with consideration of multiple clinical factors including, but not limited to (Victor M. Zaydfudim, In: Sabiston: Textbook of Surgery: The Biological Basis of Modern Surgical Practice, 21 edn. 187-222, 2022; American Society of Anesthesiologists, ASA Physical Status Classification System. 2020; Fox et al., Surgery 2014, 155: 743-53):

- Type of procedure planned (e.g., laparoscopic versus open procedure, need for postoperative drains, monitoring, or therapy)
- Urgency of the procedure
- Patient's hemodynamic stability
- Medical stabilization of comorbidities (e.g., cardiac function, diabetes)
- Likelihood of complications based on patient's medical history (e.g., bleeding, infection, thromboembolic events, fluid or electrolyte imbalance)

31:

Instruction: Ambulatory surgeries appropriate for this criteria point include intermediate risk procedures and excludes low risk procedures. Some excluded low risk invasive ambulatory procedures are (Steiner et al., In: Healthcare Cost and Utilization Project (HCUP) Statistical Briefs. Revised 2020; Kristensen et al., Eur J Anaesthesiol 2014, 31: 517-73) (this list is not meant to be all inclusive):

- Eye (lens, cataract repair)
- Ear (myringotomy)
- Nose (plastic procedures on nose)
- Breast (plastic procedures on breast)
- Dental
- Testicular or penile
- Endocrine (Lymph node or thyroid biopsies)
- Skin (superficial biopsies)
- Urological: minor (transurethral resection of the prostate)
- Musculoskeletal: minor (meniscectomy)

32:

Some excluded low risk invasive ambulatory procedures are (this list is not meant to be all inclusive):

- Eye (lens, cataract repair)
- Ear (myringotomy)
- Nose (plastic procedures on nose)
- Breast (plastic procedures on breast)
- Dental

- Testicular or penile
- Endocrine (Lymph node or thyroid biopsies)
- Skin (superficial biopsies)
- Urological: minor (transurethral resection of the prostate)
- Musculoskeletal: minor (meniscectomy)

33:

Instruction: This criteria point is for the patient with comorbidities that have been managed to achieve an optimal state prior to ambulatory surgery (when possible). These patients are at an increased risk for post operative complications and may require further monitoring post procedure (Rajan, Curr Opin Anaesthesiol 2020, 33: 724-31).

34:

Pre-op optimization refers to those patients who may require an aggressive medical management program prior to a surgical procedure (Zietlow et al., Am J Med 2022, 135: 39-48).

Pre-operative goals may include:

- Stabilization of cardiac condition
- Optimization of nutritional status (e.g., TPN)
- Maximization of oxygen delivery
- Control or elimination of infection or sepsis
- Increased granulation tissue prior to skin flap surgery
- Anticoagulation management

35:

The American Society of Anesthesiologists (ASA) classification system helps to define patients who are at an increased risk for complications during surgery. A patient that requires an ambulatory surgery or procedure, and has an ASA IV classification, is compromised by a severe comorbid condition and will require additional monitoring post-operatively (American Society of Anesthesiologists, ASA Physical Status Classification System. 2020; Mathis et al., Anesthesiology 2013, 119: 1310-21).

36:

The American Society of Anesthesiologists (ASA) classification III describes a patient with moderate to severe functional limitations from chronic illness and their perioperative risk. These limitations may prevent a patient from adequately caring for themselves post-surgery or procedure requiring additional monitoring or care (American Society of Anesthesiologists, ASA Physical Status Classification System. 2020; Foley et al., BMC Surg 2021, 21: 253).

37:

As valvular heart disease progresses, patients may develop heart failure, atrial fibrillation, exertional dyspnea, or peripheral edema. These symptoms of coronary artery disease place the patient at a higher risk for complications post ambulatory surgery (Nishimura et al., Circulation 2017, 135: e1159-e95; Nishimura et al., J Thorac Cardiovasc Surg 2014, 148: e1-e132).

38:

Nearly 20%(0.20) of patients with cirrhosis and ascites develop an umbilical hernia. Repair in this situation may be complicated by ascites leakage from trocar sites if laparoscopic surgery is done or infection if mesh is used. Elective repair after ascites control is preferred to watchful waiting (Coelho et al., World J Gastrointest Surg 2016, 8: 476-82).

39:

An example of a functional disability caused by COPD would be the inability to perform ADLs due to shortness of breath.

40:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and

reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

41:

When a patient presents for an ambulatory procedure and has uncontrolled diabetes, the patient is at higher risk of post-operative complications that may require additional monitoring (Ferri, In: Ferri's Clinical Advisor. 2022:495-509.e1).

42:

Per the American Society of Anesthesiologists, patients with a history of ESRD may require an inpatient setting for an otherwise outpatient procedure. These patients require additional monitoring due to the increased risk of post-operative complications (American Society of Anesthesiologists, ASA Physical Status Classification System. 2020; Fox et al., Surgery 2014, 155: 743-53).

43:

Dialysis includes hemodialysis or peritoneal dialysis. Both modalities are equally effective and should be provided based on patient needs and device availability (Liu et al., Cochrane Database Syst Rev 2017, 12: Cd011457).

44:

Per the American Society of Anesthesiologists, patients with a history of heart failure may require an inpatient setting for an otherwise outpatient procedure. These patients require additional monitoring due to the increased risk of post-operative complications (American Society of Anesthesiologists, ASA Physical Status Classification System. 2020; Fox et al., Surgery 2014, 155: 743-53).

45:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen.

Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections.

Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases.

Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.

Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e.g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78;

Lipska et al., Diabetes Care 2013, 36: 3535-42).

46:

Ambulatory surgeries or procedures may be affected by a patient's poor nutritional status. As an example, malnutrition can affect proper healing of wounds after surgery and may require additional monitoring or care post-operatively (Gu et al., J Arthroplasty 2019, 34: 1013-24; Hackett et al., Int J Surg 2015, 18: 184-90).

47:

An adult malnutrition diagnosis cannot be achieved with a single characteristic. Patients that are undernourished, or are not absorbing nutrients, need nutritional intervention. The American Society for Parenteral and Enteral Nutrition defines malnutrition by at least 2 of the following (Quartarolo et al., Nutr Clin Pract 2021, 36: 1068-71; White et al., JPEN J Parenter Enteral Nutr 2012, 36: 275-83):

- Unable to take in enough calories
- Documented weight loss (greater than 2%(0.02) weight loss from baseline within the last week or greater than 5%(0.05) weight loss from baseline within the last 30 days)
- Decreased muscle mass or wasting (subcutaneous fat loss)
- Decreased hand grip strength (documented loss of functional status)
- Localized or generalized fluid accumulation that masks weight loss

48:

Patients with Class III obesity, who have impaired functional mobility or lack dexterity to manage their own care, may physically be unable to reach the anatomic site of care or be unable to offload the surgical site effectively. These patients may require additional monitoring and care after an ambulatory surgery or procedure.

49:

A coagulopathy can present as a hereditary bleeding disorder such as factor VIII, IX, or von Willebrand's disease. Other conditions including, but not limited to, vitamin K deficiency, a history of disseminated intravascular coagulation (DIC), anticoagulant (heparin, warfarin) sensitivities or contraindications, or a history hemorrhage with liver disease are all considered coagulopathies (Hurwitz et al., Hematol Oncol Clin North Am 2017, 31: 1123-45). Laboratory values that are consistent with a coagulopathy include:

- Prolonged prothrombin time (PT)
- Prolonged international normalized ratio (INR)
- Prolonged partial thromboplastin time (PTT)
- Decreased platelet count

50:

Frailty is a complex condition that occurs because of the aging process and makes individuals more vulnerable to stress, leading to adverse health outcomes like impaired mobility, falls, hospitalizations, and even mortality (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub; Kwak and Thompson, Sports Med Health Sci 2021, 3: 1-10; Hoogendijk et al., Lancet 2019, 394: 1365-75). Studies have shown that older individuals identified as frail during hospital admission are at greater risk of in-hospital mortality, longer hospital stays, and functional decline at hospital discharge (Boucher et al., EClinicalMedicine 2023, 59: epub; Cunha et al., Ageing Res Rev 2019, 56: epub). Although there are many validated frailty screening tools available, there has yet to be one that is recognized as the gold standard for identifying frailty. Most of these tools are based on two models: the frailty phenotype and the deficit accumulation model. The frailty phenotype model focuses on the physical aspects of frailty (e.g., strength, walking speed, weight loss), while the deficit accumulation model identifies the number of deficiencies (e.g., functional, cognitive, comorbidities) to calculate a frailty index score (Hoogendijk et al., Lancet 2019, 394: 1365-75). However, both models have limitations in their implementation within the acute care setting, as the frailty phenotype assessment involves performing tests to evaluate the domains (e.g., grip strength, walking speed), and the deficit accumulation model is time-consuming. Therefore, frailty criteria have been developed based on literature and the two models mentioned above to help identify older adults who may have or be at risk for frailty.

51:

Frailty can lead to poor outcomes after surgical procedures. Frail older individuals have been found to be at twofold greater risk of 30-day complications and discharge to a specialized facility after surgery, and at increased

risk for postoperative delirium and 30-day mortality (Tjeertes et al., Aging Dis 2020, 11: 1276-90). If an unplanned or emergent procedure is required, older individuals with frailty may be more vulnerable to negative outcomes since their health cannot be optimized before surgery. Even low-risk emergency procedures (e.g., laparoscopic appendectomy or cholecystectomy) have been associated with a higher risk of postoperative complications, readmission, and all-cause mortality in frail older individuals (Collins et al., Surgery 2023, 173: 485-91). Therefore, it is important to consider an individual's frailty status even when undergoing an ambulatory procedure to determine the need for inpatient postoperative management.

52:

Instruction: This criterion applies to patients who have previously been diagnosed with frailty and are currently experiencing the condition. This information can be retrieved from the medical record, such as the problem list or past medical history. However, describing a patient's appearance as "frail" in a progress note cannot be reviewed in this criterion.

53:

Instruction: The following criteria should be selected based on the patient's baseline or current health status. A patient's baseline frailty state can affect their risk for severe illness. Similarly, if a patient presents with signs of frailty in the setting of an acute illness or injury, it should also be seen as a marker of illness severity (Theou et al., Clin Geriatr Med 2018, 34: 369-86).

54:

Studies have suggested that incorporating cognitive impairment evaluation with physical characteristics enhances the predictability of frailty (Avila-Funes et al., J Am Geriatr Soc 2009, 57: 453-61). More recently, a systematic review and meta-analysis found that mean Mini-Mental State Examination (MMSE) scores were significantly lower in individuals who were identified as frail by the phenotype model, which focuses on physical function, compared to non-frail and pre-frail individuals. As a result of the strong relationship between physical frailty and cognitive impairment, the authors concluded that cognitive function should be incorporated into frailty assessments (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub). Multiple frailty screening tools developed using the cumulative deficit approach assess the cognitive domain, specifically focusing on cognitive impairment or dementia (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7; Clegg et al., Age Ageing 2016, 45: 353-60).

55:

Multimorbidity refers to the presence of 2 or more chronic conditions or diseases in an individual. Chronic conditions or diseases are those that last for a prolonged period (e.g., 1 year or more) or are permanent and result in disability, decreased quality of life, or require prolonged treatment or rehabilitation (Centers for Disease Control and Prevention, About Chronic Diseases. 2022 [cited Nov 8 2023]; Nicholson et al., J Clin Epidemiol 2019, 105: 142-6; Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23). Examples of chronic conditions that have been identified in the literature include but are not limited to (Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23):

- Chronic lung disease (e.g., COPD, asthma)
- Chronic kidney disease (e.g., stage 3-4)
- End-stage kidney disease
- Diabetes
- Hypertension
- Congestive heart failure
- Inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis)
- Mental health disorders (e.g., drug or alcohol dependence, depression, bipolar affective disorder)

56:

According to a recent systematic review and meta-analysis, 72%(0.72) of frail individuals have multimorbidity. The study also found a significant association between frailty and having two or more diseases (Vetrano et al., J Gerontol A Biol Sci Med Sci 2019, 74: 659-66).

57:

According to a study conducted at a Medicare Accountable Care Organization (ACO), the presence of one functional deficit on a frailty index tool in the electronic health record (EHR) was found to be independently associated with a higher risk of hospitalization and injurious falls (Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7). When looking specifically at the association between hearing and vision deficits and frailty, a systematic review and meta-analysis found that hearing and vision impairment were associated with 2.5- and 3.0-fold higher odds of frailty, respectively (Tan et al., J Gerontol A Biol Sci Med Sci 2020, 75: 2461-70).

58:

Functional impairment refers to an individual's ability to independently perform activities of daily living (ADLs) (e.g., bathing, dressing, toileting, ambulating), instrumental activities of daily living (IADLs) (e.g., managing medications, finances, preparing meals, usage of the telephone, performing basic housework), or a deficit in hearing or vision that is not corrected by adaptive equipment (e.g., hearing aid, glasses).

59:

Malnutrition has been found to be closely associated with frailty (Lorenzo-López et al., BMC Geriatr 2017, 17: 108). This criterion can be met by one or more of the following:

- Documented weight loss, unintentional, of ≥ 10 pounds or $\geq 5\%$ (0.05) of body weight in the past year (Fried et al., J Gerontol A Biol Sci Med Sci 2001, 56: M146-56)
- Score of ≤ 11 on the Mini Nutritional Assessment-Short Form (MNA-SF) (O'Caoimh et al., Int J Environ Res Public Health 2019, 16: epub; Lorenzo-López et al., BMC Geriatr 2017, 17: 108)
- Body mass index (BMI) < 18.5 kg/m² (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7)
- Documented moderate or severe subcutaneous fat or muscle loss (Cederholm et al., Clin Nutr 2019, 38: 1-9; White et al., JPEN J Parenter Enteral Nutr 2012, 36: 275-83)

60:

Several studies have shown that individuals with polypharmacy are more likely to be frail (Gutiérrez-Valencia et al., Br J Clin Pharmacol 2018, 84: 1432-44). In a recent systematic review and meta-analysis of 66 studies, it was found that 59%(0.59) of older adults with frailty had polypharmacy, defined as the use of five or more medications at the same time. The study also revealed that the prevalence of polypharmacy was highest among older adults with frailty in the hospital setting, with a prevalence of 71%(0.71) (Toh et al., Ageing Res Rev 2023, 83: 101811).

61:

Chronic medications refer to prescription medications that are taken according to a schedule for at least 90 days, as opposed to being taken on an as-needed basis (Masnoon et al., BMC Geriatr 2017, 17: 230). This excludes topical medications, such as creams, or over-the-counter medications.

62:

This criterion captures open fractures of the fibula, radius, ulna, hand, or foot, which are not designated as inpatient but may require a staged repair.

63:

In some cases, it may not be possible to complete definitive soft tissue coverage during the initial wound debridement for an open fracture. In such situations, a staged approach to open fracture repair is required. This involves temporarily stabilizing (e.g., external fixation) the fractured bone and performing internal fixation and wound closure at a later stage (Rupp et al., Injury 2020, 51 Suppl 2: S57-s63).

64:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

65:

Acute coronary syndrome refers to any constellation of clinical symptoms that are compatible with acute myocardial ischemia. It encompasses ST elevation MI (STEMI) and Non-ST elevation MI (NSTEMI) and unstable angina.

Unstable angina presents as: new onset angina (less than 2 months), rest angina (begins at rest), increasing angina (frequency, intensity, or duration). It is defined by ECG ST segment depression or prominent T-wave inversion.

Acute myocardial infarction (AMI) is defined as a detection of the rise and/or fall of cardiac biomarkers (troponin or CPK-MB) together with evidence of myocardial ischemia. There are two types of AMI:

- NSTEMI is characterized by ECG ST-segment depression or T wave inversion and/or positive biomarkers in the absence of ST elevation
- STEMI is characterized by ECG ST-segment elevation and positive biomarkers

(Antman et al., Circulation 2004; 110(9): e82-292; Anderson et al., J Am Coll Cardiol 2007; 50(7): e1-e157; Thygesen et al., J Am Coll Cardiol 2007; 50(22): 2173-2195)

66:

Cardiac biomarkers are used to detect myocardial injury. These markers assist in the diagnosis or exclusion of a myocardial infarction and must be interpreted in conjunction with the patient's symptoms, examination, and electrocardiogram findings. Cardiac biomarkers are assessed in all patients with suspected acute coronary syndrome and should include a cardiac troponin assay. Compared to other cardiac biomarkers, troponin assays have high specificity and sensitivity when used serially to identify myocardial cell death. Troponin assays can detect the majority of myocardial infarctions within 3 hours of presentation. In some cases, detection may be delayed up to 12 hours after the onset of symptoms and repeat levels are recommended at 8 to 12 hours after symptom onset if initial tests are negative. Cardiac biomarker levels defining a positive value are determined by laboratory protocol and may vary between hospitals (Thygesen et al., Journal of the American College of Cardiology 2018, 72: 2231-64; Ibanez et al., European heart journal 2017, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

Instruction: While there are a variety of medical conditions that can lead to troponin elevation, the intent of this criterion is to capture patients with positive troponins due to acute coronary syndrome. Those patients with elevated troponins stemming from demand ischemia should be reviewed in the subset that corresponds to the source of their demand (e.g., pneumonia, heart failure, renal failure, gastrointestinal bleeding, etc.).

67:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

68:

Although the vast majority of wound care is readily handled in the post-acute setting, there are situations when a patient requires care at the acute level, such as high-frequency wound care required, prolonged length of time required performing the procedure, and the type or amount of medication required to keep the patient comfortable during the procedure. For children less than 12 years of age (or less than 30 kg), the wound care regimen may need to be adjusted to a lesser frequency and/or duration (e.g., twice per day, 10 minutes), though still requiring the current level of care, due to the patient's size, pain tolerance, fluid shifts, and vital sign lability.

To determine the appropriate discharge destination, an assessment is required of the patient and/or caregiver's ability to learn the necessary wound care regimen, complexity of the treatment regimen, and a plan for managing contributory factors. For discharge to the home setting, the patient and/or caregiver must be able and willing to participate in the care plan, or there is sufficient agency assistance available to adequately meet the patient's needs. Patient and/or caregiver instruction should include all the following (Bryant and Nix, Acute & chronic wounds:

current management concepts, 5th ed. 2015, xiv, 655 p.):

- Wound care regimen that includes a plan to monitor patient-specific contributory factors
- Adjunctive therapies (medications, nutritional plan, and supplements)
- Re-ordering medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Continued follow-up and support with health care providers

69:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

70:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

71:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics. Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

72:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

73:

Neurological deficit(s) may include:

- Altered level of consciousness
- Cognitive deficit
- Motor deficit
- Speech deficit
- Sensory deficit

Neurological disorder may include:

- Amyotrophic lateral sclerosis (ALS)
- Dementia (including Alzheimer's disease)
- Guillain-Barre syndrome (GBS)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Parkinson's disease (PD)
- Transverse myelitis

74:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

75:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., *PulmCCM* 2018; Nardi et al., *F1000Res* 2017, 6: 290; Osadnik et al., *Cochrane Database Syst Rev* 2017, 7: CD004104; Allison and Winters, *Emerg Med Clin North Am* 2016, 34: 51-62; Gregoretti et al., *Crit Care Clin* 2015, 31: 435-57).

76:

Post operative cardiac monitoring may be performed for the following surgical procedures:

- Carotid endarterectomy
- Lobectomy
- Neurosurgery
- Pneumonectomy
- Solid organ transplant
- Thoracoabdominal aortic repair
- Ventriculoperitoneal (V-P) shunt
- Wedge resection

These patients must be hemodynamically stable.

77:

Instruction: The decision to admit a patient for a surgical procedure remains the responsibility of the treating provider. Determination of the appropriate setting (inpatient versus outpatient) is a clinical decision best made with consideration of multiple clinical factors including, but not limited to (Victor M. Zaydfudim, In: Sabiston: Textbook of Surgery: The Biological Basis of Modern Surgical Practice, 21 edn. 187-222, 2022; American Society of Anesthesiologists, ASA Physical Status Classification System. 2020; Fox et al., *Surgery* 2014, 155: 743-53):

- Risk of procedure (low, intermediate, or high)
- High risk comorbid conditions (e.g., cardiac function, diabetes, renal function)
- Likelihood of complications based on patient's medical history (e.g., bleeding, infection, thromboembolic events, fluid or electrolyte imbalance)

This may differ based upon legislative and geographic variances and may impact organizational policy. Documentation of the patient's clinical condition is essential to ensure the appropriate setting and level of care required. InterQual's® *Solutions Guidelines for Surgery and Procedures in the Inpatient Setting* can be used to determine when a procedure might be appropriate for the inpatient setting.

78:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., *Cochrane Database Syst Rev* 2016: CD011826).

79:

Systemic corticosteroids can improve lung function, oxygenation and shorten recovery time and hospitalization duration in individuals with chronic obstructive pulmonary disease (COPD) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024). The administration of low-dose oral steroids demonstrates equivalent outcomes when compared to high-dose intravenous therapy for the treatment of COPD (Lindenauer et al., JAMA 2010; 303(23): 2359-2367). Duration of therapy should not be more than 5-7 days in acute COPD exacerbation (Walters et al., Cochrane Database Syst Rev 2018, 3: CD006897; Leuppi et al., JAMA 2013, 309: 2223-31).

80:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

81:

New York Heart Association (NYHA) classification for heart failure is defined as follows (Yap et al., Clinical cardiology 2015, 38: 621-8; New York Heart Association, Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis. 1964):

- Class I - No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.
- Class II - Slight limitation of physical activity. Comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.
- Class III - Marked limitation of physical activity. Comfortable at rest. Less than ordinary physical activity causes fatigue, palpitation, dyspnea, or anginal pain.
- Class IV - Inability to carry on any physical activity without discomfort. Symptoms of cardiac insufficiency or of anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.

82:

Patients with obstructive sleep apnea (OSA) undergoing invasive surgical procedures are at higher risk for respiratory desaturation, respiratory failure, cardiac events, and transfers to the critical care unit than patients who do not have OSA (Kaw et al., Br J Anaesth 2012, 109: 897-906).

83:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

84:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

85:

There have been significant improvements in immunosuppressant therapy for transplant recipients. Early protocols included azathioprine and corticosteroids, but with the addition of calcineurin inhibitors in solid organ transplantation, long-term survival and recovery has been improved. A multi-drug approach to therapy allows lower dosing of potentially very toxic individual agents. The 3 phases of immunosuppressive therapy include (Enderby and Keller, The American journal of managed care 2015, 21: s12-23):

- **Induction** - Immediately after transplant, when the risk for rejection is highest, intense antibody therapy or higher doses of maintenance therapy immunosuppressive agents are given depending on the organ being transplanted and the transplant center. Antibody therapies are T-cell depleting polyclonal (e.g.,

antithymocyte globulin (ATG)) and monoclonal (e.g., alemtuzumab or anti- CD52 antibody) or nondepleting agents (e.g., basiliximab).

- **Maintenance** - Transplant recipients remain on one or a combination of maintenance immunosuppressive therapy for the rest of their lives. Calcineurin inhibitors (e.g., cyclosporine, tacrolimus) are the mainstay of maintenance therapy with additional agents including antiproliferative/antimetabolites (e.g., mycophenolate mofetil (MMF), mycophenolic acid (MPA)), corticosteroids, mammalian target of rapamycin (mTOR) inhibitors (e.g., sirolimus), or T-cell costimulation blockers. The most common combination of these post transplant (kidney, liver, lung, heart, or pancreas) include tacrolimus, MMF/MPA, and prednisone. Immunosuppressant therapy may cause opportunistic infections, cancers and other adverse side effects. To limit these effects and increase survival and quality of life, research is underway that includes the infusion of marrow-derived autologous mesenchymal stem cells (MSCs), xenotransplantation, and advancements in regenerative medicine.
- **Rejection** - Anti-rejection therapy is given to reverse an episode of rejection and is based on the severity and type of rejection, as well as the hemodynamic consequences of rejection. Treatment includes immunosuppressant medications given for treating an acute rejection during the initial post-transplant and may consist of a medication adjustment, steroid administration, or the addition of T-cell depleting or nondepleting agents.

86:

Assessment of graft function may include laboratory or imaging studies. Serial assessment is often required during the immediate post-transplant period.

87:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

88:

Acute humoral rejection (AHR) is caused by antibodies that form within 2 to 7 days after transplant and presents with an acute loss of graft function. AHR is diagnosed by a C4D stain positive for circulating alloantibodies. Treatment is a combination of plasmapheresis (to remove antibodies) and heightened immunosuppression (tacrolimus, mycophenolate, steroids with muromonab-CD3 added) (Colvin, J Am Soc Nephrol 2007; 18(4): 1046-1056).

89:

Apheresis is a procedure that separates and removes elements in the blood (e.g., white blood cells (WBCs), platelets, stem cells, plasma bound organic compounds, or antibodies). Subsequent replacement with blood components, solvent detergent plasma (SDP), albumin, and/or normal saline may follow apheresis.

90:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

91:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

92:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

93:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance

- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Depret et al., *Ann Intensive Care* 2019, 9: 32; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138).

94:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

95:

Blood pressure is considered labile when it is extremely variable and difficult to control.

96:

Instruction: This criterion includes administration of aerosolized medication via nebulizer or inhaler. Appropriate selection of a delivery device for aerosol therapy depends on multiple factors including cognitive state or developmental stage, manual dexterity, breathing pattern, respiratory drive, and drug formulation. Efficacy is optimized when the device is used correctly and complements the patient's age, physical characteristics, and cognitive ability (Tashkin, *Int J Chron Obstruct Pulmon Dis* 2016, 11: 2585-96; Brittan et al., *J Pediatr* 2015, 166: 998-1005 e1).

97:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., *Neurocrit Care* 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

98:

Procedural sedation is the administration of a sedative, with or without analgesia, to induce an altered state of consciousness while preserving cardiorespiratory function, to allow a patient to tolerate a painful procedure. The target level of sedation should be specific to the needs of the patient, what the procedure involves, and the duration of the procedure (Godwin et al., *Ann Emerg Med* 2014, 63: 247-58 e18; National Institute for Clinical Excellence (NICE), Clinical guideline 112. NICE 2010, 37p.).

99:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., *J Thromb Haemost* 2013:).

100:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or

cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

101:

The mainstay of treatment of patients with hemorrhagic shock is blood product transfusion (e.g., packed red blood cells). Despite the general principle of minimizing the use of intravenous fluids in patients with known hemorrhage, crystalloids are often administered until blood products are available, especially in patients who are hypotensive (e.g., mean arterial pressure less than 65). Vital sign changes may underestimate blood loss in patients. Patients with a 30%(0.30) loss of total blood volume may have little to no change in systolic blood pressure (SBP) and only demonstrate modest tachycardia (i.e., heart rate greater than 100). By the time an SBP has dropped to less than 100 mmHg, there may be as much as a 40%(0.40) loss of total blood volume.

102:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

103:

Hypercalcemia (Mousseaux et al., Ann Intensive Care 2019, 9: 133; Turner, Clin Med (Lond) 2017, 17: 270-3; Maier and Levine, J Intensive Care Med 2015, 30: 235-52).

104:

Calcium levels within criteria represent total serum calcium.

105:

In hypercalcemia, saline diuresis is administered to maintain an output of 200 mL per hour.

106:

Abnormalities of calcium can have significant impact on morbidity and mortality due to its crucial role in neuromuscular and cardiac function. Patients who present with symptoms usually have an abrupt decline in ionized calcium.

Common symptoms of hypocalcemia include:

- Perioral numbness
- Carpopedal spasm of the hands or feet
- Flaccid paralysis
- Muscle weakness
- Paresthesia
- Tetany

Symptomatic hypocalcemia requires treatment with intravenous calcium (Turner et al., Endocr Connect 2019, 8: X1).

107:

Reference: (Jordan and Caesar, BMJ Qual Improv Rep 2015, 4:)

108:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

109:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

110:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

111:

Hypernatremia (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203)

112:

Hypovolemic patients should be resuscitated with 0.9% sodium chloride regardless of the serum sodium level until their vital signs are normal. Normovolemic patients with hypernatremia should receive either 0.45% sodium chloride or 5% dextrose infusion (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203).

113:

Symptoms of hyponatremia may include nausea, lethargy, headache, and muscle weakness. If hyponatremia worsens, symptoms may progress to mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, obtundation, and seizures should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Weismann et al., Endocr Connect 2016:).

114:

Symptoms of hyponatremia include:

- Coma
- Headache
- Lethargy
- Nausea
- Obtundation
- Seizures
- Stupor
- Vomiting

115:

Fluid restriction should be avoided during the first 24 hours of therapy with conivaptan or tolvaptan as it may increase the likelihood of an overly rapid correction of serum sodium and may lead to the development of cerebral edema.

116:

Hypovolemic hyponatremia may be treated with sodium chloride 0.9% (0.009). For severe hyponatremia, treatment with an infusion of sodium chloride 3% (0.03) may be required. Rapid correction of serum sodium can cause spastic quadriparesis or quadriplegia, pseudobulbar palsy, dysarthria, mutism, pupillary or oculomotor abnormalities, stupor, coma, ataxia, dystonia, osmotic demyelinating syndrome, or even death (Weismann et al., Endocr Connect 2016:).

117:

Hypophosphatemia (Megapanou et al., Drug Saf 2020, 43: 197-210)

118:

An intra-aortic balloon pump (IABP) is a mechanical circulatory support device that utilizes a catheter-mounted intravascular device positioned in the descending thoracic aorta. The balloon automatically inflates during early diastole, increasing coronary blood flow and deflates during early systole, reducing afterload as the left ventricle ejects. Indications for use include cardiogenic shock, temporary support of left ventricular function after myocardial infarction (MI), preservation of myocardial viability, or refractory ventricular arrhythmias post-MI or preoperative bridge before coronary artery bypass graft (CABG) surgery. Contraindications include aortic regurgitation and aortic dissection.

119:

Procedures designated as major procedures may require management at the Critical level of care due to the risk of decompensation or the need for intensive monitoring and interventions due to the complexity of the procedure.

Major procedures may include:

- Cranial or thoracic repair
- Spinal fusion or stabilization
- Repair or replacement of large vessels (e.g., aorta, vena cava, renals, or iliacs)
- Abdominal resection or repair (e.g., pancreas, liver, transplant, kidney, prostate, colon, small bowel, or stomach)
- Significant blood loss (in adults: greater than 750 cc; in children: ≥ 10 ml/kg)
- Procedures where significant fluid shifts are expected in patients with limited compensatory mechanisms due to medical comorbidity (e.g., heart failure, coronary artery disease, serious heart disease (e.g., cyanotic heart disease), chronic lung disease, chronic obstructive pulmonary disease (COPD), acute or chronic renal insufficiency)

120:

Malignant hyperthermia associated with anesthesia is an inherited disease manifested by a rapid increase in body temperature, which is often lethal if not promptly treated. This disease is due to abnormal calcium metabolism within the muscle fiber. Exposure to a triggering drug such as a general anesthetic, succinylcholine, or a local anesthetic, causes persistent skeletal muscle contraction. This results in hypermetabolism, including tachycardia, arterial hypoxemia, acidosis, and profound hyperthermia. According to the Malignant Hyperthermia Association of the United States (MHAUS), treatment during the post-acute phase of a malignant hyperthermia crisis includes observation at the critical level of care for a minimum of 24 hours, dantrolene administration, temperature regulation, laboratory monitoring (including potassium, creatine kinase, myoglobin, arterial blood gases, and glucose), hydration, and maintaining urine output greater than 2 mL/kg/hr. The European Malignant Hyperthermia Group (EMHG) also includes the high dependency unit and recovery unit as appropriate levels of care for observation following a crisis (Rusyniak and Sprague, Clin Lab Med 2006; 26(1): 165-184, ix; Malignant Hyperthermia Association of the United States, Emergency Therapy for Malignant Hyperthermia. 2008; Glahn et al., Br J Anaesth 2010, 105: 417-20).

121:

Malignant hyperthermia involves a sudden increase in core body temperature. Temperature regulation begins with active cooling. Once the core body temperature reaches less than 100.4-101.3°F(38.0-38.5°C), active cooling should be discontinued to avoid subsequent hypothermia. Active cooling measures include intravenous infusion of cold saline, application of ice to body surfaces, and iced saline lavage of the stomach, bladder or rectum. Measurement and monitoring of core body temperature is recommended following the malignant hyperthermia crisis (Malignant Hyperthermia Association of the United States, Emergency Therapy for Malignant Hyperthermia. 2008; Glahn et al., Br J Anaesth 2010, 105: 417-20).

122:

Necrotizing fasciitis (NF) is a life-threatening, rapidly spreading soft tissue infection resulting in progressive destruction of the fascia and subcutaneous tissue in any anatomic location. Group A hemolytic streptococci and *Staphylococcus aureus* are the most common causes. Manifestations of streptococcal NF appear within 24 hours of

the initial lesion. Localized painful erythema and edema lead to the development of bullae within 3 to 5 days that progress to necrosis of the skin, sloughing, and cutaneous gangrene. The patient may experience a high fever, altered mental status, tachypnea, tachycardia, and hypocalcemia. The cornerstone of treatment is extensive surgical debridement to remove all nonviable fascia and tissue and empiric antibiotic therapy (Endorf et al., J Burn Care Res 2009, 30: 769-75) (Compton, Clin Geriatr Med 2013, 29: 443-59).

123:

Cardiac transplantation provides the greatest survival benefit and remains the gold standard treatment in patients with heart failure (HF), but limited availability of donor hearts make ventricular assist devices (VAD) a viable alternative. A VAD is a mechanical pump that assumes the function of the damaged ventricle and helps restore normal hemodynamics and end-organ blood flow. Pumps provide either pulsatile or continuous flow and may be implanted in one or both ventricles:

- Bi-ventricular assist device (BiVAD)-replaces the function of both ventricles
- Left ventricular assist device (LVAD)-replaces the function of the left ventricle
- Right ventricular assist device (RVAD)-replaces the function of the right ventricle

VAD insertion is currently approved as:

- Bridge to recovery: in patients with reversible ventricular dysfunction after cardiac surgery or myocardial infarction (MI)
- Bridge to transplantation: as temporary support for patients with end-stage HF awaiting transplant
- Destination therapy: as a permanent cardiac replacement device for patients who are not candidates for heart transplant

There are some devices that are not portable and require the patient to remain hospitalized while the device is in place. Some VADs can be inserted percutaneously and therefore, may be used more quickly than traditional, more invasive VADs (Miller and Rogers, JAMA Cardiol 2018;; Mancini and Colombo, Journal of the American College of Cardiology 2015, 65: 2542-55; Maini et al., Catheter Cardiovasc Interv 2014, 83: E183-92).

124:

In some facilities a ventricular assist device (VAD) will be used as a bridge to heart or heart-lung transplant. This may stabilize a patient long enough to survive until transplant, but the patient is still fragile and relatively unstable.

Instruction: In this clinical setting, the patient may require Critical care until a donor organ is available. Due to the restrictions associated with these devices, the Critical level of care is required for percutaneous VADs.

125:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

126:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

127:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

128:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

129:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

130:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

131:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

132:

Instruction: Application of this criteria point is appropriate for patients who require post operative management **and** active management of a comorbid condition or risk factor.

133:

Examples of procedures considered moderate stay include:

- Appendectomy, Open
- Arthrotomy, Hip
- Cervical, lumbar, or thoracic fusion
- Cholecystectomy, open
- Colectomy, partial
- Coronary artery bypass graft (CABG)
- Craniotomy
- Exploratory Laparotomy
- Small Bowel Resection

This list is not intended to be all inclusive.

134:

Examples of procedures designated as long stay include:

- Amputation of leg or foot
- Liver transplantation
- Mitral Valve Replacement (MVR) or Repair
- Pancreatectomy (subtotal, total)
- Pancreatic resection (includes Whipple)
- Tricuspid Valve Replacement (TVR) or Repair or Resection

This list is not intended to be all inclusive.

135:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

136:

Instruction: For this criteria point, repositioning refers to adjustment of the chest tube.

137:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

138:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

139:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

140:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

141:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

142:

Bisphosphonates and corticosteroids are more commonly used in the treatment of hypercalcemia related to malignancy.

143:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

144:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

145:

Examples of potassium binding agents include but are not limited to:

- Sodium polystyrene sulfonate (Kayexalate® or Kionex®)
- Sodium zirconium cyclosilicate (Lokelma®)

- Patiromer (Veltassa®)

146:

Hyponatremia (Seay et al., Am J Kidney Dis 2020, 75: 272-86)

147:

An ileus is a blockage, partial or complete, or paralysis of the bowel that prevents the passage of intestinal contents from moving forward. It is more common in the small bowel, as opposed to large bowel. It is often the result of surgical adhesions or hernia. Small bowel ileus symptoms typically present as acute and severe, with complaints of nausea and vomiting, cramps, bloating, and the retention of stool and flatus. It can often be treated medically. Large bowel ileus usually has a milder presentation and often is due to cancer. It may also be the result of adhesions or stenoses from recurrent diverticulitis or, less commonly, is associated with a sigmoid volvulus or hernia (Vilz et al., Dtsch Arztebl Int 2017, 114: 508-18).

148:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

149:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

150:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

151:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

152:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

153:

The anticoagulation regimen may include low molecular weight heparin (LMWH), unfractionated heparin (UFH), or fondaparinux with or without warfarin, or warfarin alone. For adult patients, non-vitamin K oral anticoagulants (NOACs) (also known as novel oral anticoagulant, direct oral anticoagulant (DOAC), and target-specific oral anticoagulant) such as dabigatran, rivaroxaban, apixaban, or edoxaban, may also be used (Ortel et al., Blood Adv 2020, 4: 4693-738).

154:

In patients with a deep vein thrombosis (DVT) or pulmonary embolism (PE) without warfarin contraindication, warfarin and parenteral anticoagulation should be administered for at least 5 days and until the international normalized ratio (INR) is greater than or equal to 2.0. Home parenteral anticoagulation (overlap therapy) is recommended if the patient is discharged prior to 5 days of combination therapy (Kearon et al., Chest 2016, 149: 315-52).

155:

Warfarin dosage is adjusted according to the following international normalized ratio (INR) values (Kernan et al., Stroke 2014, 45: 2160-236; Nishimura et al., Circulation 2014, 129: e521-643; Whitlock et al., Chest 2012, 141: e576S-600S):

- Range of 2.0-3.0 with a target of 2.5 for patients with bioprosthetic valves, mechanical aortic valves, or for patients without implanted mechanical valves
- Range of 2.0-3.0 with a target of 2.5 for patients with ischemic stroke or transient ischemic attack (TIA) with a history of rheumatic mitral valve disease and atrial fibrillation
- Range of 2.5-3.5 with a target of 3.0 for patients with mechanical mitral valves or combination mitral/aortic valve implantation

156:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

157:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

158:

Instruction: When conducting a continued stay review during the immediate post op period for a short, moderate, long stay, or transplant procedure, "Routine post-op stay" criteria should be selected. Continued stay reviews which go beyond the designated LOS of any procedure require "Complication or comorbid condition" criteria to be selected to complete the review.

159:

Hypertension is defined as a systolic blood pressure (SBP) of greater than 130 mmHg or a diastolic blood pressure (DBP) of 80 mmHg or greater. Recommendations for initial management may include a conservative approach with diet and lifestyle modification with or without medications. The goal of antihypertensive therapy is to reduce morbidity and mortality. In non-urgent situations, blood pressure (BP) control is achieved in the outpatient setting. Goals for target BP level should be individualized, but a SBP less than 130 mmHg and a DBP less than 80 mmHg are desirable. Populations at high risk include patients with the following conditions (Whelton et al., Hypertension 2018, 71: 1269-324):

- Clinical cardiovascular disease or 10-year atherosclerotic cardiovascular disease (ASCVD) risk 10%(0.1)
- Heart failure
- Stable ischemic heart disease
- Chronic kidney disease
- Chronic kidney disease after renal transplantation

160:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

161:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

162:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

163:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

164:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory

status. The patient will require assistance with one or more limitations.

165:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

166:

Graft dysfunction is usually treated empirically with anti-rejection therapy while awaiting the results of the biopsy. Immunosuppressive medications are initiated based on the signs and symptoms of organ dysfunction or graft versus host disease (GVHD).

167:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

168:

Medication adjustments tend to be driven by patient response. Patients being treated for symptoms of rejection usually have a biopsy and their medications adjusted based on the biopsy results to control the rejection.

169:

All medications commonly used to prevent rejection or opportunistic infections in the post-transplant period cause some degree of side effects. One of the important tasks of the transplant team is to titrate the dosage to minimize toxicity while maximizing benefit. Often this fine-tuning is empiric. If a patient has an intolerable side effect from a particular drug, an alternative medication or different combination or dosage of medications is attempted. Tacrolimus and cyclosporine (CSA) are immunosuppressive agents that inhibit calcineurin and are given to solid organ transplant recipients to prevent and treat allograft rejection. Tacrolimus is less nephrotoxic and may be more cost-effective than CSA. However, new-onset diabetes should be closely monitored during the medication period. The dosing of CSA is more difficult to adjust appropriately in children, especially in younger/smaller children (less than 2 years). This is thought to be due to their shorter intestine, leading to less absorption of the drug, and also their faster metabolism. More time may be needed to adjust the dose and determine the appropriate medication management in this population than in older children or adults. The goal is to preserve the graft without seriously harming the patient.

170:

Treatment with low molecular weight heparin (LMWH) is recommended, rather than treatment with unfractionated heparin (UFH), in patients with acute deep vein thrombosis (DVT) or pulmonary embolism (PE) (Kearon et al., Chest 2016, 149: 315-52). Both LMWH and fondaparinux levels may be elevated in patients with renal insufficiency. Fondaparinux is contraindicated in patients with a creatinine clearance less than 30 mL/min. LMWH should be used with caution in patients weighing less than 50 kg due to unpredictable pharmacokinetics. In these patients, UFH is preferable (Minichiello and Fogarty, Med Clin North Am 2008, 92: 443-65, x).

171:

Heparin-induced thrombocytopenia (HIT) results from sensitization to the heparin molecule. Usually, it occurs between 5 to 10 days after the current heparin exposure but may occur as early as minutes to hours after exposure. HIT is a greater than 50%(0.50) serial reduction in a patient's baseline platelet count. A direct thrombin inhibitor (DTI) (e.g., argatroban, bivalirudin) is the initial treatment for patients with active HIT, a history of HIT, or those with a strong clinical suspicion of HIT who require anticoagulation. Although the American Society of Hematology (ASH) guidelines suggest the use of fondaparinux or direct oral anticoagulants (DOACs), the recommendation is conditional and based on very low certainty of evidence. In addition, these agents are considered off-label use in the treatment of HIT since they have yet to be approved by the U.S. Food and Drug Administration (FDA) for this condition. For patients who require long-term anticoagulation, warfarin should not be initiated until platelet count recovers (e.g., platelet count greater than $150 \times 10^9/L$). Additionally, a 5-day overlap with warfarin and DTI therapy is recommended (Cuker et al., Blood Adv 2018, 2: 3360-92; Guyatt et al., Chest 2012, 141: 48S-52S).

172:

Although the American Society of Hematology (ASH) guidelines for management of venous thromboembolism (VTE) recommend offering home treatment over hospital treatment for uncomplicated deep vein thrombosis (DVT), this recommendation does not apply to individuals who have inadequate home support, lack of access to medication, or a history of poor compliance (Ortel et al., Blood Adv 2020, 4: 4693-738). Patients with these risk factors may require inpatient admission for bridging or pre-treatment with parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin) prior to being converted to an oral regimen.

173:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting
- Skilled nursing services unavailable (e.g., Home Care)

174:

Instruction: Patients who have not achieved a therapeutic INR on 2 consecutive days and continue on LMWH or fondaparinux will meet this criterion.

According to the American Society of Hematology (ASH), when warfarin is being initiated, it should be overlapped with parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin (LMWH)) until a therapeutic international normalized ratio (INR) of 2.0-3.0 is achieved (Ortel et al., Blood Adv 2020, 4: 4693-738).

175:

Patients to be treated with dabigatran or edoxaban, both direct oral anticoagulants (DOACs), require at least 5 days of parenteral anticoagulation prior to initiation (Ortel et al., Blood Adv 2020, 4: 4693-738).

Instruction: This criteria point is intended for patients who require continued hospitalization for pretreatment with parenteral anticoagulation (e.g., low molecular weight heparin (LMWH) or fondaparinux) prior to starting dabigatran or edoxaban.

176:

Anticoagulation includes low molecular weight heparin (LMWH) or fondaparinux.

177:

Wound care teaching should include instruction on the following:

- Adjunctive therapies (medications, nutritional plan, and supplements)
- Follow-up plan of care
- Medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Wound care regimen (including patient-specific factors)

178:

This criterion is intended to be used for the patient who returns to the operating room (OR) in the event of a complication (e.g., bleeding, fistula repair, intra-abdominal infection, wound or ostomy revision) or staged procedure (e.g., open fracture definitive soft tissue coverage and internal fixation).

179:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

180:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.

The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

181:

Instruction: This criterion applies to patients with actual, suspected, or a history of heparin-induced thrombocytopenia (HIT).

182:

There are 2 types of thrombi which can develop in a ventricular assist device (VAD) (Bartoli et al., J Card Surg 2014, 29: 83-94):

- A red thrombus occurs when stagnating blood coagulates due to low pressure and is formed when fibrin deposits trap erythrocytes. This type of thrombus is best managed with anticoagulation medications.
- A white thrombus occurs in areas of turbulence and is composed of activated platelets and is treated with antiplatelet agents.

183:

Absolute contraindications to thrombolytic therapy include prior or suspected intracranial bleed, prior stroke, intracranial aneurysm, arteriovenous malformation, neoplasm, closed head or facial trauma, active internal bleeding, suspected aortic dissection, known bleeding diathesis, or uncontrolled hypertension (greater than 180 systolic, greater than 100 diastolic) (Bartoli et al., J Card Surg 2014, 29: 83-94).

184:

Reference: (Rutledge et al., J Heart Lung Transplant 2013, 32: 569-73)

185:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit

supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

186:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection.

To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm} (1.5 \times 10^9/L)$

$= 330/\text{cu.mm} (330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

187:

Patients with a ventricular assist device (VAD) are at risk for developing hypovolemia, bleeding, infection, thromboembolism, and arrhythmias. Stability in patients with a VAD is often patient specific. Symptoms of instability in a patient with a VAD include fluctuating vital signs, decreased VAD flows, or decreased level of consciousness.

188:

Evidence of adequate perfusion includes intact mentation, adequate urine output, warm extremities, or the presence of peripheral pulses.

189:

In continuous flow axial pumps, measurement of systolic blood pressure is not possible because there is no audible closure of the aortic valve. The sound heard during auscultation represents the mean arterial pressure (MAP). Use of a Doppler flow probe is helpful in defining blood pressure in patients with a ventricular assist device (VAD). In patients with continuous flow pumps, the goal MAP is between 70 and 90 mmHg. The use of multiple vasoactive or antihypertensive agents may be required to achieve this goal (Mancini and Colombo, Journal of the American College of Cardiology 2015, 65: 2542-55; Wilson et al., J Am Coll Cardiol 2009, 54: 1647-59).

190:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

191:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack

of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

192:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

193:

Patient and/or caregiver education and discharge instructions include the following:

- Activity restrictions
- Care of wounds, drains and tubes
- Dietary modifications and/or restrictions
- Medication management: route, dosage, and indications; recognition of adverse reactions and interactions with other medications or food
- Recognition of symptoms to report to the physician: bleeding, pain, or redness at surgical site, increased or purulent drainage, fever, and any adverse medication reactions
- When to seek medical care and the importance of adhering to the follow-up schedule

194:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

195:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer

2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

196:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

197:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

198:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

199:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

200:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

201:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

202:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

203:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

204:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

205:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its

intensity)

- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

206:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

207:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

208:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

209:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

210:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia

- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

211:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.